Network UPS Tools Release Notes

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REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.8.5.1628 2.8.5.1645- 1645+g3505443e4	2026-10-18	Current release snapshot of Network UPS Tools (NUT).	
2.8.5	2026-04-07	Some changes to docs and recipes, especially for parallel builds; introduced a way to build with private shared libraries to save space. Updated NUT SEMVER definition some more, added a normalization and comparison script. Some rounds of code-hardening project. Added reference RPM/DEB recipes to test NUT iterations on OBS. Numerous driver updates, some new ones introduced.	JK
2.8.4	2025-08-07	Fixed a few regressions introduced by release v2.8.3. Some changes to docs and recipes, especially for parallel builds. Updated NUT SEMVER definition some more. Some rounds of code-hardening project. Numerous driver updates, some new ones introduced.	JK
2.8.3	2025-04-07	Some changes to docs and recipes, libupsclient API and functionality. Updated NUT SEMVER definition and added scripting around it. Groundwork for vendor-defined status and INSTCMD buzzwords like "ECO". Fixed some regressions and added improvements for certain new device series.	JK
2.8.2	2024-04-01	Some changes to docs and recipes, libnutscan API and functionality. Added nutconf (library and tool). Fixed some regressions and added improvements for certain new device series.	JK
2.8.1	2023-10-31	Some changes to API, docs and recipes, in particular to simplify local builds and tests (e.g. to help end-users check if current NUT codebase trunk has already fixed an issue they see with a packaged installation). Revived NUT for Windows effort, further improved other OS integrations. NUT became reference for "UPS management protocol", Informational RFC 9271. Documentation files refactored to ease maintenance. More drivers and new driver categories introduced.	JK

REVISION HISTORY

NUMBER	DATE	DESCRIPTION	NAME
2.8.0	2022-04-26	Change of maintainer. Many changes to API, docs (both style and content), and recipes, with a stress on non-regression test-ability, run-time debug-ability, general codebase maintainability, as well as OS integrations (notably nut-driver-enumerator for systemd and SMF service instance maintenance). Added a lot in area of CI support and documented pre-requisite package lists for numerous platforms, and CI agent set-up. Added libusb-1.x support and many new driver categories (and drivers), and daisychain device connection support. Instant commands enhanced with TRACKING to enable protocol-based waiting for completion of a particular INSTCMD or SET operation.	JK
2.7.4	2016-03-09	NUT variables namespace updated, in particular for outlet groups, alarms and thresholds, ATS devices, and battery.charger.status to supersede CHRG and DISCHRG flags published in ups.status readings. NUT network protocol extended with NUMBER type; some API changes.	AQ
2.7.3	2015-04-22	Documentation revised, including some API changes. Added NUT DDL links. NUT variables namespace updated.	AQ
2.7.2	2014-04-17	The nut-website project was offloaded into a separate repository. FreeDesktop HAL support was removed (obsoleted in GNOME consumer). Introduced nutdrv_atcl_usb driver.	AQ
2.7.1	2013-11-19	NUT source codebase migrated from SVN to Git (and from Debian infrastructure to GitHub source code hosting). jNut binding split into a separate project. Introduced libnutclient (C++ binding), al175, apcupsd-ups and nutdrv_qx drivers, Mozilla NSS support for simpler licensing than OpenSSL, and a newer apcsmart implementation. Documentation support enhanced with a spell checker, contents massively updated to reflect project changes.	CL

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.6.5	2012-08-08	New macosx-ups driver, new implementation of mge-shut driver. NUT variables namespace updated. Docs cleaned up and revised.	AQ
2.6.4	2012-05-31	New NUT network protocol commands (LIST CLIENTS, LIST RANGE and NETVER), and socket protocol commands (ADDRANGE, DELRANGE). NUT variables namespace updated. Introduced nut-recorder tool.	AQ
2.6.3	2012-01-04	No substantial changes to documentation.	AQ
2.6.2	2011-09-15	Introduced nut-scanner tool and nut-ipmipsu driver, systemd support, and a new apcsmart implementation.	AQ
2.6.1	2011-06-01	Introduced default.* and override.* optional settings in ups.conf, an ups.efficiency report, and outlet.0 special handling.	AQ
2.6.0	2011-01-14	First release of AsciiDoc documentation for Network UPS Tools (NUT).	AQ

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1 Introduction

The primary goal of the Network UPS Tools (NUT) project is to provide support for Power Devices, such as Uninterruptible Power Supplies, Power Distribution Units and Solar Controllers.

This document intends to report high-level changes delivered by NUT project releases, as well as practical nuances for packagers and end-users who would be upgrading their NUT deployments.

It DOES NOT intend to detail the change log: it is very large and complicated to render properly, so is better served by another document artifact.

If you wish to discover how everything came together, have a look at the Project History.

2 NUT Release Notes (and other feature details)

This document summarizes the practical side of changes coming with each newer release development, as compared to the preceding release.

If you're upgrading from an earlier version, or are a package maintainer, please see also the UPGRADING.adoc file about anticipated impacts of ongoing development on existing deployments and third-party consumers.

Please note that web and source document links, product and service names listed in historic entries of past releases may no longer be relevant.

For a complete and more detailed list of changes, please refer to the ChangeLog file (generated for release archives), or to the Git version control history for "live" codebase.

2.1 PLANNED: Release notes for NUT 2.8.6 - what's new since 2.8.5

https://github.com/networkupstools/nut/milestone/13

- (expected) Dynamic Mapping Files (DMF) feature supported, to allow the driver binaries to be built once and data mappings to be loaded and modernized on the fly (porting from 42ITy project)
- (expected) Porting of patches suggested by different distribution packages
- (expected) C code clean-up/consistency (string format security, work with Boolean values, string to number conversions, etc. in the same manner)
- (expected) clean-up of libusb API variants support [#300 and follow-ups]
- (expected) CI automation for coding style
- (expected) CI automation for driver flags and variables to be certainly documented, handled in augeas lenses, nutconf classes, etc.
- (expected) CI automation for use of data points in drivers that conform to patterns defined in docs/nut-names.txt
- Second-level bullet points listed in this file will now use 4 spaces (not 3 like before) for easier initial indentation of new entries.
- Fix fallout of development in NUT v2.8.0 through v2.8.5:
  - `libnutconf`: character reads from a TCP socket now use the Windows socket API, matching string reads. Previously they failed with a bad file descriptor error on Windows. [issue #2293]
  - Now `upssched -l` preserves timer expiry timestamps beyond January 2038 on systems with 64-bit `time_t` and 32-bit `long`. [issue #3096, PR #3669]
  - PyNUT: `GetUPSCommands()` now requests command descriptions with the advertised identifiers on Python 3 and uses those identifiers when extracting the response. Returned keys and descriptions remain byte sequences. [issue #3621]
  - `nut-scanner` tool updates:

- Mutexes used in parallelized scans were not properly released on failure code paths (impacts likely all 2.8.x until now). [PR #3551]
- Third-party libraries which `nut-scanner` loads dynamically (libusb, Net-SNMP, neon, Avahi, GLib/GIO, FreeIPMI) are now looked up by the `SONAME` recorded in the shared object seen at build time (such as `libusb-1.0.so.0`) before the fully versioned file name seen then (such as `libusb-1.0.so.0.6.0`). Previously an ABI-compatible update of such a library by the OS distribution, which renamed the file but kept the `SONAME`, silently disabled that scan type until the NUT package was rebuilt. [issue #3474]
- `nut-scanner` now really tries each name it knows for a library that it loads dynamically. A failed `lt_dlopen()` attempt was remembered in the module handle, so `nutscan_load_*_library()` answered from that memory and every name after the first one was skipped (impacts all NUT 2.8.x until now). This was only visible where such a library was not found by its file name and could be resolved by the run-time linker alone. [PR #3654]
- `nut-scanner` now reports a library it could not load with a single line at the default debug level, naming the scan type that is disabled as a result. The names it tried, and why each one failed, are logged at debug level 1 instead, since a failed name followed by a working one is not a failure. [PR #3739]
- When the NUT client library was not found under its known file names and had to be loaded by the libtool default search instead, the result was stored as the availability of the XML/HTTP scan. So the `-M` scan could be disabled despite a working `libneon`, or offered without one (impacts NUT v2.8.3 through v2.8.5). [PR #3657]
  - USB drivers: fixed an out-of-bounds read of the candidate HID report descriptor lengths, which could cause a segmentation fault (regression added in NUT v2.8.5 release). [PR #3550]
  - Refactoring of `configure.ac` for NUT v2.8.5 release led to a problem when explicitly specifying same `--with-user` **OR** `--with-group` name as would be default for current platform (so one was treated as not-set and failed a sanity check). [issue #3513]
- USB drivers now prefer the longer of conflicting HID report descriptor lengths for EcoFlow devices using USB ID `3746:ffff`. This makes the complete descriptor available on affected River 3 Plus firmware while retaining the shorter length as a fallback.
- Windows serial compatibility: fixed timeout handling to return data already received as a successful short read, while safely canceling and completing pending overlapped I/O before reusing its state. [issue #3452, PR #3560]
- Introduced and intensively evolved an experimental `apcmicrolink` driver for devices with the APC Microlink protocol on serial port connections.
  - Initial version, tested against APC Smart-UPS 750 (SMT750RMI2UC). [PR #3406]
  - Added USB HID support (`port = auto`, matching the usual NUT USB-driver convention), tunneling the Microlink protocol over USB HID for models with no serial port. Tested against APC Smart-UPS 500 (SCL500RMI1UC) by a single tester; considered experimental even relative to the rest of this already-experimental driver. See apcmicrolink(8) for USB-mode caveats, including a udev rule needed to keep the kernel's `usbhid` driver from claiming the device. [PR #3579]
  - USB mode: if the Microlink tunnel does not respond during startup, the driver now starts up on standard-HID-PDC fallback data (`ups.status`/`battery.charge`/`battery.runtime`) instead of refusing to start; outlet-group data and instant commands become available automatically once the tunnel connects, with no driver restart needed. [PR #3579]
  - Added `load.cycle` and per-outlet-group instant commands (`outlet.group.N.load.*`, `outlet.group.N.shutdown`, and the `outlet.group.N.switchable` variable to distinguish switchable groups from an unswitched main bank. [PR #3579]
  - USB mode: reworked the interrupt-IN read path to use a permanently outstanding asynchronous transfer serviced by a dedicated thread, so device replies are no longer missed while the driver is between polls. Several long-run stability fixes go with it: a USB bus reset is now triggered only on a genuine disconnect rather than on a retry count, the cached page 0 contents are no longer wiped on every session (re)start, the read queue is no longer flushed on every session retry, and the authentication challenge is randomized. Also warns once when built against the untested libusb-0.1 backend, and backs off instead of spinning when the kernel's `usbhid` driver has reclaimed the interface. [PR #3579]
  - Mapped two further descriptor usages: `experimental.battery.serial` (the battery pack's own serial number, distinct from `ups.serial`) and `microlink.diag.slave_password_echo` (auth troubleshooting visibility only, not proof that authentication was accepted). [PR #3579]

  - A page 0 whose announced frame width disagrees with the width of the frame that carried it is now rejected instead of cached. Such a page redefined the length the frame parser demanded, after which nothing checksum-validated again and the session could not recover. Affects both transports. [issue #3587]
  - Further updates, the driver remains experimental, but USB sessions should hold up better, and fall back more readily to standard HID PDC data when the Microlink tunnel is unstable or unavailable. [PR #3591]
  - USB mode: the standard-HID-PDC fallback now engages whenever the Microlink tunnel stops producing usable data, not only when it fails to answer during startup. A device that answers every poll while reporting an all-zero state previously left `ups.status` empty indefinitely. The driver hands back to Microlink data once that source looks plausible again, with hysteresis so the two do not swap on alternate polls.
  - The Microlink tunnel is now polled for a full page pass per update cycle, instead of one record per `pollinterval`. The device does not acknowledge its authentication handshake until roughly 20 exchanges after the response is sent, and serves no measurement page until it does, so the slower cadence could leave every measured value reading 0 on a device that was answering normally. Publishes `experimental.microlink.diag.auth_status` and `experimental.microlink.dia` and warns when the handshake goes unacknowledged.
  - The driver now closes the Microlink session with a STOP when it exits. Without it the device could answer a later client's session request mid-walk rather than at page 0, and would not re-synchronize until re-enumerated.
  - `2:4.7.28` and `2:4.7.49` are percentages of the nominal ratings rather than absolute readings. `ups.load` is now published from the former, with `ups.realpower` and `ups.power` derived against the nominal ratings, instead of both reading two orders of magnitude low. `ups.test.result` now comes from the battery-scope usage that a self test actually updates, and `ups.test.interval` is derived from the self-test schedule enumeration.
  - Newly mapped: `input.sensitivity`, `ups.beeper.status`, `experimental.battery.firmware` and `experimer`
  - Driver-specific diagnostic variables moved from `microlink.*` to `experimental.microlink.*`.
  - Documented the Microlink descriptor format that the driver parses, for anyone extending its variable maps. See the NUT Developer Guide.
  - `outlet.group.count` and the per-outlet-group commands/variables no longer require an unswitched "main" outlet group to be present; devices with only switched groups now expose them correctly. [issue #3587]
  - `ups.beeper.status` is decoded as a bitmask instead of matching exact enum values, so devices that set additional unrelated bits alongside the enabled/disabled bit report the correct status instead of a raw number. [issue #3587]
  - `beeper.mute` is replaced by `beeper.enable`/`beeper.disable`: the register it wrote had no confirmed effect, and the actual beeper on/off control used by PowerChute is a persistent setting on the same usage `ups.beeper.status` reads, not a temporary mute/re-arm action.
  - Further contributed `apcmicrolink` driver fixes for issues reported from a Smart-UPS X 1500 (SMX1500RM2UC, firmware "UPS 16.0") on the USB transport [issue #3587, PR #3714]:
  - The calibration result register (descriptor `2:13`) no longer leaks its labels into `ups.status`: a unit whose last runtime calibration had ended "Failed" reported `OL Failed` permanently. Only a calibration in progress is published there now, as `CAL`.
  - USB mode: the zero padding that follows each Microlink record in a fixed-size HID Input report is no longer fed to the byte-stream frame scanner. Leading zeros are invisible to the Microlink checksum, so a record ending in `0xFF` also validated one byte early, was taken for a (rejected) page 0, and was lost for that poll pass; on the test unit this happened two to three times a minute.
  - USB mode: the standard-HID fallback no longer publishes a status it never read. Its snapshot counted as fresh as soon as any fallback report arrived (battery charge or runtime alone), while the "AC present" flag still held its initial zero, so a driver starting on fallback data could publish `OB` for a UPS on line power; when that data then went stale, `upsmon` forced a shutdown. A status is now published only after both status flags were decoded from a recent report. Also corrected the startup failure message shown when `hid_fallback = no`. [issue #3587, PR #3728]

- Contributed `apcmicrolink` driver follow-ups [issue #3587]:

  - `beeper.enable` and `beeper.disable` now change only the enabled/disabled flag of the alarm-setting register, leaving the alarm-delay value it also holds untouched. Writing a bare 1 or 2 zeroed the delay, and real hardware silently ignored that write, so the commands had no effect.

  - `shutdown.return`, `shutdown.stayoff` and `shutdown.reboot.graceful` now ask the UPS to apply its turn-on delay as well as its turn-off delay. Without it the shutdown still happened, but switched outlet groups did not sequence back on when mains power returned. The command is built in `apc_common.c`, which the `apc_modbus` driver shares, so its `DRIVER_VERSION` is bumped too; that driver's behavior on this path has not been checked on hardware.
  - The calibration status in `ups.status` (see above) also follows the UPS-scope register `2:4.5.13`, for devices that populate that one instead of `2:13`.
  - Instant commands such as `calibrate.start` and `bypass.start` are only advertised when the device's Microlink descriptor contains the register they write to. They used to be listed on every device, so on a model without that register (the tested Smart-UPS 500 has no runtime calibration command register) they could only ever fail. With debug enabled, the driver logs which commands it left out and why.
  - The runtime calibration status register in object `A` (`A:6.13`) is now read, as `experimental.microlink.ups.calibratio` and a running calibration there raises `CAL` in `ups.status`. It is the only calibration status the tested Smart-UPS 500 has, so that model previously reported no calibration state at all. It is kept apart from the existing `*.calibration.result` variables because it is not yet known whether it carries the same value on devices that have both.
  - Serial mode: session start sends only the INIT byte again, as the original serial implementation did. The STOP byte that the USB transport needs before INIT had also been sent on serial, where a previously working setup then failed to start.
  - When the driver gives up at startup, the message now says whether the session handshake failed or the session started but never became ready, and only mentions the standard HID fallback on USB.
  - Serial mode: the driver polls continuously again, as the original serial driver did, instead of pausing for `pollinterval` between passes since USB support was added. USB mode is unchanged.
  - Serial mode: startup is complete once the authentication response has been sent, as with the original serial driver, rather than waiting up to 15 seconds for the device to acknowledge it, during which a few missed replies aborted startup. The acknowledgement is still collected by the regular polling. USB mode is unchanged. If startup does fail after the handshake, the message now names the step that was missing.
  - Serial mode: one record is read per poll again, and data is published after each one, as with the original serial driver. Reading a full pass (about 6 seconds on a tested unit) before publishing delayed status changes such as going on battery by that much. USB mode is unchanged.
  - Serial mode: the driver no longer sends a STOP byte when it exits. The original serial driver never did; only the USB transport needs it. USB mode is unchanged.

- `apc_modbus` driver updates:

  - Fixed string join not doing zero termination. [PR #3413]
  - Decode `RunTimeCalibrationStatus_BF` into `experimental.ups.calibration.result` [PR #3416]
  - Decode `BatterySystemError_BF` as alarms. [PR #3415]
  - Decode `PowerSystemError_BF` as alarms. [PR #3415]
  - Added `modbus_retries` option to configure the number of read retry attempts on timeout errors, simplifying error recovery. [PR #3414]
  - Increased default TCP response timeout to 2000 ms. [PR #3418]
  - Three fixes for the read-retry loop added in PR #3414: the loop ignored `exit_flag`, causing `upsdrvctl` to kill the driver and wedge the device if it was killed mid-exchange; now we retry on more failure causes than just a timeout; quieter retry logging to avoid severe syslog/journal thrashing. [PR #3571, #3414]

- `bestups` driver updates:

  - Fixed Megatec Q1 parser to expect specific response format and avoid blind `scanf()` usage. [GHSA-hwqr-5pjj-3m25]

- `blazer_usb` driver updates:

  - Bounded USB reply parsing by the number of bytes actually received, in the `ippon`, `cypress` and `phoenix` transports, preventing out-of-bounds reads when a reply fills the buffer without containing the expected carriage return or a zero byte. The same defects were fixed in the corresponding `nutdrv_qx` transports, which this code predates.

- `dummy-ups` driver updates:

  - Repeater mode now marks data stale after a failed refresh, including an interrupted variable list, and restores freshness after a complete response. Runtime reconnection attempts continue as before. [PR #3648]
  - Added `authconf` driver parameter for repeater mode to control authentication configuration discovery. It accepts `default`, `none`, or a specific authconf file path. [issue #3329]

- `huawei-ups2000` driver updates:

  - Fixed a Modbus compatibility problem that made changing any variables or sending any instant commands impossible with some UPS models. This was caused by the use of `modbus_write_registers()` (Modbus command `0x10`), which is not officially supported, making any register write impossible on some models. All uses of `modbus_write_registers()` have been changed to `modbus_write_register()` (Modbus command `0x06`) to avoid this problem. [issue #3593, PR #3604].
  - Fixed a serious problem previously existed in all NUT versions that made many instant commands unusable, including `bypass.start`, `shutdown.return`, `shutdown.reboot`, and `shutdown.reboot.graceful`. Any attempt to use them would fail, with an error logged to `syslog`: `reg1 is negative`. [issue #3603, PR #3604].

- `metasys` driver updates:

  - Fixed parsing of UPS serial number string from the response, now taking care to constrain permitted lengths and offsets. [GHSA-6p5c-r2jg-r9mj]

- `nhs_ser` driver updates:

  - Modernized serial communication and added validated settings for baud rate, data bits, parity, stop bits, flow control, per-byte read timeout, and transmission pacing. Also improved serial resource handling and reconnection behavior. [issue #3510, PR #3543]
  - Reworked incoming serial framing to use the total packet length declared by byte `[1]`, instead of treating every `0xFF` as a new packet and every `0xFE` as its end. A leading `0xFF` now starts a frame only while the reader is idle; subsequent `0xFF` and `0xFE` values are preserved as payload or checksum data. The frame is completed at its declared length and accepted only when that final byte is `0xFE`. Declared lengths are also bounded by the receive buffer before further bytes are stored. This prevents valid 18-byte and 50-byte HWINFO replies containing an internal `0xFF` from being split into unrecognized fragments, allowing the driver to obtain model, firmware, serial number and battery-pack information instead of repeatedly requesting HWINFO and delaying normal polling. The raw packet captures and protocol analysis which made this correction possible were contributed by GitHub user `freechurros`. [issue #3592]
  - Applied follow-up hardware-test feedback from `freechurros`: corrected `interpret_pkt_hwinfo()` to validate the HWINFO checksum instead of the DATA checksum, exposed unrecognized packets through `debug_pkt_raw`, and published the documented lower-case `battery.charger.status` values for compatibility with Home Assistant. [issue #3592]
  - Send the long, short and compatibility hardware-discovery requests from `upsdrv_initinfo()`, checking the serial descriptor before every request. Failed or partial writes now mark data stale and invalidate the port so the normal update cycle reopens it before further communication.
  - Refresh cached model-specific data from every valid HWINFO packet and restart discovery after a serial reconnection, allowing a replaced UPS to be identified without restarting the driver.
  - Added the legacy five-byte compatibility initialization request (`FF 05 01 06 FE`) as a third fallback. The driver first makes six extended initialization attempts, then selects randomly among the extended, normal and compatibility requests so older or otherwise compatibility-dependent NHS units can answer with their supported protocol variant.
  - Consolidated the NHS model-code metadata into an indexed lookup table, preserving the existing model descriptions and nominal VA mappings while making unknown-code handling explicit.

- `nutdrv_qx` driver updates:

  - Added the `richcomm-svc` USB communication subdriver for older session-initialized Richcomm bridges. It configures the bridge before reusing the existing Armac command framing and Megatec protocol. Polling and a quick battery test were verified with an SVC V-1000F (`0925:1234`, `UPS USB MON V1.4`). [issue #3562]

- Introduced a `santak` subdriver and protocol for single-phase Santak Castle C*K units, developed and tested against a Castle C1K (`C1K(G7)`) over RS232. These speak the same Q6 dialect as `innovart31`, but are single-phase and NAK the `F` ratings query, so `innovart31` rejects them outright. Model, nominal output and the input transfer window are read from the `WH` command instead; `battery.charge`, `battery.runtime` and a pack-level `battery.voltage` come from `Q6`, none of which the previously documented `blazer_ser` mapping could provide. [issue #1039, PR #3662]
- Bounded USB reply parsing by the number of bytes actually received, in the `ippon`, `cypress`, `phoenix`, `fuji` and `gtec` transports, preventing out-of-bounds reads when a reply fills the buffer without containing the expected carriage return or a zero byte. Beyond the debug logging, the affected length also reached the reply returned to the caller in the `ippon` and `fuji` transports.
- Only claim a USB device as "supported" during discovery out of the box if `subdriver_command` was assigned (`1A86:7523` is used by CH340/341 USB chips not only in UPSes). [issue #3410]
- Added a quirk for the `cypress` subdriver, enabled by default for devices with USB ID `0665:5161` (Belkin, Voltronic, VOLT Polska), to drain stale bytes from USB buffers of the device before our driver sends queries, and use a longer timeout. If this change breaks your device, please add `cypress_drain_quirk = 0` in your `ups.conf` and let us know. [issue #2534, PR #3563]
- With the `cypress_drain_quirk` enabled, keep replies in step with commands after the UPS answered too late or sent data unasked. Such a `0665:5161` bridge holds only the newest 8 reports (64 bytes) of UPS output while the driver does not read, so the rest of a late reply was taken for the reply to the next command, and a longer reply (e.g. the 76-byte Voltronic `QGS`) lost its start. Without the quirk (as in NUT 2.8.5) this kept the driver one reply behind, reporting "data stale" until it was restarted. The driver now drains until the UPS stays quiet for longer than it pauses within a reply (up to 160 ms were seen), and logs unasked data (once, while it keeps recurring). If the device keeps sending for longer than a reply takes (3 seconds), it resets the USB device and reconnects, up to 3 times in a row, before giving up and exiting, instead of exiting right away with a misleading "Got disconnected by another driver" message. [PR #3720]
- A USB reconnection attempt which failed again (the device was closed for another reconnection) is no longer reported as "re-established", for all USB subdrivers. So with a positive `reconnect_max_tries`, such attempts now count towards that limit instead of resetting the count. [PR #3720]
- The `cypress_drain_quirk` setting (and its default for USB ID `0665:5161`) now also applies when `subdriver = cypress` is set explicitly in `ups.conf`; previously that silently turned the quirk off. [PR #3720]
- Modified `megatec` subdriver to allow `ups.firmware` with empty contents (when a device returns all-spaces in that part of the `I` query response) out of the box and not ignore the device as unsupported, by adding the `QX_FLAG_ABSENT` in the mapping table. The practical effect should be similar to specifying `novendor` in the `ups.conf` (except that the option bypasses the check altogether, so driver instances handling well behaved devices would also not show their manufacturer). [issue #3436]
- Modified `armac` subdriver's USB interrupt read logic to request up to 8 bytes per read instead of strictly requiring 6 bytes. This resolves an initialization failure with certain UPS models (e.g., SVC VP-1250-LCD) that return a 7-byte report instead of the standard 6 bytes, allowing the Megatec payload to be successfully reconstructed. [PR #3485]
- Added an `omron` USB subdriver and an `omron` protocol subdriver for the OMRON BN150T, and registered its USB ID `0590:00b7`. The transport uses a 16-byte HID Output report and validates transfer and reply lengths. The base `Q1` mapping follows `q1` but omits two status characters. Index 2 was set during stable operation on mains, both in normal mode and in ECO mode, and was clear while on battery; neither OMRON driver version handles it. Index 7, which `q1` would publish as the beeper status, is not read by either OMRON driver version. The shutdown commands use OMRON's form. The subdriver also adds extension queries for battery charge, runtime and temperature, output frequency, firmware (`ups.firmware` and `ups.firmware.aux`), serial number, battery date and nominal power and battery voltage ratings. The implementation was ported from OMRON-authored GPL driver version 1.00 distributed in the Synology DSM 7.3-86009 GPL sources and version 1.02 distributed in the QNAP QTS 5.2.3 GPL sources. Polling and the direct shutdown commands were verified on an unloaded BN150T over USB; other OMRON models and serial connections are untested. [issue #3112, PR #3554]
- Updated `voltronic` subdriver to support Microtek/Voltronic UPS devices that report `QPI` as `(PI71`, such as Microtek GI3KL 3000VA Online UPS. [PR #3655]
- In `Voltronic-QS-Hex` subdriver, the format string sanity check added in recent NUT releases caught a mismatch in `ups.load` reading. [issue #3532]
- Modified `voltronic-qs` subdriver to request the QI command, supplying NUT with battery charge, runtime, input frequency, output current, and input transfer low/high values on devices which support it (`H`, `I`, `J` protocols, e.g. Powershield PSDR800). [#3521]

  - Fixed incorrect claim for `innovart33` subdriver — the original device claiming procedure was borrowed from `innovart31` without thoroughly checking the code. [issue #3661, PR #3713]

- Introduced a new NUT driver named `ragtech` which provides support for the Ragtech "Easy Pro" family of line-interactive UPS units (also sold under the NEP, TORO, INNERGIE and OneUP brands in Brazil). Devices present a USB CDC-ACM serial interface (VID `0x04D8`, PID `0x000A`) and speak a proprietary 6-byte register-access protocol with read/write/AND/OR opcodes; the driver covers the 20 models in the family-10 device table and was validated end-to-end against an Easy 2000 TI. Shutdown-related instant commands are opt-in via the `allow_shutdown` flag because the firmware does not auto-restart on mains return. [Initial PR #3447]

  - Put the CDC-ACM port into raw mode at startup. A freshly enumerated port defaults to canonical mode, where the kernel buffers the firmware's binary replies waiting for a newline that never arrives, so the first poll timed out as "no reply to initial status poll" unless the port had been set raw externally. [issue #3500]
  - Retry the initial status poll a few times before bailing out, so the driver starts cleanly under the stock systemd units when the device has only just enumerated at boot, instead of leaning on a fatal exit and a `Restart=on-failure` cycle. [issue #3500]
  - Drain the firmware's handshake reply before the first status poll. Some models (e.g. the Easy Pro 3200 VA GT) are half-duplex and dropped the read command that was sent while the handshake reply was still in flight. [issue #3500]
  - Compute `battery.charge` as `raw / 2.55` so a fully charged battery reports exactly 100 % instead of 100.2 %.
  - Added a `va` option to override the apparent-power rating when the firmware model id is shared between product lines, correcting `ups.power.nominal`, `ups.realpower.nominal` and the computed `ups.load`. [issue #3500]

- `richcomm_usb` driver updates:

  - Completed support for NUT standard USB configuration options, and fixed some problems possible when iterating devices. [issue #1768, PR #3477]

- `riello_usb`, `riello_ser` driver updates:

  - riello_usb, riello_ser: report NAK on RE as failure [issue #3546]

- `usbhid-ups` driver updates:

  - When reconnecting, report success more visibly (not only in debug), and do not reset driver state to "quiet" when it will loop trying. [PR #3423]
  - `apc-hid` subdriver: mapped `shutdown.return` to `UPS.APCGeneralCollection.APCDelayBeforeReboot` with value "1", for newer APC Back-UPS devices (e.g. Back-UPS BX750MI) where that usage is the only shutdown-capable one. Their firmware accepts but never executes the value "10" written by the previously only matching command `shutdown.reboot`, so the built-in shutdown sequence reported success without ever powering the load off. The value "1" is what APC PowerChute writes (confirmed by decoding USB captures of the vendor software, and by live tests on another unit): the load is cycled after a fixed ~2 minute grace, the command is executed only while on battery, and output returns when wall power is present, even if AC came back during the grace period. [issue #2683, PR #3566]
  - `idowell-hid` subdriver now also supports GoldenMate 1000VA/800W LiFePO4 battery packs (`0x06da:0xffff`), which carry the same `-BMS-` firmware and HID descriptor as the existing `0x075d:0x0300` device. The shared Phoenixtec vendor ID (`0x06da`) is gated by device strings so the other devices on that ID still fall through to `liebert-hid` / `mge-hid`. [issue #3501, PR #3502]
  - `idowell-hid` subdriver now repairs a firmware bug in those GoldenMate LiFePO4 packs, which encode the four-byte Logical Maximum of `UPS.PowerSummary.RunTimeToEmpty` with reversed byte order in report 0x02. The resulting value does not fit a signed 32-bit `long`, so on platforms where `long` is 32-bit (notably Windows) it wrapped negative, and the following `UPS.PowerSummary.Voltage` item inherited the same broken maximum from HID global state. Both `battery.runtime` and `battery.voltage` then read as nonsense; platforms with a 64-bit `long` were unaffected. [PR #3555]
  - Added the `microdowell-hid` subdriver for the MicroDowell `Enterprise USB` interface (USB ID `075d:0201`), as fitted to the B.8 (800 VA). The device was previously unsupported, and its report descriptor is broken in three independent ways, all repaired by the subdriver's `fix_report_desc()` under predicates that only match the faulty encoding: `PhysicalMinimum 0` / `PhysicalMaximum 300` is emitted once and never reset, so every later item inherits it and

logical_to_physical() rescales the single-bit PresentStatus flags from 1 to 300 - the info_lkp_t tables match on the exact value 1, so no status flag was ever recognised and an online, charging UPS reported OB OFF; the nine PresentStatus usages are declared as one 9-bit Input item carrying the ACPresent usage, so that flag read true whenever any of the nine was set; and UPS.PowerSummary.Voltage is 16 bits wide but declares LogicalMaximum 255, so GetValue() masked away everything above bit 7 and 27.0 V came out as 1.4 V (the same class of firmware fault as the cps-hid entry below). ups.load and battery.runtime are deliberately left unmapped: this firmware declares PercentLoad but never populates it, and always reports RunTimeToEmpty as 0xFFFF. [issue #3712]

- usbhid-ups asked for the wrong buffer size when the maxreport tweak was enabled: refresh_report_buffer() used sizeof(rbuf->data[id]), but data[] is an array of pointers, so this yielded a pointer's width (8 bytes on LP64, 4 on ILP32) rather than the size of the buffer behind it. The tweak therefore never requested the full buffer, and because the buffer is allocated with exactly the declared report length, any report shorter than a pointer handed the USB layer a length larger than the allocation - inviting the device to write past it on every poll. The report buffer now tracks its allocated capacity explicitly alongside the declared length, and that capacity is what gets requested and clamped against. Devices already using the tweak (some APC Back-UPS models) now receive the intended full-buffer request. [issue #3712]
- idowell-hid subdriver now also repairs the Usage Page of two PresentStatus flags in those GoldenMate packs: the firmware leaves the Battery System page selected, so the intended Good and ShutdownImminent bits were decoded as AtRateTimeToEmpty and AverageTimeToEmpty and ignored. They now feed the OFF and LB status flags via the existing mappings. [issue #3015, PR #3726]
- mge-hid subdriver updated to suppress CHRG status on constant-charge mode devices when battery is fully charged. [issue #3518, PR #3519]
- mge-hid subdriver: extended the mge_model_names[] list for the Eaton 5P Gen2 series (largely guessing, feedback and PRs for adaptation to actual string values reported by devices via USB are welcome), so these devices would now report battery.voltage and battery.voltage.nominal. [#2380]
- cps-hid subdriver: widened under-declared Logical Maximums for UPS.Input.Voltage, UPS.Input.ConfigVoltage UPS.PowerSummary.Voltage and UPS.PowerSummary.ConfigVoltage on CyberPower HID devices (e.g. PR3000ELCDSL/PR3000ELCDSXL, USB ID 0764:0601), which transmit correct values but declare maximums below them, so NUT masked/clamped them to wrong readings (input.voltage 70, battery.voltage 22.4 instead of 225 / 48 V). The fixes mirror the established cps_fix_report_desc() and apc-hid patterns for this device family. [issue #3089]
- tripplite-hid subdriver now recovers the output power reading, which was published as a constant 0.0. Two defects combined: the HID ActivePower usages were mapped to ups.power, which docs/nut-names.txt defines as apparent power (VA) rather than real power (W); and on the tested SMART1500LCD the firmware declares its ActivePower item with the HID PDC unit for watts/VA (0x0000D121) while leaving the Unit Exponent at 0, so get_unit_expo() applied a factor of 1e-7 and a genuine 520 W reading arrived as 5.2e-05, which the %.1f format then rendered as 0.0. Both ActivePower usages now map to ups.realpower, matching the cps-hid and mge-hid subdrivers; ups.power is mapped to the corresponding ApparentPower usages and ups.realpower.nominal to UPS.Flow.ConfigActivePower. A new tripplite_fix_report_desc() supplies the exponent the firmware omits; it is deliberately limited to the ActivePower item, since no Tripp Lite descriptor is known to expose an ApparentPower item with the same defect. Verified on a SMART1500LCD (09ae:2012), which now reports ups.realpower: 512.0 at ups.load: 56 against its 900 W rating. Note this is a behaviour change for units that report a non-zero ActivePower: the reading moves from ups.power to ups.realpower. [issue #3580, PR #3581]
- in legrand-hid subdriver the beeper.disable instant command now sends AudibleAlarmControl=1 ("disabled") instead of 3 ("muted"), so the buzzer stays off across alarms; added beeper.mute and ups.beeper.status. [issue #3638, PR #3639]
- legrand-hid subdriver: added a fix_report_desc() hook which clears the Unit and Unit Exponent on 1-bit items of Legrand devices. Some firmware (seen on USB ID 1cb0:0035, which names itself "Reserve") leaves a Unit Exponent of -2 in effect for the PresentStatus flags, so libhid scaled a set bit to 0.01: ACPresent never read as true and the UPS was reported OB while on mains, and LB could never be raised. It can be turned off with disable_fix_report_desc. [PR #3738]
- Added a new vertiv-hid subdriver for the Vertiv Liebert GXT5 (USB ID 10af:1000). This device shares the Liebert vendor ID handled by belkin-hid, but exposes everything under UPS.PowerSummary only (no UPS.Input, UPS.Output or UPS.Battery collections), so with the previous subdriver only 8 of 83 mappings matched: ups.load was never published, battery.runtime was served in minutes, and battery.voltage read 0.4 because the firmware declares its Volt items with a Unit Exponent of 5 instead of 7. The new subdriver maps the PowerSummary usages, converts

runtime to seconds, and repairs the exponent with a `fix_report_desc()` hook (so `disable_fix_report_desc` reverts it). `belkin-hid` now explicitly declines `10af:1000` so the new subdriver is used even when `productid` is set in `ups.conf`. Tested against a GXT5 on mains only; on-battery status transitions are not yet confirmed. [issue #3613, PR #3730]

- `snmp-ups` driver updates:
  - Corrected the IETF fallback warning so it is emitted for automatic IETF selection rather than vendor-specific mappings. [issue #3585, PR #3767]
  - Fixed a NULL pointer dereference when setting exact outlet mappings without a corresponding numbered outlet template, and restored handling of exact server-side outlet variables and commands. [issue #3360, PR #3610]
  - `apc` subdriver: exposed `ambient.temperature` from universal I/O sensors used with APC Network Management Card 3. [issue #3250]
  - Extended the XPPC-MIB subdriver (enterprise 935) to expose `battery.runtime`, `battery.voltage`, `input.frequency` `output.voltage.nominal` and `ups.firmware.aux` from Phoenixtec based UPS cards that already answer the rest of the subtree. [#2608]
  - Mapped `standby(8)` from XPPC output status to `OL` rather than `OFF`, because line-interactive devices use it for inverter standby while their UPS output remains online on utility power. [issue #3512]
  - Introduced `voltronic` subdriver for Voltronic enterprise 43943 UPSes, developed and tested against a PowerShield Centurion RT 1000VA (PSCERT1000) with a PSSNMPV4 card running firmware 1.1.8.C. [issue #3478, PR #3570]
  - `apc-ats` subdriver: added `output.current`, `ups.power` and `ups.load` from the APC ATS output phase table, which also provides `output.voltage` on models without the output bank table, such as the AP4452A.
- `upsdrvctl` tool updates:
  - Previously when looping to start a driver (and initially failing), we checked if it completed the start-up during cool-down delay only when called to start multiple (all) drivers at once. There is no reason to not do so for single-driver runs — addressed with this release. [#3302]
- common code:
  - Driver help now prints identical option names, types and descriptions only once, reducing repeated entries from protocol subdrivers. Configuration handling and the parsable `-L` list are unchanged. [issue #2064, PR #3741]
  - The configuration and protocol parser now accepts a literal `#` inside double quotes, including in UPS and `dummy-ups` outlet descriptions. Unquoted `#` still starts a comment, and escaped hashes remain supported. Encoded output continues to escape hashes for older peers; default `upsc` text output is unchanged. [issues #607, #1306, PR #3633]
  - The configuration file parser now warns on `stderr` when an unquoted `#` glued to a word starts a comment. A password with a hash in `upsmon.conf` is a common case. The warning gives the line and word number and leaves out the word itself. Parsing does not change, and protocol input does not warn. [issue #3711]
  - The common state store now retains value changes which differ only in letter case, including updates to opaque strings such as `device.contact`. Variable names remain case-insensitive and immutable values remain protected from changes. [issue #3622, PR #3630]
  - POSIX daemons now warn if their real or effective UID remains zero after the common credential switch. [issue #3471, PR #3609]
  - Refactored `common::background()` method used by numerous NUT daemons to handle parent and child code paths by separately addressable methods. This got used in the shared drivers `main` code file to preclude losing track of some libusb resources when forking the daemon. [#3437]
  - Revised *dstate* machinery to track socket connections closed mid-way through a call, to avoid access after `free()`. [#3302]
  - Some operating systems provide in `sys/stdarg.h` definitions that others access through common `stdarg.h` — code updated to try including both. [PR #3577]
  - Debug logging in `dstate.c` and `upssched.c` reported sent buffers "as is", complete with a confusing blank line where `\n` ended the message. This is now passed through newly introduced `upsdebug_ascii_compact()` method for better readability and more compact logs. [PR #3525]

  - When dealing with driver `AF_UNIX` sockets (non-WIN32), use non-blocking `connect()` to ensure a timeout if something does not respond. [PR #3456]
  - Windows driver state named pipes now use explicit security attributes instead of relying on default settings from the launch context. [PR #3479]
  - Likewise for named pipes in other NUT daemons and `upssched`. [PR #3527]
  - Drivers with libusb-1.0 code path, and `nutdrv_qx` in particular: revise to fix per-close `libusb_exit()` deadlock, and escalate a persistent `LIBUSB_ERROR_OVERFLOW` state to USB reset handling. [PR #3448]
  - Some driver code paths used buffers sent via USB transfers as fixed-size chunks populated only partially with bytes we intended. These buffers are now more diligently pre-zeroed as their siblings were in other code paths, to avoid sending host stack garbage to a device. [#3529]
  - Introduced a `reconnect_max_tries` setting for NUT drivers, so they can be configured to exit after the specified number of reconnection attempts (so that the OS service management facility can completely recycle the driver program). Multiple drivers were updated to use the new facility to consistently report `reconnect_trying()` states instead of what they did before, including: `adelsystem_cbi`, `apc_modbus`, `apcsmart`, `bcmxcp_usb`, `blazer_usb`, `generic_modbus`, `nhs_ser`, `nutdrv_qx`, `powerman-pdu`, `powervar_cx_usb`, `richcomm_usb`, `riello_usb`, `tripplite_usb`, `usbhid-ups` (it is possible that for now some of them would log the situation twice, with old methods and the new, and that some would abort the program if reconnection attempt fails). [PR #3541]
  - `belkin`, `bestfortress`, `genericups`, `powerpanel`, `tripplite` drivers introduced attempts to reconnect in case of failure, instead of just exiting right away or remaining in data stale mode indefinitely, also using `reconnect_trying()` for consistent reporting. They now track serial port file descriptor validity a bit more diligently. [PR #3541]
  - The `hidparser.c` code (used primarily in USB drivers) risked a heap out-of-bounds write if a device reported a HID report descriptor with more usage items than the fixed stack in the parser could hold. A similar problem could lead to "HID path too long" messages and loss of readings from the device. [PR #3586]
  - The `hidparser.c` code was further updated to use not a fixed 50-entry "usage table" buffer for resource descriptor parsing, but one aligned with the actual size detected during device discovery. This should avoid potential truncation introduced by the fix above (and avoid the overflows it addressed) in normal device interactions. [#3588]

- NUT client libraries:

  - Introduction of `nutauth.conf` support caused some rearrangement of codebase layout in the library, and this in turn caused some fallout. Fixed `libupsclient` builds with OpenSSL or NSS when Clang treats declarations after executable statements as errors. [issue #3761, PR #3762]
  - The Perl `UPS::Nut` module now decodes quoted values and backslash escapes in query/list replies and `nutauth.conf` entries, and quotes authentication arguments. Single quotes in configuration are literal. Command descriptions now accept the server's `CMDDESC` response.
  - Complete support for actions documented in `docs/net-protocol.txt` was implemented in C++, Python and PERL bindings in-tree, and for Java in jNut nearby. Among other things, all these libraries now support `STARTTLS` and `TRACKING` to wait for server confirmation of a `SET VAR` or `INSTCMD` request, and this is tested by the NIT script. [issues #656, #1348, #1349, #1350, #1613, #1711, PR #3402]
  - Enhanced client side of `STARTTLS` dialog to follow up by a simple query (for protocol version) to verify that handshake succeeded. This change impacted also the classic C `libupsclient` library. [issue #3387, PR #3402]
  - `PyNUTClient.disconnect()` now attempts to send `LOGOUT`, closes the socket and clears its reference while the client remains alive. Such cleanup remains harmless when repeated or when socket operations fail. [issue #3619, PR #3627]
  - Updated OpenSSL code paths in the `libupsclient` C library to support features earlier only available with NSS builds, like specifying the client certificate+key, optionally with password, and pinning expected server certificates. For both backends such pinning should now honour the *certverify* setting of the `CERTHOST` entry (e.g. not abort the connection attempt if that number is *0*). Also, the `-1` value is now supported to mean "use global default setting". [issue #3331]
  - Updated SSL support in the `upsd` data server to handle `CERTREQUEST` (optional validation of clients identified by a certificate) also when built with OpenSSL, optionally using the `CERTPATH` with a collection of CA certificates (directory or a big PEM file). Also support `CERTIDENT` to provide a private key password and ensure that the certificate in `CERTFILE` has an expected subject name as an exact string, or that its CN or SAN match the provided string as a standard expression of host name (section 3.5 of RFC 1034) or IP address. [issue #3331]

  - The `libupsclient` API was extended with a `upscli_init2()` method which allows to pass the `certfile` argument needed for OpenSSL builds. [#3331]
  - The `libupsclient` © and `libnutclient` (C++) API were updated to report the ability to check `CERTIDENT` information. [#3331]
  - Introduced support for "authconf" files to store and convey NUT client authentication details. [issue #3329]
  - Added C/C++ client support for multiple independent OpenSSL contexts in the same process, or using multiple NSS client certificates with its single context, for ability to connect to NUT data servers managed by different entities (e.g. certificate authorities and client/server realms). In-tree multiple-connection clients (`upsmon`, `upsstats.cgi`, `upslog`) were updated to use this feature. [issue #3439]
  - To support the ability of multiple SSL authentication details for `upsmon` and `upsstats.cgi`, their configuration files were extended with `AUTHCONF` keyword which can point to the `nutauth.conf` file they should use, or pass the special values `default` (require a user- or system-provided `nutauth.conf` file) or `none` (skip authconf discovery entirely), matching the `-A` option semantics of command-line NUT clients. [issue #3439]

- Various clients:
  - Flush standard output and error buffers before handling clean exit rituals, to avoid losing the result of client's work in case that clean-up crashes (e.g. due to third-party code involved). [#3454]

- `upsimage.cgi` client updates:
  - Adjust scaling for small (single-digit) value ranges to avoid division by zero and not rendering anything. [issue #3469]

- `upsc`, `upscmd`, `upsrw` command-line client updates:
  - Enabled support for `nutauth.conf` files to provide credentials and/or SSL settings in the client which previously only did best-effort attempts at secure communications without an individual certificate, and only anonymously for reading. The new `-A filename` option defaults to trying to use a `nutauth.conf` file (if NOT specified here, but rather found in one of the default locations) but not failing if one is not usable; any specific values can require use of such a file (`default`) or to not even try reading one (`none`). [issues #3329, #3411]

- `upslog` client/tool updates:
  - Fixed a startup memory leak when replacing UPS name and hostname strings allocated during initial parsing or wildcard discovery. [#3771]
  - Added support for best-effort use of `nutauth.conf` files from default locations or via `-A` option, as described above. Since this client can establish multiple connections, keep in mind that currently it can only identify itself with some one (first seen) client certificate, if `CERTIDENT` settings are used. Multiple `CERTHOST` directives for specially trusted servers can be used. [#3329]

- `upsstats`, `upsset`, `upsimage` CGI client updates:
  - Added support for best-effort use of `nutauth.conf` files from default locations described above (no way to choose the location, other than by web-server environment variables for CGI calls). [#3329]
  - `upsstats.cgi` additionally has a new `AUTHCONF` configuration option that can be passed via its `hosts.conf` file, as detailed above. [#3439]
  - `upsset.cgi` now displays writable strings with maximum lengths above 127 characters without aborting or truncating the length; invalid string lengths are reported as an unknown type. [issue #3545, PR #3668]

- `upsmon` client updates:
  - Introduced support for `CERTFILE` option, so the client can identify itself to the data server also in OpenSSL builds. [issue #3331]
  - Failure of SSL connection setup (e.g. due to inability to load the NSS library, a practical regression since NUT v2.8.4 release, now that success of SSL initialization with both backends is more diligently tracked) should now not cause exit of the client if secure connection is not required by its configuration in the first place. [#3420]

  - Potentially a breaking change for existing deployments with `upsmon` and `upssched` integration: the `CMDSCRIPT` and `NOTIFYCMD` definitions were revised for the first token to mean an exact program name (possibly with spaces in the value), not a shell scriptlet. Documentation in earlier NUT releases implied that arguments could be passed here — this is no longer true. If you need to pass special arguments to a notification program, please use additional separately quoted tokens, or write a small helper script to do so and specify its path here. The `SHUTDOWNCMD` is still a scriptlet, as far as specifying command arguments is concerned, but it also newly supports parsing multiple tokens as additional arguments. [PR #3499]
  - Introduced a new notification type for unexpected clock change which is specially handled (data re-initialization), whether due to sleep and wake-up, or NTP/RTC clock changes during boot, or severe delays on a stressed system (to name a few practical cases). [issue #3405]

- `upssched` client/tool updates:

  - Fixed handling of `NOTIFYMSG` from command line if other arguments are present (e.g. debugging with `-DDDDDD`). [issues #3105, #3525, PR #3527]
  - Rewrote the WIN32 code path to spawn the timer daemon. [issue #3525]

- `NUT-Monitor` Python GUI client:

  - The PyQt5 and PyQt6 `NUT-Monitor` commands now share one Python 3 implementation. The dispatcher accepts `NUT_MONITOR` or `NUT_MONITOR_QT=6` to require a particular backend. [PR #3596]
  - Fixed Qt tray tooltips to render in plain text, not rich text which ended up as literal HTML on KDE Plasma 6; now that rich text formatting is only used for main window status labels. [PR #3430]
  - Fixed both Python 2 and 3 variants to handle automatic client reconnection (e.g. after system suspend/resume, network interruption, or NUT server restart). [issue #3509, PR #3523]
  - Fixed wrong lookup into UI widgets which broke reconnection. [PR #3616]
  - Fixed some issues with Qt6 startup and desktop file integration. [#3624]
  - Added support for `nutauth.conf` files as source of configuration about default/server/user access credentials and SSL setup. [#issue #3514]
  - Fixed TAB/SHIFT-TAB focus change order through login page. [PR #3533]
  - Updated formal version of all three client variants to `2.1.0`. [#3514, #3509]

- `upssched` client/tool updates:

  - Added a `clean_exit()` handler similar to that in `upsmon`. [PR #3499]

- `upsd` data server updates:

  - Validate complete numeric values and conversion ranges for `MAXAGE`, `TRACKINGDELAY`, `MAXCONN` and `CERTREQUEST` in `upsd.conf`, retaining the previous setting when conversion fails. Preserve public types and avoid overflow in the `MAXAGE` grace-second calculation at the maximum signed integer value. [issue #676]
  - The `LISTEN` directive in `upsd.conf` now accepts `hostname:port`, `IPv4:port` and `[IPv6]:port`, with matching support in `nutconf` and Augeas. IPv6 addresses ending in `::` also accept an appended `:port`, such as `:::3493`. Bare IPv6 addresses and the existing space-separated address and port syntax retain their meaning. [issue #2424, PR #3709]
  - If we hit "Too many open files" during configuration reload, close the oldest client connection and retry. [issue #3365]
  - If the `MAXCONN` requested in the configuration file exceeds the OS allowance on open file descriptors, fail early since the requested configuration can not be guaranteed and can mis-fire unexpectedly much later (tell the sysadmin to increase `ulimit` or set up a more conservative `MAXCONN`). If there is a separate soft and hard limit, and `MAXCONN` exceeds the soft limit, try to raise the bar. [issue #3365]
  - If SSL configuration was provided, but the server failed to apply some aspect of that, it should now abort with an explanation (and not proceed with insecure start-up like it could do before). [issue #3331, PR #3435]

- The `nut-driver-enumerator.sh` script (NDE) updates:

  - The shell/awk based parser now decodes ups.conf values using NUT token syntax, preserving quoted hashes and escaped characters and ignoring trailing comments. For example, port = "auto" # comment now properly selects USB media for nutdrv_qx. The normalized configuration output retains quotes and backslashes; changed section checksums can cause a one-time reconciliation after upgrading. [PR #3635]

- Revised dusty corners of NUT documentation:
  - Corrected client API signatures and array initialization descriptions, SNMP daisychain alarm handling and subdriver mapping references, and developer-guide protocol links and source access instructions. [PR #3745]
  - Reworked the packaging proposal around deployment roles and build contents, clarified service-managed startup and multi-token UPS status, and added support-data redaction guidance. Improved Windows archive navigation, completed self-test and Jenkins introductions, and indexed authentication, SSL and library logging manuals. [PR #3748]
  - Corrected operator guidance for communication loss, certificate expiry and device-list reloads, debug output and polling intervals. Documented the scanning thread limit, completed notification lists and linked the SSL configuration directives. [PR #3743]
  - Documented the existing --with-debuginfo and --enable-cppunit developer options, including their defaults and dependencies. [issue #2311, PR #3740]
  - Included the eight existing text cable references in the generated cable guide and user manual, preserving their diagrams and attribution. [issue #1367, PR #3742]
  - Corrected the usbhid-ups on-battery discharge option name and the belkinunv configuration guidance. Documented the existing microdowell delay settings and the nhs_ser model override and bypass-alarm settings. [PR #3744]
  - Documented the existing Modbus timeout components and defaults for generic_modbus and adelsystem_cbi, and corrected the genericups driver name in the GE/IMV/Victron cable reference. [PR #3749]
  - Documented the existing mfr and model options for liebert. Documented sysfs_dir for hwmon_ina219, and corrected the hwmon path in this driver and documented the integer range and default for its low battery threshold. [issue #2422, PR #3671]
  - Documented how to use sdcommands with a stopped driver daemon to send a supported instant command (typically for shutdown) without a locally running upsd data server. [issue #1677, PR #3672]
  - Updated the m4 macro reference for developers of the configure script in docs/macros.txt. [PR #3674]
  - Corrected the installation staging example to preserve the source tree (temporary workspace) and stop when staging prerequisites fail. [PR #3677]
  - Initialized the connection pointer in the C++ client example before exception handling. [PR #3678]
  - Corrected libupsclient API declarations, logging examples and the descriptions of SSL capability flags and raw line reads. [PR #3679]
  - Corrected nut-scanner library (libnutscan) API descriptions and the consumer example for current function names, argument types, device-option pointers and initialization. [PR #3680]
  - Corrected source retrieval examples, removed an unsupported configure script option and documented the pkg-config installation-directory default. [PR #3681]
  - Corrected protocol notation and updated the architecture documentation for individual variable requests and command tracking. [PR #3682]
  - Corrected the simulation filename and recorder command in the developer walk-through, and fixed quoting in the recorder manual example. [PR #3683]
  - Corrected nut-scanner simulation-file and configuration-directory documentation and help, and repaired inconsistent scan examples. [PR #3684]
  - Corrected upsmon notification argument handling and escaping examples, shared-timer examples and the documented FINALDELAY default. [PR #3685]
  - Corrected the PyNUT GetUPSCommands method example and the GetUPSNames return-type documentation. [PR #3686]
  - Corrected the documented connection defaults for upsc, upscmd, upsrw, upsmon and upslog clients, and preserved the remote destination in the upsc query-loop example. [PR #3687]
  - Clarified client authentication configuration, unattended credential selection and instant-command permissions. [PR #3688]

  - Corrected the `upsd` manual descriptions of state-path precedence and debug-output stream. [PR #3689]
  - Updated the `apc_modbus` manual for its implemented capabilities, timeout defaults and register-read attempt limit, and corrected the corresponding help text in the `apc_modbus` driver program. [PR #3690]
  - Updated APC Microlink support references and qualified the USB kernel-driver detachment guidance. [PR #3691]
  - Corrected standalone `belkinunv` and `genericups` driver examples to supply a driver identifier and port option, and clarified the `genericups` monitoring prerequisite. [PR #3692]
  - Corrected the GPIO driver executable name in examples and fixed a quoted value in its udev rule example. [PR #3693]
  - Added the required USB identity fields to the Gtec configuration example for `nutdrv_qx` driver, with explicit placeholders for the USB device identifiers. [PR #3694]
  - Removed unsupported quick-test duration configuration guidance from the Powervar serial and USB manuals. [PR #3695]
  - Removed an ineffective notification option recommendation from the `mge-shut` driver troubleshooting guidance. [PR #3696]
  - Corrected the `upsdrvsvcctl` command reference and removed a stale TODO reminder for its existing shutdown fallback. [PR #3697]
  - Corrected the grep pattern, directory path and displayed result prefixes in the FUSE data-walk example, and fixed a spelling error in the README. Added `scripts/fuse` files to `EXTRA_DIST` and `spellcheck` goals. [PR #3698]
  - Corrected documentation cross-references and exact variable, option, command and sample-file names in examples. Updated the spelling dictionary for verified terms. [PR #3699]

- Recipes, CI and helper script updates not classified above:

  - Introduced `AGENTS.md` for AI agents to help in a consistent manner with minimal friction. [#3608]
  - Introduced `SECURITY.md` for rules about responsible disclosure of potentially exploitable bugs. [#3470]
  - Introduced `ci_build.sh` settings and respective CI workflow settings to optionally re-use a `config.cache` file from older runs, and similar logic for NIT `nit.sh` to re-use generated certificate and private key files. Both can be enabled by `export DO_USE_NUTCI_CACHE=yes` and would store data under a `${CI_CACHE_NUT_BASEDIR}` directory (defaults to `~/.cache/nut-ci`); more fine-tuning environment variables are provided. Directory names are based on hashes of strings that identify the content, which are size-limited to some 8 characters. [issues #3108, #3390, #1711, #3526, PR #3109]
  - Introduced `tools/asciidoc-github-links.pl` to simplify `docs/Makefile.am` generation of `*.adoc-parsed` files with substituted GitHub links, and to allow that logic to be executed separately from `make` and potentially be re-used by other projects. Added support for lists of PRs/issues as used in this document, and support for GitHub Security Advisories (GHSA). [PR #3594, issue #1953]
  - Rearranged the GHA job to prepare Dist and Docs Tarballs. [PR #3460]
  - Updated `scripts/Windows/dllldd.sh` helper script to better detect DLL names not exposed with dynamic linking metadata (e.g. some Mozilla NSS dependencies were not previously discovered and bundled). [issue #3420, PRs #3429, #3432]
  - Updated a NUT CI farm worker to FreeBSD 15 and updated the recipes and documentation according to problems that were found along the way. [#3453]
  - Various recipe portability fixes for AIX, NetBSD and OmniOS at least. [#3489, #3495, #3498]
  - Refactored detection of semaphore capabilities (for parallel execution) into a new m4 macro `NUT_CHECK_SEMAPHORES`, added support for autoconf caching of the results for repeated runs. [PR #3675]
  - Jenkins used as the NUT CI farm core has bumped JDK requirements for its controllers and build agents, `docs/config-prereqs` revised.
  - `Jenkinsfile-dynamatrix` recipe revised to generate fewer RISC-V recipes specifically (very slow worker), focusing on the newest compiler and C/C++ revisions available on the worker. On the other side, this CI recipe was extended to require more "quick phase" builds to cover spellcheck on many distributions, as their base dictionaries like different things. [PR #3757]
  - Further rearrangement of AppVeyor CI builds and security hardening of GitHub Actions used to prepare the dist tarballs and report on progress there. [PRs #3753, #3754, #3756]

- CI build matrix scenarios (Jenkins autotools builds, `ci_build.sh` with `BUILD_TYPE=default-withdoc*`) now pass `--disable-spellcheck`, so their `make check` no longer runs (and fails on) the documentation spellcheck. The GitHub Actions dist tarball worker checks the spelling explicitly, with both a current and an old (7.1, as on CentOS 7) version of the `aspell-en` dictionary. Local builds keep the default of `configure --enable-spellcheck=auto`. [PR #3752]
- The `make shellcheck` (and `make check-scripts-syntax`) recipe can now check the POSIX shell scripts with each interpreter listed in the `SHELL_PROGS` environment variable (default is still the system `/bin/sh`), and hides warnings about the legacy command substitution syntax (e.g. from `ksh93`) to avoid noise that obscured other problems. [issues #1058, #1059]
- Added a "GHA-07: ShellCheck" GitHub Actions workflow which checks the shell scripts changed by a pull request, and fails only for findings of "warning" or "error" level that are new compared to the base branch (many existing scripts have known findings), with annotations on the changed lines and a list in the job summary. The same check can be run locally with `perl tools/shellcheck-ci.pl`. A new `.shellcheckrc` file disables the checks for deliberate portability idioms of NUT scripts (backticks, `x"$var"` comparisons, `expr`).
- The `make spellcheck` recipes now report unknown words with the file name, line and column where they were found (rather than byte offsets in the `aspell` output), and in GitHub Actions also as annotations on the source lines (once per unknown word, despite the retries) and in the job summary.
- The `configure` script now also looks for `aspell` filter resources under `lib64` (e.g. on CentOS 7) and in directories named by just the major and minor version (e.g. `aspell-0.60` for `aspell` 0.60.6.1). [issue #2068]

2.2 Release notes for NUT 2.8.5 - what's new since 2.8.4

https://github.com/networkupstools/nut/milestone/12

- common code:
  - Introduced `setproctag()` and `getproctag()` (see examples in `upsmon`) to help track the log messages from massively-forking NUT daemons. [#3084]
  - Introduced `NUT_DEBUG_PROCNAME` environment variable support to optionally log also the process name (or however the program chose to identify itself). This may be useful when multiple NUT daemons log into the same file or console, without syslog to prefix the name into each line (e.g. in tests, single init script systems like Home Assistant, etc.) [PR 3368]
  - Extended with plural `checkprocnames()` and `compareprocnames()`, as well as `sendsignalpidaliases()` and `sendsignalfnaliases()`, for binaries that expect to have one of several names at the moment (e.g. when renaming a program to "**-old" or promoting a "**-new" into the stable name as a default implementation). Drivers united by `main.c` framework introduced a `upsdrv_tweak_prognames()` hook method to let them manipulate the array of `prognames[]` aliases. [PR #3101]
  - USB drivers could truncate strings received from devices as a fallout of PR #2604 merged in NUT v2.8.3; this should now be fixed. [PR #3211, issue #3201]
  - `libusb0` and `libusb1` integrations now try both "Resource Descriptor Length" values they could identify on some devices: sometimes the first (or otherwise preferred) option may be not the correct one, so try the other too. [issue #3136]
  - Improved high-verbosity debug tracing of NUT `libhid` and `usbhid-ups` to help make sense of data processing issues (HID paths, strings). [#3201]
  - Further removal of unbounded `strcpy()` and `sprintf()` usage. [#3253]
  - For state tree methods, introduced `difftime_st_tree_timespec()` to abstract platform-dependent definitions of `st_tree_timespec_t`. [PR #3156]
  - Introduced global variables for last changed timestamp and value of `battery.charge`, integrated with `status_commit()`, so all drivers should now report a `CHRG` or `DISCHRG` status, and verify whether reality agrees with a status reported by device/protocol (if any). This is also a step in direction of eventually providing a common `runtimecal` style fallback to all drivers. [issue #3146, PR #3156]
  - Generalize `elapsed_since_timeval()` code, originally repeated in several modbus drivers. [issue #1938, PR #3321]
  - Introduced `NOTIFY_STATE_EXTEND_TIMEOUT` option and relevant variables for `upsnotify()` method, so we can tell systemd to not kill the service unit if its shutdown takes long (see also the `SHUTDOWNEXIT` option in the `upsmon` client). [issue #3003, PR #3151]

  - Daemons running with systemd integration tended to report "logged the systemd watchdog situation once, will not spam more about it" even if those "logged" messages were at an invisible verbosity level. [issue #3157, PR #3151]
  - Systemd units for some daemons were revised to honour the `MODE` setting from `nut.conf`, if available, to gracefully not-start when explicitly not-requested (e.g. do not start `upsd` in `MODE=netclient`), even though the unit is nominally enabled; this should make packaging and providing service pre-sets more simple. [issue #837, PR #3233]

    In a way, this should also fix fallout of a change delivered in NUT v2.8.4 release, where `upsmon` could have started with `MODE=none` in `nut.conf`, but the `nutshutdown` script would bail out quickly and quietly. [PR #3008] * Fixed thread-safety of IP address printout in `libupsclient` method `upscli_tryconnect()` (practical bug seen in `nut-scanner` parallel scans). [issue #3234] * The `upsdrvquery` code base was revised to handle SIGPIPE without crashing (if the consumers did not neuter the signal themselves), although this behavior is optional by `upsdrvquery_NOSIGPIPE` Separated an `upsdrvquery_ping()` operation from earlier code, so it can be directly used by consumers (in-tree this is drivers and `upsdrvctl`). [issue #3276, PR #3277] * Revised `void*` pointer casting into specific types for values returned by `xmalloc()` and friends, or iterated in loops via `something->next`, as `clang-21` now warns about not-casting void pointers (formally valid, practically uncertain if that code matches the developer's intention), and (better safe than sorry) added `NULL`-return checks although we should not normally get into those code paths with the `x*()` methods which should bail out and abort the program with a common error message when an allocation fails. [#3293] * We have long abstracted `(s)size_t` value printing in debug logs etc. using `PRIuSIZE` and similar macros. They fell back to `"%zu"` when not provided otherwise, but not all platforms actually deliver these format characters. Now it is evaluated at `configure` time (to check that the characters may be used), and if not — during `nut_stdint.h` parsing to fit known `int`/`long`/`long long` types. [#3300] * Added new API methods and defined bitmap values for `libupsclient` C binding to query and report SSL capabilities of the current library build (none, OpenSSL, Mozilla NSS): `upscli_ssl_caps_descr()` and `upscli_ssl_caps()`. Updated common NUT clients to report this info in their detailed help banners. Done similarly for `upsd`. The NIT (NUT Integration Test) suite piggy-backs on this to add run-time dependent tests of SSL capability. Added `upscli_upslog_set_debug_level()` and `upscli_upslog_get_debug_level()` methods to facilitate NUT debugging for clients built with shared NUT private libraries. [issues #3328, #1771, #2800, PR #3330] * Revised OpenSSL handshake to support retries requested by the library, so it at least works reliably on different platforms within the scope it has now. [issue #3331, PR #3344] * Extended `PyNUTClient` to support `STARTTLS` and optional server/-self identification, added NIT tests for that ability. [issues #1711, #3329, #1600, PR #3352]. * Ported C client library `libupsclient` support logic for SSL connections into C++ client library `libnutclient` and added tests for it. [issues #1599, #1711, PR #3353] * Added support for command-line debugging with `-D` option into `upsc`, `upscmd` and `upsrw` command-line clients, and the `nutconf` tool. Revised handling of `-D` and/or `NUT_DEBUG_LEVEL` environment variable in other programs. Extended public API of libupsclient and libnutscan to diligently arrange debug-logging when at run-time those libraries actually have different copies of the methods and data involved. [#3378]

- NUT for Windows specific updates:
  - Revised detection of (relative) paths to program and configuration files near the currently running program in NUT for Windows builds, including limited support for UNC paths (ignoring a localhost `\\?\` prefix). [issues #3207, #3063, #3065, #3219]
  - Augmented WIN32 `nut.exe` launcher with `status` and `restart` actions, and general ability to debug-log the activities during service management operations. (Re-)registration of the "Network UPS Tools" service should now populate a nice description of it. The `-U` (uninstall) action should now also try to stop the service first. [PR #3235]
  - Using `nut.exe -N` for testing and pressing *Ctrl+C* should now cause the started daemons to be killed off. Previously they would linger and e.g. preclude subsequent experiments with the service wrapper. Console close events are ignored, so there is a way to indefinitely keep the daemons started by test-mode wrapper running (kill via Task Manager). [#3312]
  - Revised WIN32 `WSAStartup()` and registration of `atexit(WSACleanup)` to only be done once per program (and cleanups to be always registered); this impacts the C `libupsclient` and C++ `libnutclient` libraries (and so most NUT clients and tools which use the former, and `nutconf` which uses the latter), the `apcupsd` driver, `upsd` data server, and `nut-scanner` tool. [PR #3237]
  - Documented setup of Microsoft IIS to host the NUT CGI programs. [#3207]

- `asem`, `bestfortress`, `bestuferrups`, `bicker_ser`, `everups`, `metasys`, `masterguard`, `mge-utalk`, `oneac`, `phoenixcontact_modbus`, `pijuice`, `powercom`, `powervar_cx_ser`, `usbhid-ups`, `victronups` driver updates:
  - Moved code to align with NUT driver architecture, so `upsdrv_initups()` would focus on preparing communications, and `upsdrv_initinfo()` would actually talk to the device and set initial critical data values. [#1962]

- `apcsmart` driver updates:
  - Add support for dual-byte command for APC SPM series (and potentially SPRM SRVS series). [PR #3327]
- `apcupsd-ups` driver updates:
  - Abandoned use of obsolete `gethostbyname()` in favour of `getaddrinfo()`. Extended to be IPv6-capable along the way. [#1209]
- `apc_modbus` driver updates:
  - Fixed register range in a logging string.
  - Add outlet group timers.
  - Fixed outlet group command handling to dynamically detect available groups (MOG, SOG0 - SOG2) in the device. Commands and variables are now only exposed for outlet groups actually present on the UPS. Global `load.*` and `shutdown.*` commands now correctly target all available outlet groups. [PR #3395]
- Introduced a new NUT driver named `meanwell_ntu` which provides support for the Mean Well NTU series hybrid inverter and UPS units. [PR #3206]
- Introduced a new NUT driver named `must_ep2000pro` which provides support for the MUST EP2000Pro compatible, line-interactive UPS models which talk serial-protocol modbus protocol. [PR #3228]
- `nhs_ser` driver updates:
  - Driver source code includes a table to map baud rates (defined by the OS as C macros) to numbers and strings more useful to the driver program. It was discovered that not all operating systems define all the baud rates referenced by this table, so now several higher speeds would be compiled conditionally. [issue #3163]
- `nut-ipmipsu` driver updates:
  - Fixed an issue where multiple calls to clean-up crashed the program. Special thanks to maintainer of libfreeipmi for helping catch it. [#3193]
- `nutdrv_qx` driver updates:
  - Define an internal `QX_FLAG_MAPPING_HANDLED` to check if the subdriver code (mapping table) and the data found from device walk sit together well — namely, that we successfully use reasonably many of the existing mappings. Suggest how user can help improve the driver if too few data points were seen. [PR #3095]
  - The `qx_process_answer()` method was extended to raise `errno=ETIMEDOUT` (rather than `EINVAL`) for short reads that are zero length (and/or that `errno` value was set in caller's context already). This was used to make `Voltronic` protocol initial connections more reliable when the device controller lags upon first contact (is booting itself?) [issue #3276, PR #3283]
  - New `nutdrv_qx_innovatae` adds support for Ippon INNOVA TAE series which are similar to Q1 except in reporting of nominal `voltage`, `current`, `frequency` data (that along with `battery_voltage_reports_one_pack` and `override.battery.packs=12` settings yields reasonable `battery.voltage.*` data points). [issue #3137, PR #3139]
  - `nutdrv_qx_masterguard` sub-driver adds a way to call battery test until the battery charge is low, using a new `test.battery.start.low` instant command name (registered in `docs/nut-names.txt`). The difference between the "battery test until battery low" and "deep battery test" is that the latter is actually a battery calibration test and needs some preconditions to be met, i.e. the load must be between 30% and 100% and the starting battery capacity is greater than 99%. The "Battery test until battery low" can be started anytime. Tested on Masterguard A1000 and A2000 units. [#3135]
  - `nutdrv_qx_masterguard` sub-driver fix `masterguard_add_slaveaddr` to return the correct command length, resolving a bug for all commands using the `,XX` slave address placeholder. This also enables support for the `test.battery.sta` (`TUD`) command. [#3141]
  - `nutdrv_qx_masterguard` sub-driver adds a `clear.fault.record` instant command to clear the fault record. Also adds two experimental variables: `battery.ageing.factor` (Battery ageing factor in promille) and `battery.calibratio` (Battery calibration factor in percent). Tested on A1000 and A2000 units. [#3181]

  - Hides QX_FLAG_NONUT variables from syslog unless the debug level is raised. [issue #3190, PR #3198]
  - Added support for battery.charge in hunnox subdriver. [#3254]

- `powerp-bin` and `powerp-txt` driver updates:

  - Their `upsdrv_initinfo()` methods did not explicitly reference the `INSTCMD` and `SETVAR` handler methods (it is unknown whether this actually broke anything, or the needed values were deduced from subdriver definition then). For consistency, these pointers are now assigned. [#1962]

- `riello_ser` and `riello_usb` driver updates:

  - Since the beginning, these drivers fenced availability of **either** `load.*` **or** `shutdown.return` instant commands based on current power situation (which was also unknown when the `drivername -k` mode was used to shut down the system). There seems to be little reason for such separation, so it is now avoided and the current device status is queried when starting to handle the UPS poweroff, as a means to verify communication ability. [#1962]

- `snmp-ups` driver updates:

  - Define an internal `SU_FLAG_MAPPING_HANDLED` to check if the subdriver code (mapping table) and the data found from device walk sit together well — namely, that we successfully use reasonably many of the existing mappings. Suggest how user can help improve the driver if too few data points were seen, or if the `mibs=auto` detection only found the fallback IETF mapping. [PR #3095]
  - Quiesced "unhandled ASN 0x80 received from ..." messages when we fall back to using the numeric value from last OID section. [issue #1358, PE #3337]
  - Added new vendor extension mappings for Vertiv Liebert UPS, tested with PSI5-750MT120 UPS model and IS-UNITY-DP (Firmware 8.4.7.0) SNMP card. [issue #3296, PR #3299]

- `tripplite_usb` driver updates:

  - Added support for Tripplite protocol 3017 (mostly ASCII). [issue #2258, PR #3093]
  - Fixed default `unit_id`, apparently broken by changes in NUT v2.8.1. [issue #3191, PR #3192]

- `tripplitesu` driver updates:

  - Added configurable command delay to prevent read timeouts. [PR #3273]

- Introduced a new NUT driver named `nut-upower` which provides support for monitoring UPS and battery devices managed by the UPower daemon via D-Bus. This also introduces a new category of drivers that talk to operating system services to obtain power state information. [PR #3279]

- `usbhid-ups` driver updates:

  - Declared support for APC with USB ID `051d:0005` (details may evolve as the devices become better known). [issue #3047, PR #3081]
  - Extended iDowell subdriver with data points for GoldenMate UPS using LiFePO4 batteries (tested with GoldenMate LIPO 1500VA/1000W). [issue #3015, PR #3265]
  - Improved support for Eaton 5S1500LCD and 5S1600LCD (US version). [#2380]
  - The driver used to report `CHRG` status if the device was not "fully charged", however the latter status was only queried and reported by some (not all) subdrivers and probably not by all devices out in the field. Now we fall back to check that if the "fully charged" status is not defined at all, consider the `battery.charge` value (100 or not). [issue #3146]
  - For `ups.status` processing, do not assume that `!online==offline`, it may well be that the device just did not report anything (e.g. if it does not really talk USB HID). [#3080]
  - Suggest iterative testing with explicit `subdrivers` if collected data looks wrong. [#2058, #3061 et al]
  - Check if the subdriver code (mapping table) and the device report descriptor sit together well — namely, that we do not unwittingly ignore any useful reports, and that we successfully use reasonably many of the existing mappings. Suggest how user can help improve the driver if too few data points were seen. [#3082, #3095]

  - Default to `usb_hid_rep_index=1` for Liebert PSI5 and PowerSure PST devices with ID `10af:0002`. [#3345, #1028, #1252 etc.]
  - `cps-hid` updated with Replace Battery information. [PR #3268]
  - `openups-hid` had `nobattery` definition inverted, causing alarms — now fixed to use the same mapping helper as other subdrivers. [issue #3246]
  - `powercom-hid` subdriver improved with a `powercom_hack_voltage_lkp` to query `battery.voltage`. [issue #2766]
  - `powercom-hid` subdriver now honours configured `offdelay` and `ondelay` driver options in `ups.conf`, and prefers them (if present) over the values received from the UPS itself (like `ups.delay.shutdown` and `ups.delay.start`), to fix implementations of shutdown/stayoff/startup instant commands. [PR #3288]
  - Improved handling of transient `LIBUSB_ERROR_IO` failures during polling. Some devices (CyberPower, etc.) have firmware bugs causing random I/O errors on certain HID reports. The driver now skips failing reports and continues polling rather than triggering expensive reconnection attempts. True disconnections are still detected via other error codes or when all polls fail. [issue #3116]
  - `arduino-hid` subdriver enhanced with `ups.load`, `input.voltage` and `output.voltage` readings where supported (e.g. Ugreen US3000). [#3281]
  - Introduced a `winhid` extension in NUT for Windows builds, to directly work with Windows HID API so that extra driver like WinUSB or Zadig is not needed. Note that currently this backend follows a read-focused scope: it supports device discovery and matching by identifiers such as `vendorid` and `productid`, input/feature reads, and interrupt-report fallback handling. Feature-report writes are not implemented in this phase, so `setvar`/`instcmd` operations may be unavailable. [PR #3335]

- `nut-scanner` tool updates:
  - Fixed `nut-scanner` search for "simulation devices" to not use only the built-in NUT configuration path on all platforms, but to also consider `NUT_CONFPATH` and other fallback locations, like other code does. [PR #3249]
  - Enhance debug-logging of dynamic library loading with information about any missing method in the library discovered at run-time, if lack of such prevents us from using that library, and blocks scanning of corresponding protocol and/or media to discover possibly supported devices. [PR #3310]
  - Introduced `nut-scanner` support for new `nut-upower` driver. [PR #3293]

- `upsd` data server updates:
  - Sometimes "Data for UPS [X] is stale" and "UPS [X] data is no longer stale" messages were logged in the same second, especially on busy systems. Now we allow one more second on top of `MAXAGE` setting to declare the device dead, just in case fractional/whole second rounding comes into play and breaks things. [issue #661]
  - Fixed `LISTEN *` handling for `upsd.exe` in NUT for Windows builds. [PR #3237]
  - Extended processing of `MAXCONN` setting to allow larger values than the operating system allows, by only waiting for that amount of Unix sockets or Windows `HANDLE` objects at a time, and moving on to another chunk. The system-provided value can be further limited by `NUT_SYSMAXCONN_LIMIT` environment variable (e.g. in tests). For the other side of the coin, the NIT script now supports an optional `DUMMY_UPS_SWARM_COUNT` environment variable which can specify the amount of additional drivers to spawn for the sake of stress-testing, and `UPSLOG_SWARM_COUNT` for clients. [#3302]
  - Extended processing of `CERTREQUEST` setting to handle numeric or specific string values, to match both ways of reading ambiguous documentation. Added `configure --with-ssl-client-validation` toggle to expose the macro previously meant to be passed via `make` command line. [PR #3330, but beware issue #3329]

- `upsdrvctl` tool updates:
  - Make use of `setproctag()` and `getproctag()` to report parent/child process names. [#3084]
  - The global-level `debug_min` setting in `ups.conf` now also applies to bumping the debug verbosity of `upsdrvctl` tool. [#3276]
  - If we fail to stop a driver by signal (e.g. PID file was not saved), retry to `INSTCMD driver.exit` it by socket protocol. [#3277]

  - The NUT Integration Testing suite now also involves `upsdrvctl` to run one driver instance indirectly, helping make check-NIT` catch portability issues with different builds of the tool. [#2800, #3292]

- `upslog` tool updates:

  - Updated `help()` and failure messages to suggest `-m '*,-'` for logging of all known local devices to stdout. [#3083, #2756]

- `upsmon` client updates:

  - The `SHUTDOWNEXIT` option was not handled properly, and the requested long delays did not happen in practice. [issues #2133 and #3084, PR #3086]
  - Make use of `setproctag()` and `getproctag()` to report parent/child process names. [#3084]
  - Clarify the warning about lack of `POWERDOWNFLAG` in configuration. [#3391]
  - Introduced a `SHUTDOWN_HOSTSYNC` notification message, to report that the primary `upsmon` initiated the shutdown and has some secondaries to wait for first. [#3084]
  - Make sure an `FSD` notification is issued for each UPS when this primary `upsmon` instance sets it (and does not return to usual data processing loop to see and report it like secondaries do). This allows a `NOTIFYCMD` such as `upssched` on the primary to handle the pending power outage (e.g. begin stopping heavy services) even while `upsmon` waits for the secondaries to complete their shutdowns and log out of the `upsd` data server. [issue #3003, PR #3110]
  - Tickle service watchdog timer (if any) so the `nut-monitor.service` or equivalent unit is not killed off due to quietness during shutdown. [#3003]

    Note
    If using `upssched` and monitoring multiple UPSes, consider setting up a `START-TIMER-SHARED` rule with a short (approx. 1 second) timeout to group several `FSD` notifications into one executed action. [PR #3097]

- `upsc` has now optional JSON output [issue #3172, PR #3178]

- `upscmd` and `upsrw` clients would now report that the "OK" response only means that the server accepted the query, if wait mode ("TRACKING") was not specified, and what users could do about it (e.g. use the `-w` option or `export NUT_QUIET_OK_NOT` to hide the warning). [issue #3383]

- `upsset` CGI tool should now recognize `RANGE NUMBER` and `NUMBER` types. [#3164]

- `upsstats` CGI tool updates:

  - Now has JSON output mode via `?json` (or `&json`) CGI query parameter. New `@TREELINK_JSON@` keyword added and HTML templates updated to expose these JSON documents when browsing. [issue #2524, PRs #3171, #3249]
  - The `@STATUS@` directive now optionally supports separators (not only the hard-coded `<br/>` of the older versions. [issue #2525, PR #3318]
  - Handle `device.model` in addition to `ups.model` in upsstats HTML templates. [#3180]
  - Introduced a `@NUT_UPSSTATS_TEMPLATE@` command which the HTML template files now MUST start with (safety check that we are reading a template). [issue #3252, PR #3249]
  - (Experimental) Custom templates other than `upsstats{,-single}.html` can now be specified as CGI parameters, if locally permitted via `hosts.conf`. Modern-looking templates were provided. [issues #2524, #3316, PR #3304]

- `upssched` tool updates:

  - Previously in PR #2896 (NUT releases v2.8.3 and v2.8.4) the `UPSNAME` and `NOTIFYTYPE` environment variables were neutered for the timer daemon, since their values at the moment it first started were irrelevant when actual timers fired. Lack of those values was however also against the documented expectations. Now the values set when the `upssched` client is called and ends up updating the timer daemon would be passed into the timer. Also a `START-TIMER-SHARED` operation was added, to track a single named timer for possibly many devices and/or event types — in this case they would be comma-separated in the environment variable values. [issue #3092, PR #3097]

  - The `CANCEL-TIMER` action no longer stops at first hit for the timer name, but goes on (in case there are duplicates). A new `CANCEL-TIMER-MATCHED` was added, which also cares to match `UPSNAME` and `NOTIFYTYPE` values (if passed). [#3097]
  - Introduced support for `DEBUG_MIN` setting via `upssched.conf`. [#3097]
  - Introduced `upssched -l` mode to list currently tracked timers. [#3097]
  - Make use of `setproctag()` and `getproctag()` to report parent/child process names. [#3084]
  - Introduced optional passing of `NOTIFYMSG` text (normally originating from `upsmon` which calls `upssched`) as an environment variable into the ultimately executed `CMDSCRIPT` processes. [#3105]

- The `nut-driver-enumerator.sh` script (NDE) updates:

  - Revised info/error/warning/debug message emission so they go to `stderr` and have a consistent look. Revised some typos along the way. [issue #3194]

- `configure` script updates:

  - Added an option to `--enable-shared-private-libs` to deliver NUT common libraries as shared objects used by different NUT binaries, rather than linking just the used bits into each binary. This has some potential both for mayhem (so disabled by default) and for significant reduction of installation footprint. [issue #2800]
  - For ages, most recipes for building NUT had customized the `sysconfdir` to be `/etc/nut`, which is not exactly the **system** configuration directory. This is finally deprecated, with new `--with-confdir` configuration option taking over the role of a full path to configuration files (by default `${sysconfdir}${confdir_suffix}`), and new `--with-confdir-suffix` allowing to specify just `/nut` or `/ups` that would be tacked onto the default `${sysconfdir}` to resolve the `${confdir}`. Default behavior should be same as with previous builds: if `sysconfdir` is customized (or `prefix` is kept at built-in default), the `confdir_suffix` will default to empty; otherwise it assumes the value of `/${PACKAGE_NAME}` to become `/nut` in most cases. A `--with-confdir-examples` option was also introduced, to help distributions that place `*.conf.sample` files into docs or other locations. [#3131]
  - The `configure` script would now probe (if it can) the operating systems for more user and group account names, such as `upsmon`, `nutmon`, `ups`, `nut` (last hit wins, separately for user and groups accounts) settling on one of those if detected instead of `nobody` (and optionally `nogroup`). It would also warn if `nobody` or `nogroup` end up being used for a build. [#3173]
  - Default `PIDPATH` is now more strictly `/var/run`, unless building on a system conforming to FHS-3.0 standard where that location is absent or is a symlink, while `/run` exists and is a true directory. [#3099]
  - Made default NUT CGI URI (used in HTML templates, which may be edited later on a NUT deployment to customize for specific web server setup) configurable `--with-cgi-uri`. [#3207]
  - Introduced `--with-python{,2,3}-modules-dir` to specify PyNUT(Client) module installation location (for module-named dir to be created under it), if not bundling with NUT-Monitor UI app. By default the respective interpreter's *site-packages* or *dist-packages* location will be used, so you may have to adjust search paths for any other values. [#3062]
  - The script would now warn the caller if several different Python program paths or module locations were discovered (at least one automatically), in case this was not something the callers wanted — just so they know. [#1792]
  - New supported values were introduced for `--with-python{,2,3}=auto-prio=NUM`: this allows to auto-prioritize some **one** available Python interpreter. Default settings were changed from plain `auto` (allowing for multiple `python*` interpreters and `site-package` locations to be used), to prefer a `python3` if available, else `python2`, else `python`. [#1792]
  - The `configure` script should now try harder to report specifically the "purelib" location as `PYTHON*_SITE_PACKAGES`. [#1209]
  - Added a `--with-systemdsysusersdir=PATH` option to have systemd pre-create the user and group accounts for NUT, before packaging or manually executed commands do (this is particularly important for immutable-image operating environments). [PR #3262]
  - Streamlined handling of `--with-*` and `--enable-*` options using NUT *m4* macros; fixed reporting help for options whose defaults are evaluated. [issue #3049, PR #3140]

  - Automatic `--with-devd` could not detect the need to install a FreeBSD devd configuration snippet file, because it expected a source that should not have been there (probably a copy-paste typo in NUT v2.8.0). Also fixed a missed compiler family dependency when handling `--enable-strip`, a few logical mismatches of Solaris/illumos packaging options, and default report vs. setting of `--enable-force-nut-version-header`. Rephrased and updated many help messages. The `configure` script logs should now clarify where CFLAGS/LIBS/LDFLAGS values come from (*pkg-config*, *default*, user provided *confargs* etc.) [#3140]
  - Added `configure` script options for *libregex* tuning, just so that one is on par with our other optional dependencies. [#3140]
  - Changed `--enable-extapi-enphase` to not default to `auto` and then get installed on every system capable of using it; this is a niche capability for users of specific devices via a cloud portal. [follow-up to PR #2813 for issue #2807]
  - Fixed `m4/ax_realpath_lib.m4` up with more ways to try and gauge the build-time library name from linker/compiler, to improve the chances that `nut-scanner` would find them. [#3293]
  - Added an option to (primarily) `--disable-threading` for systems with detected but broken `libpthread` support, or to test alternate code paths during development or in CI. [#3300]
  - Adjusted C++ header search path on Termux when `-isystem` is involved, as noted with NSS builds. [issues #1599, #1711, PR #3353]

- Recipes, CI and helper script updates not classified above:

  - Fixed CI recipes for PyPI publication of PyNUT(Client) module to also include the source distribution (was posted for NUT v2.8.1 and v2.8.2 tagged releases, but absent for v2.8.3 and v2.8.4). [#3056]
  - A new version of BSD `make` in NetBSD 11 (Beta) aka `bmake` in some other operating systems, complains about some of our sub-`$(MAKE)` calls despite the explicit `+` prefix in corresponding lines, possibly because that call happens not directly in a rule definition, but in a substituted shell code snippet. The recommended workaround was to add a "Special Source" called `.MAKE` to let the tool know that we expect sub-makes in that recipe. However, **other** `make` implementations do not know about this token, so its support is tried and detected at `configure` time. [PR #3300]
  - Updated `make spellcheck` to help avoid asciidoc admonition blocks with visually invalid sentences (after rendering as a box in HTML or PDF). [#3077]
  - Added a `make dist-docs` goal to generate, collect and tarball all document types (`man`, `html-single`, `html-chunked`, `pdf`) that we may have enabled for the current build configuration and available tooling. [#1400]
  - Further developer workflow speedup with `make spellcheck-interactive-quick` which should now not re-check source texts that were okay with the previous dictionary contents, in case some new terms have to be added. [PRs #3186, #2871]
  - Added `make .libs-dev-PATH` recipe to generate a file which developers or CI scripts can source to prefer freshly built NUT libraries and some tools in their `LD_LIBRARY_PATH` and `PATH`. [#3379]
  - Introduced `make distcheck-completeness` goal to verify that our distribution archives can exactly reproduce themselves. [#2829]
  - Added a GitHub Actions CI job to generate, upload and recycle `make dist` and `make dist-docs` tarballs so they are easier to obtain for people and other CI systems (which might not want to follow a Git repository). Links to these artifacts can be seen in workflow page (or its logs), and added as a line in GitHub Checks list associated with a commit. Note that the GitHub Action storage keeps artifacts for at most 90 days. [#1400, #2829]
  - Revised CI and deliverable scripts, and Makefile recipes, to not use the verbatim `grep -E` (loudly preferred by newer systems, but may be absent on older ones) after all, nor use `egrep` (loudly disliked by newer systems). Instead, use what `configure` script detected for the generated files (or ones made from templates), and use a similar detection in standalone scripts. Also revised the use of `grep -q`, `id -u`, `diff -u`, etc. which are not ubiquitous, and of `test -e` which is not only absent in some older shells, but can cause them to abort processing the script immediately. Also the `if ! condition` syntax is not supported everywhere (or the `!` operator generally). [#3099, #1660]
  - Generally improved shell script portability (including `Makefile.am` embedded recipes) when mixing back-quotes and double-quotes. Namely, the classic Solaris `ksh` implementation (also used in current versions of Solaris and illumos based distributions) was strict about following (an older revision of) the POSIX standard here, and some use-cases misbehaved with our older script spelling. [issue #3196]
  - Revised `drivers/Makefile.am` to not always build `libdummy_serial.la`, but only if it is needed by serial-only or modbus drivers. Introduced a `libdummy_usb.la` to only build USB layer sources once and share the results between several USB-capable drivers. [#2800]

  - Revised `tools/gitlog2version.sh` helper script (logic added into a new `tools/semver-compare.sh`) with a mode to expand NUT SEMVER components into wide zero-padded numbers, so it is easier to alphanumerically compare different releases regardless of version component lengths in digits (and differences in their amounts). You can request an inverse operation with `NUT_VERSION_STRIP_LEADING_ZEROES=true`. Added tests to cover different shell interpreter platforms (piggy-back on the `tests/nut-driver-enumerator-test.sh` script), and made sure that outputs of legacy-mode processing (with `NUT_VERSION_DEFAULT` string provided by caller or saved in a tarball) are consistent with git-mode. The new `tools/semver-compare.sh` helper can be used directly to expand and strip version strings, sort and compare multiple versions as one simple operation. [issue #3055, PRs #3213, #3217]
  - Dropped the `compile` script from Git sources. It originates from automake and is added to work area (if missing) during `autogen.sh` rituals anyway. It is still distributed as part of `make dist` tarball. [#1209]
  - Extended `ci_build.sh` and `build-mingw-nut.sh` so that certain values of `NUT_SSL_VARIANTS=[yes, no, auto, ssl, nss, openssl]` can be used with generic builds to test a specific code path and not only the auto-detected one. [#1711]

- Upstreamed reference packaging recipes (DEB, RPM) from the 42ITy project which can be used with OBS (Open Build Service by SUSE), both to support end-user testing of NUT development, and to have a yet another NUT CI farm player with a distinct opinion on code quality. Many of the changes listed above happened thanks to this effort, and due to concerns raised by OBS build agents with various Linux distributions. [#1209, #1316, #3144]

- Updated `docs/*.txt`: add asciidoc comments with links to nut-website rendered contents of most interesting pages. [#3095]

- HTML renditions of NUT man pages and books (or chunked chapters thereof) now include the AnchorJS script to provide permalinks to sections. [#3185]

- Source directory `data/html` was renamed to `data/htmlcgi` in order to better reflect its contents and purpose, compared to documentation-oriented `html*` directories (and recipe variable names). The static HTML pages now also suggest a NUT icon (installed from `docs/images`). This may impact some packaging recipes which do not rely on `make install` alone. [#3049, #3249]

- Fixed man page naming for `nutdrv_siemens-sitop(.8)` (dash vs. underscore) to match the driver program name. Follow-up from slightly botched renaming in original contribution. [PR #545]

- The NUT Integration Testing suite (NIT) script, if started as `root`, can now consult its run-time situation vs. `BUILTIN_RUN_AS_USE` and `BUILTIN_RUN_AS_GROUP` environment variables, and if those accounts do not exist (e.g. running in a packaging build root), a different value like "nobody" or "nogroup" would be defaulted for tests. [#1209]

- Revised use of NUT macros for USB Vendor IDs listed in different drivers, so the names would be better exposed in generated `udev.rules` and similar files. [PR #3139]

2.3 Release notes for NUT 2.8.4 - what's new since 2.8.3

- Bug fixes for fallout possible due to "fightwarn" effort in 2.8.0+:

  - In `usbhid-ups` sources, introduced optional `HU_FLAG_PARAM_REQUIRED` for `setvar()` or `instcmd()` handling (and a `HU_TYPE_CMD_PARAM_REQUIRED` shortcut) for setting in the mapping table flags, to specify variables or instant commands that require an argument (either from caller or a non-`NULL` default in the run-time table after device data discovery); if the flag is not set, a zero value is assumed. Incomplete code was a regression of NUT v2.8.3 causing some instant commands to fail. [#2860, #2955]

- Fix fallout of development in NUT v2.8.0 and/or v2.8.1 and/or v2.8.2 and/or v2.8.3:

  - Fixed a regression in recipes of NUT v2.8.3 release (as compared to v2.8.2), where `configure --with-docs=all` no longer failed a run of the `configure` script when some of the required rendering tools were not in fact available. [#2842, fixed by #2921]
  - Some recipe improvements in earlier releases led to `make check` always running a spelling check (if tools are available), even if the explicit `configure --disable-spellcheck` option was used. Now it would not run if disabled (e.g. to speed up CI builds in scenarios that focus on other aspects of the code base), although developers can still use the explicit `make spellcheck*` goals, when tools are in fact available. [#2973]

  - A change in `Makefile.am` recipes to evaluate some driver names in the `DRIVERLIST` variables inspected by `configure` script, rather than having all their names hard-coded like before, led to inability to `configure --with-drivers=dummy-ups` [#2825, #2927, fixed by PR #2929]
  - A problem noted with `upsdrvquery` (since NUT v2.8.1) message logging at high debug verbosity levels (5+) with very large blocks of content has exposed a deficiency in variable-argument handling, and specifically adaptive resizing of the output buffer or truncation of logged inputs (which is something NUT code tried to do since the beginning of time), and could lead to "segmentation fault" crashes on some platforms. [issue #2948, PR #2963]
  - Documentation build recipes overly zealously pre-processed source files, which was not applicable for each and every document type we have (e.g. binary images for illustrations); this caused grief with some toolkits. [issue #2989]

- common code:
  - Revised common `writepid()` to use `altpidpath()` as location for the PID file creation, if the default `rootpidpath()` is not accessible (e.g. daemon was not initially started as `root`). Likewise updated short PID file based signal sending to consult both locations. [#1717]
  - Linux may report a `/proc/X/exe` symlink with an embedded "(deleted)" suffix, if the binary was removed (or replaced) since the running process started. This confused our code which verifies that when it is sending a signal to a PID, that PID does reflect the expected NUT program. [#3021]
  - Refactored NUT "common" sources to reference `nut_version.h` macros from a smaller C source file, to minimize the compilation unit size impacted by development iterations. [issue #2097]
  - Common code hardening: added sanity-checking for dynamically constructed or selected formatting strings with variable-argument list methods (typically used with log printing, `dstate` setting, etc.) [#2450, #3016]

- Warn if `%n` formatting string is used — it is deprecated in some newer distros due to security concerns.
  - Refactored repetitive implementations of `inet_ntopSS()` (nee `inet_ntopW()` in `upsd.c`) and `inet_ntopAI()` methods into `common.c`, so now they can be re-used or expanded more easily. [#2916]

- `upsd` updates:
  - Fixed two bugs about printing the "further (ignored) addresses resolved for this name": the way to extract IP address string was not portable and misfired on some platforms, and the way to print had a theoretical potential for buffer overflow. [#2915]
  - Print arguments of a processed command into the debug log, to help track down what unsupported queries are about, etc. (but only endeavor to spend time, RAM and CPU on this if debug verbosity is high enough). Hide the sensitive commands' parameters unless verbosity is unusually high. [#3023]

- `upsdrvquery` API updates [#2969]:
  - Added `upsdrvquery_oneshot_conn()` for issuing one-shot queries using an existing `udq_pipe_conn_t *` connection. The caller manages the lifecycle of the socket connection, and the function includes a best-effort call to restore broadcast mode after the query to return the connection as it was.
  - Added `upsdrvquery_oneshot_sockfn()` for initiating one-shot queries using a socket filename. Shares internal logic with the existing `upsdrvquery_oneshot()`, which uses a UPS and driver name, respectively.
  - Introduced `upsdrvquery_restore_broadcast()` to explicitly restore broadcast mode (`BROADCAST 1`) on a connection, helping return it to a consistent and talkative state.
  - Revised connection ownership handling: internal functions like `upsdrvquery_prepare()` and `upsdrvquery_request()` no longer close connections they do not own. Responsibility for cleanup is now delegated to the caller to avoid unintended side effects and better align with expected usage patterns.

- common driver code:
  - Update reports of failed socket file creation, to help troubleshooting some error cases in the field. [#2959]
  - Removed workarounds trying to migrate legacy driver raised `ALARM` status tokens into modern `alarm_*` function logic. Rather, we keep supporting them as separate from the modern logic, seeing as `upsmon` does not care where the token itself was raised for its notifications. Driver-code related test-cases were updated to reflect these changes. [issue #2928, PRs #2931 and #2934]

  - Introduced some macros in `drivers/upshandler.h` for common syslog level definitions and message wording for beginning and failing `instcmd()` or `setvar()` operations consistently in different drivers. As a related change, operations that intend to turn off or restart the load, or can do that by side effect (e.g. calibration if batteries are old or dead), would explicitly `upslogx(LOG_CRIT,...)` by default before commencing. [#2957]
  - Fixed a couple of ancient memory leaks: one "shared" during driver program initialization, and one specific to `dummy-ups` wind-down. [#2972]
  - Added a `suggest_NDE_conflict()` method so drivers which lack access to the expected device can consistently suggest that this may be because of running both an NDE-wrapped service unit and a manually launched driver program at the same time. Currently added to `libusb{0,1}.c` code, but may later be expanded to e.g. serial drivers and other media, when their behavior in such situations gets identified. [follow-up to issue #477, PR #3041]

- `apc_modbus` driver updates:

  - The time stamp and inter-frame delay accounting was fixed, alleviating one of the problems reported in issue #2609. [PR #2982]
  - Fix missing variables due to mismatching format string. [PR #3013]

- `bcmxcp` driver updates:

  - The latching on to a previous replace battery status was fixed, with its alarm state variable now correctly being reset; previously a factually replaced battery did not clear the alarm and the whole driver needed to be restarted. [issue #2999, PR #3002]

- `clone`, `clone-outlet`, `nhs_ser` driver and `nutdrv_qx_ablerex` subdriver updates:

  - Refactored to follow modern handling of status and alarm conditions, aligning with current driver design practices. This includes fixing copy-paste related issues in alarm reporting and removing some alarm messages that should instead be reflected as status flags. [#2936]

- `dummy-ups` driver updates:

  - A new instruction `ALARM` was added for the `Dummy Mode` operation of the driver, enabling simulation of UPS alarm states more closely in line with modern, real-world UPS driver implementations. This follows the updated principle of keeping alarm states decoupled from the `ups.status` variable, with alarms now raised via common alarm functions rather than direct manipulation. [issue #2928, PR #2936]

- `nutdrv_qx` driver updates:

  - Added support for "preprocess"/"process" methods called from mapping tables to report back to the driver that an argument value was not supported, so `setvar()` or `instcmd()` can not proceed safely and should return `STAT_SET_CONVERSION_FAI` or `STAT_INSTCMD_CONVERSION_FAILED`. [#3017]
  - Introduced `innovart33` protocol support for Ippon Innova RT 3/3 topology UPSes. [#2938]
  - Updated `megatec` protocol for more detailed responses to `I` query which may return `ups.serial` (after a shorter `device.mfr`) and the `battery.runtime` (after a shorter `device.model`). Note that the expected response is shorter than in other dialects (38 vs. 39 bytes), so if this change breaks anything for your UPS that reported the values above correctly (e.g. the `ups.firmware` version becomes shorter or none of these are reported), please let NUT developers know. [#2980]
  - Revised `voltronic` protocol to suppress alarm "UPS is in ECO Mode", using "buzzword mode" settings more correctly than in the previous iteration, shipped in NUT v2.8.3 release (as PR #2750 for issue #2708). [issue #2494]
  - Introduced a `voltronic-axpert` subdriver for Voltronic Axpert inverters which speak the P30 protocol, currently in a highly experimental state: with initial support for query commands, but most values are "hidden" from default NUT builds by being defined in `experimental.*` namespace, and should also be enabled by `configure --with-unmapped-data-poi` Development was based on work done in the Voltronic Sunny subdriver in https://github.com/nickma82/nut/tree/nutdrv_qx_voltronic-sunny_rebased%2Bcommand [#1407]

    Still TODO: - Implement commands to write to the inverter. - Remove commented code originating from `sunny` driver. - Further testing. - Understand how to map values we can read/set with those devices to the NUT standard vocabulary at `docs/nut-names.txt` (or extend it by discussion and agreement with community — notably to track a `pv.*` namespace for photovoltaic, separately from `ups.*` etc.) - Update docs (manpages, acknowledgements...)

- `phoenixcontact_modbus` driver updates:
  - Added more settings that can be tuned — support for shutdown variables, UPS mode selector, PC reset delay after main power recovers, and automatic switch to battery mode (and back) if main power is below or above a defined threshold (see the new "Configurable Values" section in the man page). They can be configured via `default.*` values in `ups.conf`. [#2986]
- `pijuice` driver updates:
  - Converted to NUT standard use of `status_set()` with single-token values. [issue #2708]
- `snmp-ups` driver updates:
  - Added support for "fun"/"nuf" methods called from mapping tables to report back to the driver that an argument value was not supported, so `setvar()` or `instcmd()` can not proceed safely and should return `STAT_SET_CONVERSION_FAILED` or `STAT_INSTCMD_CONVERSION_FAILED`. [#3017]
  - Fixed `ups.test.date` to be semi-static in `apc-mib` mapping, so it would be queried more than once per driver up-time. [issue #3011]
  - Fixed debug-logging around `SU_FLAG_STATIC` entries to clarify when they get skipped. [issue #3011]
- `usbhid-ups` driver updates:
  - Added support for "fun"/"nuf" methods called from mapping tables to report back to the driver that an argument value was not supported, so `setvar()` or `instcmd()` can not proceed safely and should return `STAT_SET_CONVERSION_FAILED` or `STAT_INSTCMD_CONVERSION_FAILED`. [#3017]
  - `hid_ups_walk(HU_WALKMODE_INIT)`: report if exactly one of "fun" or "nuf" dynamic value mapping methods is defined in a one-line table, and this may preclude reads/writes of that variable. [#2956]
  - The `cps-hid` subdriver's existing mechanism for fixing broken report descriptors was extended to cover a newly reported case of nominal UPS power being incorrectly reported due to an unrealistically low maximum threshold, as seen with a EC850LCD device. [issue #2917, PR #2919]
  - Further revision of "ECO mode" related code in `mge-hid` subdriver, following up from work started for NUT v2.8.3 release. [PR #2956]
  - Added APC BVKxxxM2 and BKxxxM2-CH to list of devices where `lbrb_log_delay_sec=N` may be necessary to address spurious LOWBATT and REPLACEBATT events. [PR #2942, PR #3007, issue #2347, issue #3006]
  - The `powercom-hid` subdriver did not handle delayed shutdown commands properly, at least as of RAPTOR RPT-1500AP LCD models. The RPT series seem to have same USB protocol like Smart KING Pro series, and that only supports a specific set of possible delays for Shutdown commands (certain fractions or counts of whole minutes) — which should now be handled if `powercom_sdcmd_discrete_delay` flag is set. [issue #3000]
- New NUT drivers:
  - Introduced a `ve-direct` driver for Victron Energy UPS/solar panels monitoring. Most specific reported values are in an `experimental.*` namespace, as a community we need to come up with standard naming for those via `docs/nut-names.txt` [#440]
  - Introduced a `nutdrv_hashx` driver for numerous devices from Ablerex, Atlantis Land, Epyc, Infosec, ION, PowerWalker, Right Power Technology, Salicru, UPS Solutions and other vendors (originally shipped with a "PowerMaster+", "PowerMaster" or "PowerGuide" software companion suite). This seems to be a protocol developed by Cyber Energy for serial-port devices, subsequently used by different vendors in their own products or re-branded Cyber Energy creations. [#2940]
  - Introduced a `failover` driver for monitoring multiple UPS driver sockets and seamless switching out of UPS data in a failover situation, includes support for end-to-end tracked instant commands and also variable updating. [#2962]
  - Introduced USB (`powervar_cx_usb`) and Serial (`powervar_cx_ser`) drivers for Powervar CUSPP protocol, tested with GTS (USB) and UPM (USB, Serial) models. [#2988]
- The `nut-driver-enumerator.sh` script (NDE) updates:

  - Now NDE internally tracks dependency of one driver on another one that should be locally running to serve the "original" data points (`clone`, `clone-outlet`, `dummy-ups`, `failover`). It should create "soft" dependencies between respective service instances to order their start-up sequence. [#2962]
  - Fixed NDE to not consider "masked" systemd units as non-existent or as syntactically failed instantiated unit names. [#3033]

- NUT Monitor GUI:

  - Ported Python 3 version to Qt6, now shipped alongside Qt5 for systems with either or both, maximizing compatibility with old and new setups. [#2946]

- `upsmon` client:

  - Clearer debug logging of `SHUTDOWNCMD` and `NOTIFYCMD` that would be used (or warnings that none was set); flush output buffers after these messages and after each main loop cycle, so any emitted text is seen in a timely manner. [issue #3003, PR #3008]

- The `nutshutdown` script (end-game integration for UPS power-off in case of FSD initiated by `upsmon`) was updated to consider `MODE=none` set in `nut.conf` and bail out quietly. [issue #2935, PR #3008]

- Manual page recipes and contents:

  - Introduced handling (possibly rewriting) for man page section "Overviews, conventions, and miscellaneous" (commonly number 7), to deliver support for `man nut` queries (NUT overview manual page also created). [#2945]
  - A new `configure --with-docs-man-dir-as-base` option was introduced so that directories for man page sections can now be automatically named as either "base" number of the section (e.g. `man1`) or by full section name (`man1m`), as different OS distributions have different preferences in this regard. [#2950]
  - Option to `configure --enable-docs-man-for-progs-built-only` was added, to differentiate NUT builds that deliver man pages for only built programs (legacy default) or for all of them (as needed for docs sites). [#2976]
  - Option to `configure --enable-docs-changelog` was added, specifically to allow developer iterations to not waste CPU time rebuilding the huge `ChangeLog*` files whenever their Git index changes. [#3019]
  - Options to `configure --with-docs-changelog-start` and/or `configure --with-docs-changelog-end` were added to allow developers to customize the size of `ChangeLog*` files when they are generated. Default starting value is `auto` which applies the legacy default `v2.6.0` to release/pre-release builds, or when local Git version info could not be retrieved, and the most-recent release tag (or `master` as fallback) for usual build iterations. Default ending value is `HEAD` for the current git commit at the moment the ChangeLog is (re-)generated. Balancing against the option to not build `ChangeLog*` files at all, this couple allows quicker builds that exercise all relevant recipe code paths. [#3019]

- Extended the `gitlog2changelog.py` helper script to report start/end commits actually used, and to allow callers to tweak them better (not only `HEAD` for the end of range); this may be of interest to other projects which use this script. Allow `configure` to disable generation of either certain `ChangeLog*` rendering formats or completely, to speed up developer iterations (much time is wasted when dev-testing new code, due to git index changes if NUT was configured to build with documentation). [#3019]

- The `BUILD_TYPE=default-all-errors ci_build.sh` script handling was revised to simplify code, and to default in CI builds to a quicker mode which randomly mixes the selected SSL, USB and UNMAPPED variants (and relies on the dozens of NUT CI farm runs per iteration to likely cover all possible combinations), which should roughly halve the CI build times. Default activity for developer builds should remain as it was — to try each such "axis" sequentially. [#2973]

- Revised generation of links to external manual pages in HTML rendering of NUT manual pages (previous recipe iterations left DocBook XML `ulink` tag "as is", which was not understood by web browsers). [follow-up to PR #2797]

- Made the distro-dependent URL template for man pages configurable. [follow-up to PR #2797]

- Revised `make install-as-root` to fall back to legacy ways of enabling services, if `systemctl preset-all` fails (assumed due to a systemd 252 bug). [#3022]

- Added a `make check-parallel-builds` recipe to help troubleshoot recipes in sub-directories, and improved buildability of existing NUT sources starting from scratch there. This is a workflow useful for NUT development (e.g. to focus only on drivers, or tests, or nut-scanner) but not so much for end-user packaging where everything builds from the root directory. [PR #3030, follows up from PR #2825, highlights why issue #2584 better be solved]

- Revised `appveyor.yml` to run CI builds faster (forfeit MSYS2 ecosystem updates and some other steps) and more likely fit in one-hour allocation. Also have it install `mingw-w64-x86_64-python-pyqt6` so the `NUT-Monitor` application can get packaged (would need a capable Python run-time though). [#3046]

2.4 Release notes for NUT 2.8.3 - what's new since 2.8.2

https://github.com/networkupstools/nut/milestone/11

- Fix fallout of development in NUT v2.8.0 and/or v2.8.1 and/or v2.8.2:
  - Move of `NUT_DEBUG_LEVEL` and "-D" CLI option handling to start of driver programs for issue #2259 in NUT v2.8.2 release misfired with regard to data-dump mode (it no longer caused foreground by default). [#2408]
  - The `nut-driver-enumerator.sh` improvements misfired in v2.8.2 release with an overlooked bit of shell syntax, and caused `nut-driver@upsname` instances to not auto-restart when `ups.conf` is edited. [#682, #2410]
  - Addition of "NUT Simulated devices" support to `nut-scanner` in v2.8.2 broke detection of (in-)ability to find and query "Old NUT" servers via `libupsclient.so` (internal flag got always enabled). [#2246]
  - A fix for `upsmon` v2.8.1 setting of `OFFDURATION` [PR #2108, issue #2104, revisiting PR #2055, issue #2044] was overly zealous and impacted also the `OB` state in cases where communications to the data server were severed and `DEADTIME` setting was not honored. [PR #2462, issue #2454]
  - Using `drivername -c reload` (e.g. facilitated by `nut-driver-enumerator` script and service when editing `ups.conf`) led to disconnected Unix sockets and a tight polling loop that hogged CPU. While the underlying bug is ancient, it took recent development to hit it as a practical regression. [issue #1904, issue #2484]
  - Fallback `localtime_r()` and `gmtime_r()` for some platform builds where a `*_s()` variant was available was not handled correctly. [PR #2583]
  - A recently introduced `allow_killpower` did not actually work as an `ups.conf` flag (only as a protocol command). [issue #2605, PR #2606]
  - The ability of two copies of the driver program to talk to each other with `upsdrvquery.c` code was not complete for the case of indefinite `select()` wait timeout. Now `upsdrvquery_read_timeout()` fixed private use of `struct timeval={-1,-1}` as a trigger to `select(..., NULL)`, as logged in one part of code and not handled in the other, for the indefinite wait [#1922, #2392, #2686, #2670]
  - The `disable_fix_report_desc` option introduced for `usbhid-ups` driver since NUT v2.8.1 was not applied for early dialog with the device while its report descriptors were being discovered. Now this flag, as well as `interruptsize` and `interruptonly`, are considered before we first try to open the USB device handle. [#1575, #1512]
  - In `cps_fix_report_desc()` we intended to fix-up input and output voltages in certain cases against high voltage transfer, we only fixed-up one of them. [#1245]
  - `upsd` should now handle `TRACKING` value of `STAT_CONVERSION_FAILED` introduced in NUT v2.8.2 for the socket protocol (between driver and data server), by returning "ERR INVALID-ARGUMENT", so there is no change for the network protocol definition. [#2182]
  - The `configure --enable-inplace-runtime` option added in NUT v2.8.1 should now also try to detect and set default values for the `--with-drvpath`, `--with-cgipath`, `--datadir` and `--libdir` options to more closely match a packaged setup and avoid confusion with e.g. two incompatible NUT client libraries in system default search path. [#2895]
- Large parts of the NUT User Manual and NUT Developer Guide were relocated into the new NUT Quality Assurance and Build Automation Guide (maintained in `docs/qa-guide.adoc`), accompanied by new chapters written and detailed for this subject; the chapter about `ci_build.sh` script became a separate `ci_build.adoc` document included into the new document. Overall, this guide intended to help the current and future maintainers of NUT itself, as well as to inspire any other projects that investigate similar solutions. [#2832]
- SEMVER, know thyself!

  - Development iterations of NUT should now identify with not only the semantic version of a preceding release, but with git-derived information about the amount of iterations that followed (if available): the three-number "semver" would be seen on release snapshots, while other builds would expose the added components: one with the amount of commits on the main development trunk since the preceding release which are ancestors of the built code base, and in case of feature development branches — another component with the amount of commits unique to this branch (which are not part of the development trunk yet). This allows to produce more relevant (monotonously growing) version identifiers for packages and similar artifacts, with more meaningful upgrades via development snapshots, eventually. A copy of the current version information would be embedded into "dist" archives as a `VERSION_DEFAULT` file, among provisions for packager tuning. [#1949]
  - Documentation about this would be maintained in `docs/nut-versioning.adoc`
  - SMF manifests and systemd units now refer to man pages and their online variants under `NUT_WEBSITE_BASE` dependent on codebase maturity (development or release snapshot); many programs now display such references in their command-line usage help, method `suggest_doc_links()` was introduced for this purpose. [issue #722, PR #2733]

- A technologically and practically interesting revamp of NUT mesh of automake (`Makefile.am`) recipes was completed, allowing for a more parallelizable build routine on multi-CPU machines — utilizing more cores and completing in less "wall-clock" time that the standard `SUBDIRS` driven approach — when running `make -j (N)` from the project root directory to build everything enabled by the `configure` script.

  This was tested with several (GNU, BSD, Sun) implementations of the "make" program on the few dozen platforms that NUT CI farm tests on. Notably, GNU make 4.x and newer seems to process parallel high-level goals and sub-`make` runs better than the competition (including GNU make 3.x).

  It is not a radical rewrite like some other research suggested, and so retains the general structure and certain benefits and flexibility of that standard `automake` approach, including developer build workflows with a bespoke `Makefile` in every significant directory. This also retains (and builds upon) the benefits of older work done in NUT, for builds in one directory to depend on libraries and other artifacts built (once) in another.

  Overall, NUT CI farm build times got 25%+ shorter (which is important as some scenarios had hit the 1-hour timeout imposed by providers of free CI hosting coupled with the weak machines provided in their free layer), and we suppose this is an interesting case for other projects to draw inspiration from for their recipe refactoring. [PR #2825]

- As an aid for developers and maintainers, a new spell-checking recipe was added to first run non-interactive spelling checks in parallel, and **only** if something fails — run an interactive check to edit the text and/or the dictionary file. The `make spellcheck` rule now also benefits from the rewritten recipes, as detailed above, to visit directories with text files in parallel. Overall, these changes may save time on multi-CPU systems, if compared to a sequential walk of all texts (or their directories) as was done before. [#2871]

- The `make dist` goal now takes more care to require availability of the man pages to put into the prepared distribution archive. These may come either from the current build, or inherited from its sources (if using a tarball initially) on a platform without tooling required for man page generation.

  This requirement compromises usability of `make distcheck` on platforms without such tools from sources without pre-built man pages (e.g. builds from git), so a couple of new goals were introduced in PR #2842: - `make distcheck-fake-man` generates placeholder files named like pre-built man pages for any missing files, just for the purpose of constructing a sane-looking dist archive to `distcheck` strictly otherwise; - `make distcheck-ci` is routed to `distcheck` or `distcheck-fake-m` based on build circumstances (ability to build man pages or presence of pre-built pages, or lack of either); - Similarly, `make dist-ci` is routed to provide a strict or faked tarball; - Earlier defined goals like `distcheck-light` or `distcheck-valgrind` now take advantage of these mechanisms to also produce usable dist archives for their relaxed or purpose-specific tests.

- Revised behaviors for the `upsnotify()` common code introduced in recent NUT releases (integrating with service management frameworks, etc.):

  - It was a bit cryptic when it reported a **failure to notify** (e.g. when a NUT program was not running as a service currently), fixed now to report human-friendly text instead of internal enum codes. Follow-up to [issue #1590, PR #1777, PR #2136]
  - Drivers should no longer print warning messages about not-initializing the notification subsystem because not running as a service when they are either started explicitly to show the help message, or when their CLI arguments are fatally wrong (no UPS name, no `port`, invalid trailing keywords...)

  - NUT programs generally should default to not "spam" about lack of known notification technology if our first message to be suppressed is already about stopping that program. This might help nag distros into getting a service framework, or integrating theirs with NUT, but is generally annoying to end-users where there's little they can do about it (other than suppressing the message with `NUT_QUIET_INIT_UPSNOTIFY` envvar).

- Drivers, `upsd`, `upsmon`: reduce "scary noise" about failure to `fopen()` the PID file (which most of the time means that no previous instance of the daemon was running to potentially conflict with), especially useless since in recent NUT releases the verdicts from `sendsignal*()` methods are analyzed and lead to layman worded situation reports in these programs. [issue #1782, PR #2384]

- Drivers started with the `-FF` command-line option (e.g. wrapped into the systemd units to stay "foregrounded" **and** save a PID file anyway) should now also handle an existing PID file to interact with the earlier instance of the driver program, if still running (e.g. started manually). [#2384]

- Drivers executed to force an UPS shutdown (with `-k` CLI option) should now try harder to kill off a daemonized sibling, if it still runs (and did not handle a `driver.killpower` INSTCMD well). [#2666]

- Extended instant commands for driver reloading with a `driver.exit` command for a protocol equivalent of sending a `SIGTERM`, e.g. when a newer instance of the driver program tries to start. [#1903, #2392]

- A new `NUT_QUIET_INIT_BANNER` envvar (presence or "true" value) can now prevent the tool name and NUT version banner from being unilaterally printed out when NUT programs start. [issues #1789 vs. #316; #2573]

- Drivers would now report the socket they are listening on, and server would report full path to the driver socket it tries to connect to. A new `NUT_QUIET_INIT_LISTENER` envvar (presence or "true" value) can prevent the socket name from being unilaterally printed out when NUT drivers start. [#2764]

- The `upsdrvctl` tool improvements:

  - It should now warn if executed on systems where NUT was built with support for service management frameworks like systemd or SMF, so nut-driver service units prepared by `nut-driver-enumerator` would conflict with manually-executed driver programs. This warning can be hushed by exporting a `NUT_QUIET_INIT_NDE_WARNING` environment variable with any value.
  - Extended `upsdrvctl` with a `list` operation (or `-l` option) to report manageable device configuration names (possible `<ups>` arguments to `start`, `stop` etc. operations), or to confirm a single name that it is known, and a `status` operation for more information. [#2567]
  - Fixed support of `maxstartdelay` at the level of driver section in `ups.conf`; added support of `maxretry` and `retrydelay` at this level; bumped the default `maxstartdelay` from 45 to 75 seconds to accommodate for longer device initialization (e.g. due to support of more Megatec Qx dialects by `nutdrv_qx`). [#2885, #2888]

- `riello_ser` updates:

  - Added `localcalculation` option to compute `battery.runtime` and `battery.charge` if the device provides bogus values [issue #2390, following in the footsteps of #1692, #1685 done for `riello_usb`] (similar to `runtimecal` in some other drivers, may be refactored to that configuration and logic model in later NUT releases)

- `apcsmart` updates:

  - Revised code to use `strncpy()` and avoid potential overflows that are possible with `strcpy()` used before. [PR #2564]
  - Lost communications led to a logging flood, should not anymore. In fact, the driver should try fully reconnecting upon getting into a prolonged data stale condition. [issue #704, PR #2564]

- `nutdrv_qx` updates:

  - Added Visench C1K (using serial port converter with USB ID `1a86:7523`) as known supported by `nutdrv_qx` (Megatec protocol) since at least NUT v2.7.4 release. [#2395]
  - Introduced `innovart31` protocol support for Ippon Innova RT 3/1 topology UPSes. [#2712, #2798]
  - Introduced `q2` and `q6` protocol support; currently also based/tested on Innova devices, but other models than RT 3/1. [#2798]
  - Introduced a `gtec` subdriver and protocol, tested over USB with a Gtec ZP120N device. [#2818]

  - Fixed `hunnox_protocol()` to honour the optional `novendor` setting for devices that are confused by such query, e.g. DEXP LCD EURO 1200VA. [#2839]
  - Extended Voltronic protocol to support longer numbers as remaining `battery.runtime` value. [#2765]

- GPIO drivers:

  - Extended to support library API of not only libgpiod v1.x releases, but also v2.x; introduced a NUT `WITH_LIBGPIO_VERSION` C macro (in `config.h`) to differentiate the library variants. [issue #2833]

- New NUT drivers:

  - `bicker_ser`: added new driver for Bicker 12/24Vdc UPS via RS-232 serial communication protocol, which supports any UPS shipped with the PSZ-1053 extension module. [PR #2448]
  - `liebert-gxe`: added new driver with support for Liebert GXE Series UPS (serial or USB posing as a serial port). [#2629]
  - `nhs_ser`: added new driver for numerous NHS Nobreaks, senoidal line — UPS models with serial port, made by NHS Sistemas Eletronicos LTDA and popular in Brazil. Currently this driver only builds on Linux. [#2692]
  - `phoenixcontact_modbus` driver: Introduced Phoenix Contact QUINT4-UPS/24DC management (only new modbus addresses). [#2689, #2716]

- Added `scripts/external_apis` with an example script integrating a non-native protocol with NUT (as live-stream input for `dummy-ups` NUT driver to publish further); that example can be installed using `configure --enable-extapi-enphase`. [issue #2807, PR #2813]

- `usbhid-ups` and `netxml-ups` updated to handle "No battery installed!" alarm also to set the `RB` (Replace Battery) value in `ups.status`. This may cause dual triggering of notifications (as an `ALARM` generally and as an important `REPLBATT` status in particular) in `upsmon`, but better safe than sorry. [#415]

- `usbhid-ups` updates:

  - Support of the `onlinedischarge_log_throttle_hovercharge` in the NUT v2.8.2 release was found to be incomplete. [#2423, follow-up to #2215]
  - Added support for `interrupt_pipe_no_events_tolerance=N` setting to optionally prevent UPS lockup, indicated by continuous "Got 0 HID Objects" situation as a clue, by reconnecting on stale data. Note that while some devices just report information upon subsequent poll and just have nothing urgent to declare with an USB interrupt, others (e.g. APC BXnnnnMI) were seen to lock up until a full connection restart. [#2671, #2681]
  - Added support for `lbrb_log_delay_sec=N` setting to delay propagation of `LB` or `LB+RB` state (buggy with APC BXnnnnMI devices circa 2023-2024). This may work better with flags like `onlinedischarge_calibration` and `lbrb_log_delay_without_calibrating` for some devices. [issue #2347, PR #2565]
  - General suggestion from `possibly_supported()` message method for devices with VendorID=`0x06da` (Phoenixtec), seen in some models supported by MGE HID or Liebert HID, updated to suggest trying `nutdrv_qx`. [#334]
  - MGE HID list of `mge_model_names[]` was extended for Eaton 9E, 5PX and 5SC series (largely guessing, feedback and PRs for adaptation to actual string values reported by devices via USB are welcome), so these devices would now report `battery.voltage` and `battery.voltage.nominal`. Also a device from 5S series (5S1200AU) was tested, although it identifies as an "Ellipse PRO" in USB metadata. [#2380]
  - Added `ups.beeper.status` support for Masterpower MF-UPS650VA using the MGE HID subdriver. [#2662]
  - Added basic support for EcoFlow River 3 Plus and Delta 3 Plus models. [issue #2735, PRs #2740, #2837]
  - Added support for `0x09D6:0x0001` devices using the MGE HID subdriver assuming devices made by KSTAR (alternately using MGE vendor ID). [#2661]
  - `powercom-hid` subdriver sent UPS shutdown commands in wrong byte order, at least for devices currently in the field. A toggle was added to set the old behavior (if some devices do need it), while a fix is applied by default: `powercom_sdcmd_byte_o` [PR #2480]
  - `cps-hid` subdriver now supports more variables, as available on e.g. CP1350EPFCLCD model, including temperature. [PRs #2540, #2711]

  - Loudly suggest to set `pollonly` flag and default a shorter `pollfreq` for CPS devices, to try avoiding device-driven timeouts. [#1689] Also adjust default `offdelay` and `ondelay` to reasonable values, and warn the users with CPS devices if their configured values are not multiples of 60. [#432, #1394]
  - In `cps-hid` subdriver, `cps_fix_report_desc()` method should now handle mismatched `LogMax` ranges for input and output voltages, whose USB Report Descriptors are wrongly encoded by some firmware versions. [#1512]
  - In `cps-hid` subdriver, try to fix frequency scaling based on the values we see from the device and/or configuration overrides (low, nominal, high) so `499.0 Hz` reading that comes from some firmware versions gets reported properly as `49.9Hz`. [#2717]
  - USB parameters (per `usb_communication_subdriver_t`) are now set back to their default values during enumeration after probing each subdriver. Having an unrelated device connected with a VID:PID matching the `arduino-hid` subdriver prevented use of an actual `usb-hid` device due to changes made to this struct during probe. [#2611]

- USB-capable drivers generally:
  - ...could earlier log `(nut_)libusb_get_string: Success` due to either reading an empty string or getting a success code `0` from libusb. This difference should now be better logged, and not into syslog. [#2399]
  - ...now can benefit from a new `nut_usb_get_string()` method which can do a fallback `en_US` query for devices which report a broken "langid" language identifier value. This notably manifested in inability to query the device Manufacturer, Model and Serial Number values with some buggy device firmware or hardware. [PR #2604, issues #1925, #414]
  - Currently this was tested to fix certain device discovery with the `usbhid-ups` driver; but should also apply out of the box to same discovery logic in `blazer_usb`, `nutdrv_qx`, `riello_usb` and `tripplite_usb` drivers.
  - Also applied to `nut-scanner` and `libnutscan`. [issue #2615]
  - More work may be needed for other USB-capable drivers (`richcomm_usb`, `nutdrv_atcl_usb`) and for general code to collect string readings and other data points, and to configure the fallback locale or choose one if several are served by the device. [issues #2613, #2614, #2615]
  - ...should now be more likely to succeed with iterative detection of an UPS interface on a composite USB device or when looking at devices with non-default interface/endpoint/config numbers. [PR #2611]
  - ...should now accept a `LIBUSB_DEBUG=INTEGER` setting in `ups.conf` (as well as an environment variable that can be generally set via `nut.conf` or service unit methods or init script), to enable troubleshooting of LibUSB itself. [issue #2616]
  - ...should now not log "insufficient permissions on everything" alone when some devices were accessible but just did not match — clarify that case in the next line, when applicable. [PR #2699]
  - ...should now track the fact of `assumed_LogMax` (typically when firmware encoding logic is wrong, and `-1` is resolved by parser). [#1512, #1040]

- `snmp-ups` updates:
  - Added support to monitor BayTech RPC3-NC PDUs, with `baytech-rpc3nc-mib` serving same basic data points as were available in `baytech-mib.c`, but checking for a different model OID subtree and different OIDs for the device model information. [#2779]
  - Fixed `netvision-mib`: sync `netvision_output_info` with currently available `SOCOMECUPS-MIB.txt`. [#2803]

- `mge-utalk` driver will no longer set non-standard status values `COMMFAULT` and `ALARM` (for a specific status bit); instead, it will set modern `ups.alarm` with values `COMMFAULT` and/or `DEVICEALARM` (and raise an `ALARM` in `ups.status` for either, as standard alarms go). [#2708]

- Introduced a new driver concept for interaction with OS-reported hardware monitoring readings. Currently instantiated as `hwmon_ina219` specifically made for Texas Instruments INA219 chip as exposed in the Linux "hwmon" subsystem of its "sysfs" interface (and talking I2C under the hood), this approach seems to have good potential to expand into covering more devices and perhaps platforms. [#2430]

- Introduced `ECO` status concept for "ECO mode" (or "High Efficiency" mode, or "Energy Saver System"...) as named and defined by hardware vendors. One common aspect is that this is a balance of electrical efficiency vs. robust outage protection (which may be overkill for IT equipment whose PSU can survive several milliseconds on capacitors alone) which can be selected at run-time. Previously such choice was made at the time of purchase, with the UPSes only supporting some one protection strategy. [issue #2495, PR #2637]

  - Updated documentation, end-user clients (CGI, NUT-Monitor UI);
  - Updated `upsmon` client with ability to report entering and exiting the ECO mode if reported by the driver;
  - Initial implementation for Eaton devices with `usbhid-ups` driver.

- Introduced handling for the `ALARM` status, which already existed as a common denominator for devices seen with active `ups.alarm` variables. UPS devices in an `ALARM` status are generally considered volatile and may be considered critical/dead by the `upsmon` client earlier than in other statuses (e.g. in no-communication situations). It has to be noted that there is no common standard for what constitutes an alarm and such alarm states were also previously observed for less severe reasons. This depends on the manufacturer/device-specific implementation in the driver. [issues #415, #2657, PR #2658]

  - Updated documentation, end-user clients (CGI, NUT-Monitor UI);
  - Updated `upsmon` client with ability to report entering and exiting the ALARM status if reported by the driver;
  - Updated `upsmon` client with setting to toggle whether an `ALARM` status can prompt the UPS to become critical in certain situations.

- The `upsmon` client can now also report entering and exiting the `OVER` (UPS overloaded), `TRIM` and `BOOST` (adjusting for bad input voltage) states. A setting `OVERDURATION` was introduced to define a timeout after which a non-communicating UPS that was last seen in state `OVER` should be considered critical (or not). [PR #1074, issue #2877]

- Revised `upssched` timer handler that can be called from `upsmon` as its `NOTIFYCMD` to not report confusing environment variable values of `NOTIFYTYPE` and `UPSNAME` from the original call when a timer eventually fires — these values are irrelevant at that distant future. The NIT (NUT Integration Tests) suite was extended to configure and call this tool, facilitating its development and troubleshooting. Also the `upssched` timer daemon part can now save its PID file (so that NIT can terminate it after tests). [#2890]

- New `libupsclient` API methods added:

  - `upscli_str_add_unique_token()` and `upscli_str_contains_token()`, to help C NUT clients process `ups.status` and similarly structured strings same way as NUT core code base. [#2852, #2859]
  - `upscli_connect()` was previously always blocking; now this is sort of optional, with new `upscli_set_default_connec` able to change the implicit timeout from default zero (meaning blocking) to a positive value (or back to 0). Several NUT clients (`upsc`, `upscmd`, `upsrw`, `upslog`, `upsmon`, `upsimage`, `upsset` and `upsstats`) were updated to default with a 10-second timeout in case of name resolution lags or unresponsive hosts (notably a problem with `upsmon` contacting many remote systems at once). The `NUT_DEFAULT_CONNECT_TIMEOUT` environment variable can be used to modify this timeout for all clients. Further new methods here include `upscli_get_default_connect_timeout()` to retrieve a copy of the last stored timeout, and `upscli_init_default_connect_timeout()` to initialize the value from a number of sources with different priorities. [#2847]
  - Symbols exported from `libupsclient` now include `nut_debug_level*` so that NUT clients can be usefully debugged (e.g. using `NUT_DEBUG_LEVEL` environment variable). [#2847]

- Several NUT clients including `upscmd`, `upsrw`, `upsimage`, `upsset`, `upsstats`, and `upslog` (during reconnection), did not `UPSCLI_CONN_TRYSSL` so went plaintext even when secure connections were possible. Fixed to at least try being secure, same way as `upsc` does for a long time. [#2847]

- `upsmon` updates:

  - It was realized that the `POWERDOWNFLAG` must be explicitly set in the configuration file, there is no built-in default in the binary program (the settings facilitated by the `configure` script during build "only" impact the `upsmon.conf.sample`, init-scripts and similar files generated from templates). [issue #321, PR #2383]
  - Added an `OBLBDURATION` (seconds) setting to optionally delay raising the alarm for immediate shutdown in critical situation. [#321]
  - Optimized `parse_status()` by not checking further strings if we had a match; report unexpected tokens in debug log. [#415]
  - Revised internal `do_notify()` method to support formatting strings with two `%s` placeholders, to use if certain use-cases pass any extra information (e.g. not just "we have alarms" but their values too). [#415]

  - Introduced handling for "unknown" `ups.status` tokens, reporting them as "OTHER" notification type (whenever the set of such tokens appears or changes) or "NOTOTHER" when they disappear. [#415]

- `upslog` updates:

  - Added support for limiting the loop count. Using in NIT (NUT Integration Test) suite for double profit (checking the tool and fallback in NIT).
  - If you use the legacy CLI options for single-system logging (`-s <system>` and `-l <logfile>`) along with newer tuple(s) for multiple-system logging (repeatable `-m <system,logfile>`), previously the single-system options were overridden by the tuple(s); now they become part of the list.
  - Internally, changed from use of shared global variables to query one UPS at a time, populated from the new list of tuples during each loop cycle, to passing and using the new tuple structures directly.
  - The `upsname` in the `system=upsname[@hostname[:port]]` parameter may be an asterisk `*` to query for devices currently served by the hostname.
  - Same log file may safely be used in different logging tuples (it is then recommended to use `%UPSHOST%` in a custom formatting string).
  - Fixed printing of `%UPSHOST%` when multiple systems are being logged.
  - A `%t` for a TAB character can now be used in the formatting string.
  - Added `-N` to prefix `%UPSHOST%%t` before the format string (whether default or custom). Useful when logging many systems into same target.
  - Added `-D` for debugging (and foregrounding by default), like with other NUT daemons.
  - Added systemd and SMF service integration. [#1803]

- More systemd integration:

  - Introduced a `nut-sleep.service` unit which stops `nut.target` when a system sleep was requested, and starts it when the sleep is finished. This helps avoid NUT shutting down a woken-up system just because its power state was critical before the sleep (called as a `SHUTDOWNCMD` implementation by the end-user), and a next-read timestamp was not seen (deemed to be a stale UPS, meaning lost communications during critical state, so must go down ASAP). While not as elegant as native systemd "inhibitor interface" support, this approach does work. [#1833, #1070]
  - Introduced support for the "inhibitor interface" as well (should be available on systems with systemd version 183 or newer) for a better handling of the time jump specifically in the `upsmon` client via new `Inhibit()` method in `common.c`. [#1070]
  - As an extension of the logic introduced above, hopefully now `upsmon` would behave better in face of any significant and unexpected clock jumps (on POSIX builds so far), even if they are not suspend/hibernate events (or they were but we could not have an inhibit lock). Now they should be handled similar (avoid stale UPS data and rash decisions) for summer/winter time change on non-UTC deployments, a debugger suspending the `upsmon` process, etc. [#2597]
  - Introduced delivery of default systemd presets (lists of enabled/disabled units). [#2721]
  - A `nut-udev-settle.service` was introduced to replace dependency on the `systemd-udev-settle.service` which is deprecated and causes warnings on some systems. It was shown to benefit NUT use-cases however. [#2638]

- `gamatronic` driver revised for safer memory operations; this was reported to have fixed a Segmentation Fault seen in earlier NUT releases with some of the devices supported by this driver. [#2427]

- `upsd` updates:

  - `upsd_cleanup()` is now traced, to more easily see that the daemon is exiting (and/or start-up has aborted due to configuration or run-time issues). Warning about "world readable" files clarified. [#2417]
  - Failure to `LISTEN` on an invalid host name (e.g. `localhost:3493` or `1.2.3.4/24`) is now logged in a more actionable manner. [#2665]

- `nut-scanner` updates:

- The tool relies on dynamic loading of shared objects (library files) orchestrated at run-time rather than pre-compiled, to avoid excessively huge package footprints. This however relies on knowing (or sufficiently safely guessing) the library file names to use, and short `libname.so` is not ubiquitously available. With the new `m4` macro `AX_REALPATH_LIB` we can store and try to use the file name which was present on the build system, while we search for a suitable library. [#2431]

  Note
  A different but functionally equivalent trick is done for `libupsclient` during a NUT build. * Fixed support for IPv6 addresses (passed in square brackets) for both `-s` start/`-e` end command-line options, and for `-m cidr/mask` option. [issue #2512, PR #2518] * Newly added support to scan several IP addresses (single or ranges) with the same call, by repeating command-line options; also `-m auto{,4,6}` can be specified (once) to select IP (all, IPv4, IPv6) address ranges of configured local network interfaces. An `/ADDRLEN` suffix can be added to the option, to filter out discovered subnets with too many bits available for the host address part (avoiding millions of scans in the extreme cases). [issue #2244, issue #2511, PR #2509, PR #2513, PR #2517] * Implemented parallel scanning for IPMI bus, otherwise default scan for all supported buses with `-m auto` takes unbearably long. [#2523] * Bumped version of `libnutscan` to 2.6.0, it now includes a few more methods and symbols from `libcommon`. [issue #2244, PR #2509] * Do not actively suggest `vendor(id)`, `product(id)`, and `serial` options for `bcmxcp_usb`, `richcomm_usb` and `nutdrv_atcl_usb` drivers for now [#1763, #1764, #1768, #2580]

- All drivers should now support the optional `sdcommands` setting with a site-local list of instant commands to handle `upsdrv_shutdown()`, which may be useful in cases when the driver's built-in commands (or their order) do not meet the goals of particular NUT deployment. This can also help with shutdown endgame testing, using a mock command like starting the beeper (where supported) to verify that the UPS communications happen as expected, without compromising the load connected to the UPS.

  Also defined `EF_EXIT_SUCCESS` and `EF_EXIT_FAILURE` in `include/common.h` to avoid magic numbers in code like `set_exit_flag(-2)`, and revised whether it is getting set at all in "killpower" vs. other cases, based on new `handling_upsdrv_shutdown` internal flag.

  Note
  During this overhaul, many older drivers got their first ever supported INSTCMD such as `shutdown.return`, `shutdown.stayoff` or `load.off`. Default logic that was previously the content of `upsdrv_shutdown()` methods was often relocated into new `shutdown.default` INSTCMD definitions. [#2670]

- Common code:

  - `upscli_splitname()` should now recognize `upsname:port` typos (missing the `@hostname` part) and error out gracefully.
  - Introduced a `NUT_DEBUG_SYSLOG` environment variable to tweak activation of syslog message emission (and related detachment of `stderr` when backgrounding), primarily useful for NIT and perhaps systemd. Most methods relied on logging bits being set, so this change aims to be minimally invasive to impact setting of those bits (or not) in the first place. [#2394]
  - `root`-owned daemons now use not the hard-coded `PIDPATH` value set by the `configure` script during build, but can override it with a `NUT_PIDPATH` environment variable in certain use-cases (such as tests). [#2407]
  - Allow drivers to set `STATEPATH` via `ups.conf` to match `upsd` custom configuration ability; the data server would prefer the value from `ups.conf` over the one in `upsd.conf`, if both are present. Note that `NUT_STATEPATH` environment variable trumps both. [issue #694]
  - Introduced a check for daemons working with PID files to double-check that if they can resolve the program name of a running process with this identifier, that such name matches the current program (avoid failures to start NUT daemons if PID files are on persistent storage, and some unrelated program got that PID after a reboot). This might introduce regressions for heavily customized NUT builds (e.g. those embedded in NAS or similar devices) where binary file names differ significantly from a `progname` string defined in the respective NUT source file, so a boolean `NUT_IGNORE_CHECKPROCNAME` environment variable support was added to optionally disable this verification. Also the NUT daemons should request to double-check against their run-time process name (if it can be detected). [issue #2463]

  - Introduced `m4` macros to check during `configure` phase for the platform, and a `nut_bool.h` header with `nut_bool_t` type to use during build, to avoid the numerous definitions of Boolean types and values (or macros) in the NUT codebase. [issue #1176, issue #31]
  - Custom `distcheck-something` targets did not inherit `DISTCHECK_FLAGS` properly. [#2541]
  - Added `status_get()` in NUT driver state API, to check if a status token string had been set recently, and to avoid duplicate settings; fixed `status_set()` for multi-token arguments. [PR #2565, issue #2708]
  - Local socket/pipe protocol introduced a `LOGOUT` command for cleaner disconnection handling. [#2572]
  - Codebase adapted to the liking of `clang-18` and newer revisions of `gcc-13+` whose static analyzers on NUT CI farm complained about some imperfections after adding newer OS revisions to the population of build agents. [#2585, #2588]
  - New checks in `clang-19` brought new findings about mismatched formatting strings and `int`-ish parameters of respective methods. Overall, had to change formatting strings in some cases, variable types in others (e.g. flags or notification types do not make sense as signed) and added casting in a few places that remained, because:

- `%x` style formatting requires an `unsigned int` variable
- Numeric literals and macros are `int` by default
- Results of math with unsigned types like `uint16_t`, done in some cases, are up-scaled into `int` by default
- `char` values, `unsigned` or not, seem to be also up-scaled into `int`
- Updated `docs/nut-names.txt` with items defined by 42ITy NUT fork. [#2339]
- Various recipe, documentation and source files were revised to address respective warnings issued by the new generations of analysis tools. [#823, #2437, nut-website issue #52]
- Fixed `configure` script to use default (target-specific) values of `CFLAGS`, `LIBS` etc. when probing relevant settings for each third-party dependency; as a consequence, on systems that support building for many targets, we check relevant build-ability for that target and not for the building system itself. [issue #2673, PR #2675]
- Fixed dynamic linking of Mozilla NSS on systems like Solaris/illumos, where the shared objects are not packaged into the common RPATH. [issue #2674, PR #2675]
- `lib/*.pc.in`: propagate `-R/PATH` to NUT library installation location (by default not in system prefix) to help third-party clients link with us automatically. [related to issue #2782, PR #2865]
- Added `scripts/valgrind` with a helper script and suppression file to ignore common third-party problems. [#2511]
- Fixed `configure --with-valgrind=PATH` vs. detection of its usability; fixed some portability issues with detection of usability per se, tried `--with-valgrind=auto` the default to auto-detect and use the feature (in tests) wherever possible, but too many NUT CI farm agents disagreed; so for now the default is `no`. [#2823]
- When drivers dump collected data (during troubleshooting), flush `stdout` buffer immediately for sane logging (especially on Windows). [PR #2699]
- Revised `nut.exe` (the NUT for Windows wrapper for all-in-one service) to be more helpful with command-line use (report that it failed to start as a service, have a help message, pass debug verbosity to launched NUT programs...) and add a man page for it. [issue #2432, PR #2446]
- The `scripts/Windows/build-mingw-nut.sh` helper script was extended to use `nut_build_${ARCH}` and `nut_instal` directories by default, with the older `nut_build` and `nut_install` short names becoming just a symbolic link to the latest executed build: this should help compare the differences of 32/64-bit builds, without them stepping on each other's toes.
- NUT binding for Python and the `NUT-Monitor` Python UI client updates:
  - The `PyNUTClient` module should no longer rely on presence of a `telnetlib` module in the build or execution environment (deprecated in Python 3.11, removed since Python 3.13). [issue #2183, PR #2792]
  - The PyPI distribution of the `PyNUTClient` module tarball should now use a lower-cased file name (and immediate versioned directory name inside) to match the requirements of PEP-0625. The Python module name (and its directory) should remain camel-cased. [#2773]

  - Added man page for the `NUT-Monitor` Python UI client.
  - The `NUT-Monitor` Python UI client itself was revised to report the `PACKAGE_VERSION` and `NUT_WEBSITE_BASE` strings in the "About" dialog contents; localization support for the dialog and some other resources was revised to work in Py3Qt5 variant of the script. [#722]

- Documentation recipe updates:

  - Enabled installation of built single-file PDF and HTML (including man page renditions) under the configured `docdir`. It seems that previously they were only built (if requested) but not installed via `make`, unlike the common man pages which are delivered automatically. [#2445]

    Note
    The `html-chunked` documents are currently still not installed.

  - Added support to `./configure --with-doc=man=dist-auto` to use the distributed manual page files if present; only fall back to (re-)building them if we can. [#2473]
  - Added a `make distcheck-light-man` recipe to require verification that the manual page files can be built using the prepared "tarball" archive. [#2473]
  - Revised the documentation building recipes, with the goal to avoid building the `ChangeLog` products and their intermediate files more than once (but still react to `git` metadata changes during development), and to sanity-check the resulting final document (currently only for `html-single` mode).

    As part of this, the `CHANGELOG_REQUIRE_GROUP_BY_DATE_AUTHOR` setting was added (for `make` calls and used by `tools/gitlog2changelog.py.in` script), and it defaults to `true` allowing for better ordered documents at the cost of some memory during document generation. [#2510] * Updated man page generation with `configure` script options to specify that manual section codes on the target platform differ from (Linux-based) defaults hard-coded into page sources; this should allow to simplify NUT packaging recipe maintenance in those diverse distributions (no more need to update patches for changed or added documentation sources). * Lines in first section of NUT configuration report (which can optionally remain as `config.nut_report_feature.log` and be installed into shared documentation of a NUT package) are now better grouped as miscellaneous features and detection results, then drivers and programs/tools. [#2676]

- Added a `common/Makefile.am` build product for a new internal library `libcommonstr.la` which allows a smaller selection of helper methods for tools like `nut-scanner` which do not need the full `libcommon.la` nor `libcommonclient.la`. [#2478, #2491]

- Added a `drivers/Makefile.am` build product for a new internal library `libserial-nutscan.la` to simplify `tools/nut-` recipes. [#2490]

- Build of `snmp-ups` and `netxml-ups` drivers now explicitly brings linker dependency on chosen SSL libraries. [#2479]

- Introduced `configure --with-modbus+usb` option to require an USB-capable libmodbus, and defaulted a couple of specific situations as if this was required (implicitly): `configure --with-modbus --with-usb` and either `--with-drive` (actually implies `--with-modbus`) or `--with-modbus-includes=... --with-modbus-libs=...` as a way to avoid surprises with custom NUT builds aiming to have an USB-capable `apc_modbus` driver (currently this requires a custom-built libmodbus). Also fixed (re-)detection of libmodbus RTU USB support with static libmodbus builds. [#2666]

- Drivers built with libmodbus (`phoenixcontact_modbus`, `generic_modbus`, `huawei-ups2000`, `socomec_jbus`, `adelsystem_cbi`, `apc_modbus`) should now report whether the library is linked dynamically or statically — this can help in troubleshooting (especially of `apc_modbus` which may be using a custom build of the library not delivered by the operating system). [#2897]

- Brought keyword dictionaries of `nutconf` and `augeas` NUT configuration file parsers up to date; restored automated checks for `augeas` lenses. [issue #657, issue #2294]

  Note
  Some known issues remain with augeas lens definitions, so currently they should be able to parse common simple use-cases but not certain types of more complex configurations (e.g. some line patterns that involve too many double-quote characters) which are valid for NUT proper. [#657]

- Cross-builds using only a host implementation of `pkg-config` program should now ignore host `*.pc` files and avoid confusion.
- NUT CI farm build recipes, documentation and some `m4`/`configure.ac` sources updated to handle a much larger build scope on MacOS. Also migrated the builders to Apple Silicon from x86 (deprecated by CircleCI). Disabled `HOMEBREW_NO_AUTO_U` to gain 40 min per build at cost of slightly older environment. [#2502, #1579]
- Introduced a simple experiment to expose NUT client readings as filesystem objects via FUSE, in `scripts/fuse/execfuse-nut` now. [#2591]
- Introduced `make install-as-root` to create directories not directly populated by `make install` and NUT build artifacts, apply permissions and (on some platforms) restart services involved with NUT. [#1298]

2.5 Release notes for NUT 2.8.2 - what's new since 2.8.1

https://github.com/networkupstools/nut/milestone/10

- Fix fallout of development in NUT v2.8.0 and/or v2.8.1:
  - dstate machinery: a segmentation fault (null pointer dereference) was possible with `INSTCMD` processing of commands without parameters nor `TRACKING` identifier. [#2155]
  - USB bus number detection for libusb-1.0 builds was overly zealous and wrongly considered zero values as an error. [#2198]
  - `upsmon` recognition of `CAL` state could linger after the calibration activity was completed by the hardware, which led to mis-processing of shutdown triggers. Also, notification was added to report "finished calibration". [issue #2168, PR #2169]
  - `upsmon` recognition of `OFF` state as a trigger for FSD (forced shut down) criticality considered also the input line state, which may be an independently evolving circumstance. [issue #2278, PR #2279]
  - `upsmon` support for `POLLFAIL_LOG_THROTTLE_MAX` did not neuter the applied setting when live-reloading configuration, so commenting it away in `upsmon.conf` did not have the effect of resetting the logging frequency to default. It also did not reset the counters to certainly follow the new configuration for existing faults. [issue #2207, PR #2209]
  - `upsmon` support for `POLLFAIL_LOG_THROTTLE_MAX` had an off-by-one error (e.g. reporting "Data stale" or "Driver not connected" every 30 sec with `POLLFAIL_LOG_THROTTLE_MAX 5` and `POLLFREQ 5` settings). [#2207]
  - Drivers running with non-default user account (e.g. with `user=root` in their configuration) failed to apply group ownership and permissions to their Unix socket file for interaction with the local data server. [#2185, #2096]
  - Dispatcher script `scripts/python/app/NUT-Monitor` referenced `py3qt3` instead of the correct `py3qt5`. It also tries to check both `py2gtk2` and `py3qt5` implementations verbosely, even if one is not installed. [#2199, #2201]
  - Set the `DesktopFileName` in `scripts/python/app/NUT-Monitor-py3qt5`, this binds the application with the desktop file and allow the Open Desktop compatible implementation to display the proper icon and application name. [#2205]
  - Original recipe for `apc_modbus` strictly required USB support even if building NUT without it. [#2262]
  - Builds requested with a specific C/C++ language standard revision via `CFLAGS` and `CXXFLAGS` should again be honoured. [PR #2306]
  - Allow requesting detailed debug builds (with disabled optimizations for binaries to best match the source code) for supported compilers using `configure` script option `--with-debuginfo`. Note that default autoconf behavior usually embeds moderate optimizations and debug information on its own. [PR #2310]
  - A fix applied among clean-ups between NUT v2.7.4 and v2.8.0 releases backfired for `usbhid-ups` subdriver `belkin-hid` which in practice relied on the broken older behavior; more details in its entry below. [PR #2371]
- nut-usbinfo.pl, nut-scanner and libnutscan:
  - Library API version for `libnutscan` was bumped from 2.2.0 to 2.5.0 during evolution of this NUT release.
  - USB VendorID:ProductID support list files generated by the script for different OS frameworks now include a comment with other possibly compatible driver names, where the respective file format allows for comments.

  - Added the concept of `alt_driver_names` in `nutscan_device_t` structure for ability to suggest a comment with other possibly compatible driver names in configuration snippets generated by `nut-scanner`; practical support implemented for USB connected drivers.
  - Added the concept of commented-away suggested option values `comment_tag` and a method to `nutscan_add_commented_c` instead of hacks in prepared config data which broke some use-cases. [#2221]
  - Command-line option `-U` for USB scan can now be specified several times to increase the detail level about hardware link to the device (this was previously always suggested, but may be not reliable if USB enumeration gets changed over time). [#2221]
  - Added generation of FreeBSD/pfSense quirks for USB devices supported by NUT (may get installed to `$datadir` e.g. `/usr/local/share/nut` and need to be pasted into your `/boot/loader.conf.local`). [#2159]
  - nut-scanner now avoids creating ambiguous `nutdevN` device section names when called separately to scan different media buses (one at a time). Now the "bus" name would be embedded (e.g. non-colliding `nutdev-usb1` and `nutdev-snmp1`). [#2247]
  - nut-scanner can now discover NUT simulated devices (`.dev` and `.seq` files) located in your sysconfig directory, and prepare configuration sections with the simulation driver (currently `dummy-ups`). [#2246]
  - nut-scanner now reports `dummy-ups` as driver when scanning NUT "bus" with Old or Avahi method. [#2236, #2245]

- upsd: Fixed conditions for "no listening interface available" diagnosis to check how many listeners we succeeded with, not whether the first one succeeded or not. If not all requested (non-localhost) listeners were available, default to fail the daemon start-up attempt; support for an `ALLOW_NOT_ALL_LISTENERS` setting was added to control this behavior. [#723]

- NUT CI improvements:

  - Added publishing recipes for PyNUT client bindings for NUT, so it ends up in the PyPI repository. [#2158]
  - Added support for new `ccache` namespace concept, where possible. [#2256]
  - Fixed an issue for builds configured `--without-usb`. [#2263]
  - Added a fallback for `libgd` discovery (for CGI etc. builds). [#2287]
  - Made `aspell` TeX module detection more reliable. [#2206]
  - Fixed recipes for completely out-of-tree builds to pass with documentation generation and checking on all tested "make" implementations. [#2318]
  - Various other recipe and documentation clean-up efforts. [#2284, #2269, #2261]

- main driver core codebase:

  - Help users of drivers that can be built to support optionally USB and other media (like `nutdrv_qx` built for serial-only support), and built in fact without USB support but used for USB devices, with some more information to make troubleshooting easier. [issue #2259, PR #2260]
  - Driver programs with debug tracing support via `-D` CLI option and/or the `NUT_DEBUG_LEVEL` environment variable now check those earlier in their life-time, so that initialization routine can be debugged. [#2259]
  - Multiple USB-capable drivers got options to customize `usb_config_index` `usb_hid_rep_index`, `usb_hid_desc_inde` `usb_hid_ep_in` and `usb_hid_ep_out` hardware connection settings via `ups.conf` options. This is treated as experimental, not all code paths may be actually using such values from `struct usb_communication_subdriver_t` rather than hard-coded defaults. Discovery of correct values is up to the user at the moment (using `lsusb`, internet search, luck...) [#2149]

- nut-driver-enumerator (NDE) service/script:

  - The optional daemon mode (primarily useful for systems which monitor a large and dynamic population of power devices) was enhanced with a `--daemon-after` variant which parses the configuration once before daemonization and this has a chance to fail while not forked off, as well as to allow only completing the service unit initialization when everything is actually ready to work (so further dependencies can start at the proper time). [#682]
  - Also applied other optimizations to the script implementation. [#682]

- powerpanel text driver now handles status responses in any format and should support most devices. [#2156]

- tripplite_usb driver now allows any device to match if a particular Unit ID was not specified in `ups.conf`. [PR #2297, issues #2282 and #2258]
- snmp-ups driver:
  - added support for Eaton EMP002 sensor for ATS16 NM2 sub-driver. [#2286]
  - mapping table updates for apc-mib sub-driver. [#2264]
- usbhid-ups driver:
  - `arduino-hid` subdriver was enhanced from "initial bare bones" experimental set of mapped data points to support some 20 more mappings to make it more useful as an UPS driver, not just a controller developer sandbox. [#2188]
  - `cps-hid` subdriver now supports devices branded as Cyber Energy and built by cooperation with Cyber Power Systems. [#2312]
  - `belkin-hid` subdriver now supports Liebert PSI5 devices which have a different numeric reading scale than earlier handled models. [issue #2271, PR #2272, PR #2369] Generally the wrong-scale processing was addressed, including a regression in NUT v2.8.0 which led to zero values in voltage data points which NUT v2.7.4 reported well [#2371]
  - The `onlinedischarge` configuration flag name was too ambiguous and got deprecated (will be supported but no longer promoted by documentation), introducing `onlinedischarge_onbattery` as the meaningful alias. [#2213]
  - Logged notifications about `OL+DISCHRG` state should now be throttled (see the driver manual page for more details) [#2214, #2215]:
- If `battery.charge` is available, make the message when entering the state and then only if the charge differs from that when we posted the earlier message (e.g. really discharging) and is under `onlinedischarge_log_throttle_hovercharge` value (defaults to 100%);
- Also can throttle to a time frequency configurable by a new option `onlinedischarge_log_throttle_sec`, by default 30 sec if `battery.charge` is not reported by the device (should be frequent by default, in case the UPS-reported state combination does reflect a bad power condition).
- nutdrv_qx driver:
  - Fixed handling of `battery_voltage_reports_one_pack` configuration flag introduced in NUT v2.8.1. [originally by PR #1279; fixed by PR #2324, issue #2325]
- Various code and documentation fixes for NSS crypto support. [#2274, #2268]
- Laid foundations for the SmartNUT effort (aiming to integrate drivers with some other backends than the networked NUT data server process).
- Eaton contributed recipes and scripts used to create the IPP for Unix bundle (aka Eaton IPSS Unix or UPP), a freely available value-added packaging of NUT distributed as the UPS software companion for OSes where their more complex UPS monitoring/management tools had not been ported. This allows for delivery of NUT packages with an interactive installer and some system integration scripts (events, notifications, status, shutdown daemon…), and was contributed to the NUT upstream project by Eaton — provided "as is" at the moment, and may later serve as foundation or inspiration for new NUT features. [#2288]
- nutconf (C++ library and tool to read and manage NUT configuration files) was started in the open by Eaton employees and used in the IPP installer, but the code lingered in a side branch. It was now brushed up to our common best practices and added to the main codebase. As of this import, there are known deficiencies in Windows platform support, as well as some un-awareness about configuration key words which appeared in NUT since 2013. [#2290]
- The `tools/gitlog2changelog.py.in` script was revised, in particular to convert section titles (with contributor names coming from Git metadata) into plain ASCII character set, for `dblatex` versions which do not allow diacritics and other kinds of non-trivial characters in sections. This can cause successful builds of `ChangeLog.pdf` file on more platforms, but at expense of a semi-cosmetic difference in those names. [PR #2360, PR #2366]

2.6 Release notes for NUT 2.8.1 - what's new since 2.8.0

https://github.com/networkupstools/nut/milestone/8

- "UPS management protocol", Informational RFC 9271 published by IETF at https://www.rfc-editor.org/info/rfc9271 and the IANA port number registry was updated accordingly at https://www.iana.org/assignments/service-names-port-numbers/-service-names-port-numbers.xhtml?search=3493 (even though this RFC is not formally an Internet Standard)
- NUT documentation files were rearranged, renaming some to `*.adoc` pattern to facilitate automatic rendering in GitHub and IDE GUIs, and adding recipe support for GitHub issue/PR links. This `NEWS` file is now proper asciidoc rendered into `release-notes.pdf` (and HTML versions). [issue #1953, PR #2048] Internally, the documents would use a new way to define cross-linking to other pages and their chapters, to facilitate different renderers (including GitHub UI), and file names created for "chunked HTML" documentation format will no longer have the "chapter number, section number" format which is not easy to maintain over time with independent builds of documentation in NUT and the actual and historic snapshots for nut-website for example. Chapter/Section names will be adapted to produce "chunked HTML" file names instead. Documentation links rendered in GitHub UI should point to the HTML pages served by a current iteration of the NUT website. [PR #226, PR #669]
- A new `configure --enable-spellcheck` toggle should add spelling checks to `make check` (by default, if tools are available) to facilitate quicker acceptance of contributions. [#2067]
- Published a new maintainer GPG key to sign tags and release artifacts, and possibly git commits as well, as part of solution for issue #1410. You can pull it from common OpenPGP servers with the following command:

```
:; gpg --recv-key DE0184DA7043DCF7
gpg: key DE0184DA7043DCF7: public key "Jim Klimov (Doing FOSS
         since last millennium) <jimklimov@gmail.com>" imported
gpg: Total number processed: 1
gpg:               imported: 1
```

  as part of https://github.com/networkupstools/nut/issues/1410 solution.

- Bug fixes for fallout possible due to "fightwarn" effort and other evolution in NUT v2.8.0 release:
  - The `upsdebugx()` and similar methods were converted to macros in #685 to avoid useless data manipulations and requests for logged information, whose results would be ignored instantly because the debug level is too low. As issue #1455 and PR #1495 found, in two cases the called commands did "meaningfully" modify data — so without debug logs the program misbehaved. A known regression for `upscode2` driver; might be or not be a problem with `upsd` server in NUT v2.8.0 release, fixed for NUT v2.8.1.
  - A table in `cyberpower-mib` (for `snmp-ups` driver) sources was arranged in NUT v2.8.0 release in a way that precluded the driver logic from looking at all of its entries. Also a fix for instant command definitions had in fact broken them due to other development. Regressions fixed for NUT v2.8.1 [#1432, #2029]
  - A change for file-change detection in `dummy-ups` driver for NUT v2.8.0 release misfired on some platforms. Regression fixed for NUT v2.8.1 [#1420]
  - Fixed building of NUT man pages when just a few drivers are selected by `configure` script for custom builds [#1467]
  - Now that `upsdrvctl` can pass debugging level through to the launched driver(s), they would by default stay in the foreground. This can complicate (or simplify, when intentional) the management of service instances. Now there are explicit `upsdrvctl` options for choosing this (`-F`/`-B`), although default behavior is retained. Note that explicit foregrounding mode also keeps `upsdrvctl` tool from exiting and would not wait for one driver to complete initialization before starting another in case of mass-management loop to start all drivers (without specifying the single device) [#1759, #1806, #1875]
  - The `apcsmart` and `apcsmart-old` handled invalid data too zealously and aborted instead of skipping over it, like they did before [#2015]
  - A bit maths optimization in `riello_ser` and `riello_usb` misfired [#2137]
  - Something about compile-time macros or other warnings-related refactoring seems to have confused the MGE SHUT (Serial HID UPS Transfer) driver support [#2022]
  - Some warnings were not detected by the tools or build scenarios used earlier, and only got addressed now

- An issue was identified which could cause `libupsclient` parser of device and host names to crash upon bad inputs (e.g. poorly resolved environment variables in scripts). Now it should fail more gracefully [#2052]
- New `configure --enable-inplace-runtime` option should set default values for `--sysconfdir`, `--with-user` and `--with-group` options to match an existing NUT deployment — for users who are trying if a custom build of recent codebase solves their practical issues. For "quick tests", a shortcut operation `./ci_build.sh inplace` was added [#1714]
- State tree structure and methods (including "dstate" wrapper for common driver internals) was enhanced with time-stamping of last modification (setting, changing, deleting the value or some fields in an entry): this allows to detect stale information in a centralized fashion [#2010]
- We lacked log information about changes of chroot jail (uncommon) and of UID/GID (everywhere), which makes troubleshooting harder (e.g. lack of access to config files or USB device nodes). Now we have it [#1694]
- A `NUT_DEBUG_PID` envvar (presence) support was added to add current process ID to tags with debug-level identifiers. This may be useful when many NUT daemons write to the same console or log file. [#2118]
- huawei-ups2000 is now known to support more devices, noted in docs and for auto-detection [#1448, #1684]
- nutdrv_qx updates:
  - a `battery_voltage_reports_one_pack` driver option was added for devices which "natively" report a `battery.voltag` for a single battery pack or cell, not for the whole assembly [#1279]
  - the `voltronic_qs_protocol` should now accept both "V" (as before) and newly "H" dialects, which otherwise seem interchangeable [#1623]
  - the `armac` subdriver was enhanced to support devices with a different response pattern than previously expected per initial contribution. It was tested to work with Vultech V2000 and Armac PF1 series. [#1978]
- nutdrv_qx and blazer updates:
  - extended default ranges for max battery voltage when guessing [#1279]
- sms_ser, a driver for SMS Brazil UPS Protocol 1Phase, was introduced. NOTE: It may later become a subdriver under nutdrv_qx. [#2090]
- usbhid-ups updates:
  - added support for `subdriver` configuration option, to select the USB HID subdriver for the device manually where automatic match does not suffice (e.g. new devices for which no `vendorid`/`productid` pair was built into any driver, or for different-capability devices with same interface chips, notably "phoenixtec/liebert" and "mge") [#1369]
  - cps-hid subdriver now applies same report descriptor fixing logic to devices with ProductID 0x0601 as done earlier for 0x0501, to get the correct output voltage data [#1497]
  - apc-hid subdriver now also supports ProductID 0x0004 [#1429]
  - ever-hid subdriver reported a `powerfactor` without a namespace (bug in 2.8.0 release), fixed to `outlet.powerfactor`
  - the `usbhid-ups` driver should now reconnect if `libusb` returned a memory allocation error [#1422] (seen as "Can't retrieve Report 0a: Resource temporarily unavailable"), which can cause practical problems in the field — the driver otherwise interpreted the situation as `ups.status` being `OL OFF` and cut the power supply.
  - powercom-hid subdriver: fixed `UPS.Battery.ManufacturerDate` to map to `battery.mfr.date` (not `battery.date` which is the maintenance replacement date) [#1644]
  - added `onlinedischarge_calibration` option for UPSes that report `OL+DISCHRG` when they are in calibration mode [#2104]
- riello_usb updates:
  - added `localcalculation` option to compute `battery.runtime` and `battery.charge` if the device provides bogus values [#1692, #1685] (similar to `runtimecal` in some other drivers, may be refactored to that configuration and logic model in later NUT releases)

- powercom driver should now try harder to refresh data from device [#356]
- tripplite_usb driver now supports configuration of `upsid` to match the specific device (not all firmware/hardware models support this) [#2075]
- apcupsd-ups:
  - improvement for `POLL_INTERVAL_MIN` from PR #797 was buggy [#2007]
  - fix to clean obsoleted readings (if any) AFTER getting new info from an `apcupsd` daemon, to avoid the gap when NUT driver knows nothing [#2007]
- apc_modbus driver was introduced, to cover the feature gap between existing NUT drivers for APC hardware and the actual USB-connected devices (or their firmwares) released since roughly 2010, which deprecated standard USB HID support in favor of Modbus-based protocol which is used across the board (also with their network management cards). The new driver can monitor APC UPS devices over TCP and Serial connections, as well as USB with a patched libmodbus (check https://github.com/EchterAgo/libmodbus/commits/rtu_usb for now, PR pending). [#139, #2063]
  - For a decade until this driver got introduced, people were advised to use apcupsd project as the actual program which talks to a device, and NUT apcupsd-ups driver to relay information back and forth. This was a limited solution due to lack of command and variable setting support, as well as relaying of just some readings (just whatever apcupsd exposes, further constrained by what our driver knows to re-translate), with little leverage for NUT to tap into everything the device has to offer. There were also issues on some systems due to packaging (e.g. marking NUT and apcupsd as competing implementations of the same features) which required clumsy workarounds to get both installed and running. Finally, there is a small matter of long-term viability of that approach: last commits to apcupsd sources were in 2017 (with last release 3.14.14 in May 2016): https://sourceforge.net/p/apcupsd/svn/HEAD/tree/
- dummy-ups:
  - Added an `repeater_disable_strict_start` option to disable the driver exiting upon encountering any kind of error at startup (as repeater). This option should allow for collective `upsdrvctl` startup despite individual target UPS to be repeated or `upsd` not having come up yet. [#2132]
  - Revised detection of file path (for "dummy" mode) which misfired under some conditions, and unified several implementations. [#2118]
- NUT for Windows:
  - Ability to build NUT for Windows, last tackled with a branch based on NUT v2.6.5 a decade ago, has been revived with the 2.8.x era codebase [#5]. It is known that at this time some features are not complete, for more details see https://github.com/orgs/networkupstools/projects/2/views/1
  - Cross-builds of NUT for Windows using Linux and MinGW (and many custom built dependency packages, as documented in the scripts/Windows/README.adoc file) are now regularly tested on NUT CI farm with moderate integration via custom build script `scripts/Windows/build-mingw-nut.sh` [#1489]
  - Semi-native NUT for Windows builds with MSYS2/MinGW x64 environment are now regularly tested on AppVeyor, with the same `ci_build.sh` script and `Makefile` checks as used across the board for local developer builds, Linux/illumos/FreeBSD/OpenBSD/... on dedicated NUT CI farm on Fosshost, and MacOS on CircleCI [#1552]
- snmp-ups updates:
  - Fixed detection for device agents which wrongly return the sysOID value as a string instead of an OID [#1710]
  - Clearer messages about skipping MIBs during driver initialization [#2037]
  - IETF MIB mapping updated for data points where negative readings are invalid [#1558]
  - Added SNMP subdriver "apc-epdu-mib" for APC easy PDU support [#1674]
  - Added SNMP subdriver "eaton-pdu-nlogic-mib" for nLogic (rebranded Eaton) support [#1698]
  - Added SNMP subdriver "hpe-pdu3-cis-mib" for HPE G2 Metered & Switched PDU initial "unitary" support (no daisychain support yet); also note that due to SNMP v1 implementation limitations on this device, you should prefer SNMP v3 to get both read and write rights [#1713]

  - Fixed processing loop for large SNMPv2/SNMPv3 responses where one item in the middle has a type error [#1682]
  - Better manage the slight nuances (especially in `ups.status`) between Eaton UPSs, and rename mibs from `pw` to `eaton_pw_nm2` and from `pxgx_ups` to `eaton_pxg_ups` [#1715]
  - Fixed the long standing "Warning: excessive poll failures" issue, that was tied to non-existent OIDs, not well handled in some parts of the driver [#1716]
  - `baytech-mib.c` subdriver: fixed `baytech_outlet_status_info[]` set of valid outlet status values [#1871]
  - `cyberpower-mib.c` subdriver: support devices which report the shorter Vendor OID as their sysOID, e.g. "CyberPower PowerPanel Personal" [#1997] and support more data points including hardware status alarms [#1982]

- The `bestfortress` driver shutdown handling was fixed to use a non-trivial default timeout [#1820]
- The `optiups` driver only gave accurate voltage information with 120VAC models and assumed a 12V battery when calculating capacity. There is a protocol command that gives a (fixed) voltage which correlates with the voltage selection DIP switches on the back of the UPS, taking into account whether it is a 120 or 240VAC model. Likewise, now the battery capacity fix is applied globally, based on whether or not the battery voltage is greater than 20V. [#2089]
- GPIO drivers [#1855]:
  - Added a new category of drivers, using GPIO interface to locally connected devices (currently limited to 2018+ Linux libgpiod, but its architecture was designed to support more OSes with their equivalents - PRs welcome)
  - `generic_gpio_libgpiod` driver using `libgpiod` backend was added (defaults to be required on Linux, optional on other platforms)
- Added support for `make install` of PyNUT module and NUT-Monitor desktop application [#1462, #1504]
- Regular CI coverage for NUT codebase enhanced with CircleCI running some scenarios on MacOS, might add Windows in the future. Fixed some build issues for MacOS that had crept into NUT v2.8.0 release [#1415, #1421]
- NUT software-only drivers (dummy-ups, clone, clone-outlet) separated from serial drivers in respective Makefile and configure script options [#1446]
- Fixed support for common USB matching options ("vendor", "device", "bus", etc.) for `riello_usb` and `richcomm_usb` [#1763] and updated man pages of all USB drivers using these options to include the same description [#1766]
- Added a "busport" USB matching option (if supported by the hardware, OS and libusb on the particular deployment, it should allow to specify physical port numbers on an USB hub, rather than logical "device" enumeration values, and in turn — this should be less volatile across reboots etc.) [#2043]
- Added an `allow_duplicates` flag for common USB matching options which may help monitor several related no-name devices (although without knowing reliably which one is which... better than nothing) [#1756]
- The `nut-scanner` program should now suggest same configuration fields as those used by common USB matching options in (most of the) drivers, e.g. adding "device" to the generated configuration section [#1790]
- Stuck drivers that do not react to `SIGTERM` quickly are now retried with `SIGKILL` [#1424]
- Each driver should now report its `driver.state` to help readers determine whether it is initializing, reconnecting, or running regular loops [#1767]
- Code which resolves full paths to libraries should now consider the common environment variable `LD_LIBRARY_PATH` as a preferred possible override to built-in paths (note that most operating systems advise against setting this variable unless troubleshooting, although other systems rely on it) [#805]
- Debug information tracing methods like `upsdebugx()` should now be less limited in the sizes of messages that they can print, such as path names that may be quite long. Note that the OS methods manipulating the strings, and receivers such as logging systems, may still impose limits of their own.
- The `nut-scanner` usage and debug printouts now include the loadable library search paths, to help troubleshooting especially in multi-platform builds; pre-filtering of the built-in paths was introduced (to walk only existing and unique directory names) [#317]

- The nut-scanner program was updated to fall back to loading unresolved library filenames, hoping that `lt_dlopen()` implementation on the current platform would find library files better [#805]
- Detection of `libltdl` in `configure` script updated with fallback code to find it on systems that deliver the library to `/usr/local/lib` (e.g. on FreeBSD) [#1577]
- An explicit `configure --with-nut-scanner` toggle was added, specifically so that build environments requesting `--with-all` but lack `libltdl` would abort and require either to install the dependency or explicitly forfeit the tool (some distro packages missed it quietly in the past) [#1560]
- The `nut-scanner` program should now by default warn about serial numbers which do not make much sense (are duplicate, empty, all same character, etc) [#1810]
- Existing openssl-1.1.0 support added for NUT v2.8.0 release was tested to be sufficient without deprecation warnings for builds against openssl-3.0.x (but no real-time testing was done yet) [#1547]
- upslog: Added support for logging multiple devices with one call to the program [#1604]
- Documentation to integrate NUT USB driver startup with `usb_resetter` script has been contributed to `scripts/usb_resette` (the script itself is tracked externally on GitHub), along with a configuration example for Linux+systemd [#1887]
- Some fixes applied to Solaris/illumos packaging and SMF service support [#1554, #1564]
- Some fixes for builds on older OSes with less functional default system shell interpreters - now `autogen.sh` supports a `CONFIG_SHELL` envvar to inject its value into generated `configure` script [#1736]
  - Note that you may have to install additional tools (possibly from third-party FOSS packaging efforts) to prepare and build the NUT codebase, and/or prefer non-default system provided implementations (e.g. to use the XPG4 `grep` with `-E` support on Solaris as detailed in https://github.com/networkupstools/nut/issues/1736 comments)
  - Build environment configuration notes in docs/config-prereqs.txt file refreshed to cover building of current NUT codebase in CentOS 6 [#1804] and Solaris 8 [#1736, #1738]
- `configure` script, reference init-script and packaging templates updated to eradicate `@PIDPATH@/nut` ambiguity in favor of `@ALTPIDPATH@` for the unprivileged processes vs. `@PIDPATH@` for those running as root [#1719]
- `configure` script enhanced: `--with-unmapped-data-points` option allows to build SNMP and USB-HID subdrivers with entries discovered by the scripts which generated them from data walks, but developers did not rename yet to NUT mappings conforming to docs/nut-names.txt standards [#1699]
- PyNUT.py version bumped to 1.5.0 with some improvements:
  - `ListClients()` method fixed (was broken in many ways), and is now CI-tested [#549]
  - `DeviceLogin()` method added (mostly as aid to CI-test `ListClients()` in a practically relevant manner, so far)
- nutclient C++ library:
  - added `listDeviceClients()` and `deviceGetClients(dev)` to `Client` classes, and `Device::getClients()` to match PyNUT capabilities [#549]
  - published artifacts may include a `libnutclientstub` which is an implementation of a NUT TCP client in C++ with in-memory data store.
- upsclient C library:
  - added support for `NUT_QUIET_INIT_SSL` environment variable to hide the infamous "Init SSL without certificate database" warning [#1662]
- The `upsd.conf` listing of `LISTEN` addresses was previously inverted (the last listed address was applied first), which was counter-intuitive and fixed for this release [#2012]
- The `upsd` configured to listen on IPv6 addresses should handle only IPv6 (and not IPv4-mappings) to avoid surprises and insecurity; it will now warn if a host name resolves to several addresses (and will only listen on the first hit, as before in such cases) [#2012]

- A definitive behavior for `LISTEN *` directives became specified, to try handling both IPv4 and IPv6 "any" address (subject to `upsd` CLI options to only choose one, and to OS abilities). When both address families are enabled, the `upsd` data server will first try to open an IPv6 socket asking for disabled IPv4-mapped IPv6 address support (if the OS honors that), and then an IPv4 socket (which may fail if the IPv6 socket already covers it anyway); in other words, you can end up with one or two separate listening sockets. [#2012]
- sstate (server state, e.g. upsd) should now "PING" drivers also if they last reported themselves as "stale" (and might later crash) so their connections would be terminated if really no longer active [#1626]
- Clarified documentation in codebase according to end-user feedback [#1721, #1750 and others over time]
- upsmon client changes include:
  - Several fixes for `upsmon` behavior [#1761, #1680...], including new ability to configure default POWERDOWNFLAG location — packagers are encouraged to pick optimal location for their distributions (which remains mounted at least read-only late in shutdown) and a new optional POLLFAIL_LOG_THROTTLE_MAX setting [#529, #506]
  - Also `upsmon` should now recognize `OFF` and `BYPASS` flags in `ups.status` and report that these states begin or end. The `OFF` state usually means that an administrative action happened to power off the load, but the UPS device is still alive and communicating (USB, SNMP, etc.); corresponding amount of power sources we `MONITOR` are considered not being "fed" for the power value calculation purposes. The `BYPASS` state is now treated similarly to `ONBATT`: currently this UPS "feeds" its load, but if later communications fail, it is considered dead. This may have unintended consequences for devices (or NUT drivers) that do not report these modes correctly (e.g. an APC calibration routine seems to start with a few seconds of "OFF" state), so the reported status is only considered as a loss of feed if it persists for more than `OFFDURATION` seconds. [#2044, #2104]
  - Introduced `SHUTDOWNEXIT no` configuration toggle for systems which require a long time to stop their workload such as virtual machines. Since the disconnection of a "secondary" client is treated by the "primary" system as permission to proceed with its own shutdown and power-off for the UPS, the original (now merely default) behavior to call `SHUTDOWNCMD` and immediately exit could be counter-productive. An optional delay can also be introduced. [#2133]
  - Note there were other changes detailed below which impacted several NUT programs, including `upsmon`.
- Extended Linux systemd support with optional notifications about daemon state (READY, RELOADING, STOPPING) and watchdog keep-alive messages [#1590]
  - Normally **inability** to send such notifications (e.g. lack of systemd or similar framework on the particular platform) would be reported once per daemon uptime on its console log, to help troubleshooting situations where such lack of notifications can cause automated service restarts. These messages can be hidden by setting `NUT_QUIET_INIT_UPSNOTIFY=true` environment variable in init-scripts on platforms where such frameworks are not expected. [#2136]
- Extended Linux systemd units with aliases named after the daemons: `nut-server.service` as `upsd.service`, and `nut-monitor.service` as `upsmon.service` (so simple `systemctl reload upsd` can work) [#1777]
- Extended driver-server socket protocol with `BROADCAST (num)` keyword, and a `NOBROADCAST` as a shortcut for `BROADCAST 0`. This allows clients to toggle whether they want to receive `send_to_all()` updates from a driver, or only answers to requests they send [#1914]
- Added support for `make sockdebug` for easier developer access to the tool; also if `configure --with-dev` is in effect, it would now be installed to the configured `libexec` location. A man page was also added. [#1936]
- Numerous daemons (`upsd`, `upsmon`, drivers, `upsdrvctl`, `upssched`) which accepted `-D` option for debug setting previously, now can also honour a `NUT_DEBUG_LEVEL=NUM` environment variable if no `-D` arguments were provided. Unlike those arguments, the environment variable does not enforce that daemons run in foreground mode by default [#1915]
  - Note that unlike some other NUT daemons, `upssched` with enabled debug does not stop reporting on `stderr`! [#1965]
- A bug in `upssched` was discovered and fixed, where it ran a tight loop stressing the CPU; it was presumably introduced between NUT v2.7.4 and v2.8.0 releases [#1964, #1965]
- Implemented generic support for INSTCMD and SETVAR use-cases shared by all drivers, and in particular to see and change active debug verbosity using the driver-server and server-client protocol (at higher priority than CLI or config file choices) per [#1285], e.g.

```
# Set verbosity level 6:
:; upsrw -s driver.debug=6 UPS

# Set verbosity level 0 to disable the noise (even if debug_min is set):
:; upsrw -s driver.debug=0 UPS1@localhost

# Un-set the protocol override, honour CLI or config-file settings again:
:; upsrw -s driver.debug=-1 remoteUPS@1.2.3.4
```

+ and a `driver.killpower` instant command (for safety, must be unlocked by `driver.flag.allow_killpower` protocol setting or `allow_killpower` configuration flag), which is now the first choice for `driver -k` operations [#1917, #1923]

- Implemented basic support for `ups.conf` reloading in NUT drivers, with a `driver.reload-with-error` instant command (more commands and signal handling may be available depending on platform), with a goal of changing inconsequential settings like `debug_min` for a running driver. This can also benefit the drivers on systems managed by real-time `nut-driver-enumerator` and for simpler changes the drivers get only reloaded and not redefined and restarted. Reload signals should also be reasonably supported with `upsdrvctl` tool. Relevant CLI options for `-c CMD` handing were added to drivers and `upsdrvctl`, although their availability may vary between operating systems [#1903, #1914, #1924]
- Drivers should now accept `SIGURG` (or `SIGWINCH` on systems that lack the former) on POSIX platforms to dump their current state information and move on. Such report goes to `stdout` of the driver process (may be disconnected when background mode is used) — this can help with troubleshooting [#1907]
- Recipes and `main.c` code were enhanced to produce a `libdummy_mockdrv.la` helper library during build (not intended to be installed nor distributed), in order to facilitate creation of test programs which behave like a driver [#1855]
- Further revision of public headers delivered by NUT was done, particularly to address lack of common data types (`size_t`, `ssize_t`, `uint16_t`, `time_t` etc.) in third-party client code that earlier sufficed to only include NUT headers. Sort of regression by NUT 2.8.0 (note those consumers still have to re-declare some numeric variable types used) [#1638, #1615]
- The `COPYING` file was updated with licenses and attribution for certain source code files and blocks coming from the Internet originally [#1758]
- The `tools/gitlog2changelog.py.in` script was revised, in particular to generate the `ChangeLog` file more consistently with different versions of Python interpreter, and without breaking the long file paths in the resulting mark-up text [#1945, #1955]
- The "NUT client for VMware ESXi" project (by René Garcia) got its build recipes published on GitHub at https://github.com/-rgc2000/NutClient-ESXi [#1961]

2.7 Release notes for NUT 2.8.0 - what's new since 2.7.4

Note
Earlier discussions (mailing list threads, GitHub issues, etc.) could refer to this change set (too long in the making) as NUT 2.7.5.

- New (optional) keywords for configuration files were added, so existing NUT 2.7.x builds would not accept them if some deployments switch versions back and forth — due to this, semantically the version was bumped to NUT 2.8.x.
- Add support for openssl-1.1.0 (Arjen de Korte)
- libusb-1.0 API support in addition to libusb-0.1 API [#300]
- Add support for `DISABLE_WEAK_SSL=true` in upsd.conf to disable older/weaker SSL/TLS protocols and ciphers: when NUT is built against relatively recent versions of OpenSSL or NSS it will be restricted to TLSv1.2 or better. For least-surprise, currently defaults to `false` and complains in log [PR #1043]

- Add support for `ALLOW_NO_DEVICE=true` (as an upsd.conf flag or environment variable passed from caller of the program), to allow starting the data server initially without any device configurations and reloading it later to apply config changes on the fly [PR #766]
- Add support for `debug_min=NUM` setting (ups.conf, upsd.conf, upsmon.conf) to specify the minimum debug verbosity for daemons. This allows "in-vivo" troubleshooting of service daemons without editing init scripts or service unit definitions.
- Improve support for upsdrvctl for managing of numerous device configs, including default "maxretry=3" and a "nowait" option to complete the "start of everything" mode after triggering the drivers and not waiting for them to complete initializing. This matters on systems that monitor from dozens to hundreds of devices.
- Drivers support a new value for `synchronous` setting, which is the new default now: `auto`. Initially after driver start-up this mode acts as the older default `off`, but would fall back to `on` in case the driver fails to send reports to `upsd` by overflowing the socket buffer in async mode — so the next connections of this driver uptime would be synchronized (potentially slower, but safer — blocking on writes to the data server). This adaptation would primarily impact and benefit devices with many (hundreds of) data points, such as ePDUs and daisy chains. [issue #1309, PR #1315]
- Daemons such as upsd, upsmon, upslog, and device drivers previously implied that enabled debugging (or upslog to stdout) means foreground running, otherwise the daemon was always sent to the background. Now there are explicit options for this (`-F`/`-B`), although default behavior is retained. This change is used for simplified service unit definitions.
- Improvements for device discovery or driver "lock-picking", including general support for:
  - "Standalone" mode (`-s` option), to monitor a device which is not detailed or mentioned in ups.conf
  - `NUT_ALTPIDPATH` and `NUT_STATEPATH` environment variables to override the paths built into the driver binary [PR #473 and #507]
  - "Driver data dump" mode (`-d` option), to poll a device for one or few (*update_count*) loops, report discovered values (dump the data tree in upsc-like format), and exit. This complements the `nut-scanner` for finding and identifying devices.
- support for new devices:
  - IBM 6000 VA LCD 4U Rack UPS; 5396-1Kx (USB)
  - Phoenix Contact QUINT-UPS model 2320461 (Modbus)
  - Tripp-Lite SU3000LCD2UHV (USB; protocol 1330)
  - Emerson Avocent PM3000 PDU (SNMP)
  - HPE ePDU (SNMP)
- nutdrv_qx: enhanced estimation of remaining battery runtime based on speed of voltage drop, which varies as they age [PR #1027]
- nutdrv_qx: several subdrivers added or improved, including:
  - "snr" subdriver with USB connection, for SNR-UPS-LID-XXXX [PR #1008]. Note that end-users should reference explicitly the `snr` subdriver in their `ups.conf` settings because of USB chip using the same values of VendorID/ProductID as fabula_subdriver, fuji_subdriver, and krauler_subdriver.
  - "hunnox" subdriver, as a dialect of earlier "fabula" [PR #638] adds support for Hunnox HNX-850 with USB connection and reported to work for Powercool, Iron Guardian, ARES devices and possibly many others from discussions linking to the pull request which introduced the driver.
  - "phoenixtec" subdriver for Masterguard A and E series, device series A700/1000/2000/3000(-19) and E40/60/100(-19). [PR #975]
  - "ablerex" subdriver provided by the OEM vendor, note that it replaces "krauler_subdriver" as default handler for VID:PID 0xffff:0x0000 [PR #1135]
  - Legrand HID defined and handled by "krauler_subdriver" by default [PR #1075, issue #616]
  - add new "armac" subdriver, tested with Armac R/2000I/PSW, but should support other UPSes that work with "PowerManagerII" software from Richcomm Technologies from around 2004-2005 [PR #1239, issue #1238]

- microsol-apc (starting at version 0.68 as derived from solis 0.67): adding support for newer APC Back-UPS BR hardware, such as APC Back-UPS BZ1500, BZ2200BI and BZ2200I [PR #994]
- pijuice: added new i2c bus driver for PiJuice HAT, a battery UPS module for the Raspberry Pi systems [PR #730]
- huawei-ups2000: added new driver for USB (Linux 5.12+ so far) and Serial RS-232 Modbus device support of Huawei UPS2000/2000A (1kVA-3kVA) series, and possibly some related FSP UPS models. [PR #954]
- socomec_jbus: added new driver for modbus-based JBUS protocol over serial RS-232 for Socomec UPS (tested with a DIGYS 3/3 15kVA model, working on Linux x86-64 and Raspberry Pi 3 ARM). [PR #1313]
- adelsystem_cbi: added new driver for ADELSYSTEM CBI2801224A, an all-in-one 12/24Vdc DC-UPS, which supports the modbus RTU communication protocol [PR #1282]
- generic_modbus: added new driver for TCP and Serial Modbus device support. The driver has been tested against PULS UPS (model UB40.241) via MOXA ioLogikR1212 (RS485) and ioLogikE1212 (TCP/IP), and configuration allows to map custom registers and addresses to NUT events [PR #1052]
- genericups: added support for FTTx battery backup devices, and new signal type mappings for the contact closure pins interpretation (RB for replace battery, BYPASS for disconnected battery, and "none" or NULL for signals to ignore) [PR #1061]
- add devices to HCL/DDL:
  - APC Back-UPS CS (USB)
  - CPS CP1500EPFCLCD (USB)
  - CPS EC350G, EC750G (USB)
  - CPS PR2200LCDRT2U (SNMP)
  - Eaton ATS 16 and 30 (SNMP)
  - Eaton 5E2200VA (USB)
  - Eaton 9PX Split Phase 6/8/10 kVA (XML/USB/SHUT)
  - Eaton 9PX (XML/USB/SHUT)
  - Eaton Ellipse PRO 650 VA (USB)
  - Ippon Back Comfo Pro II 650/850/1050 (USB)
  - Numeric Digital 800 (USB)
  - Opti-UPS PS1500E (USB)
  - Powercool 350VA to 1600VA (USB)
- C++11 support in nutclient library and cppunit tests
- Added C++ testing mock for TcpClient class (nutclientmem/MemClientStub: data stored in local memory) [PR #1034]
- Dual Python 2 and 3 compatibility in development scripts; ability to run build activities and resulting built NUT programs on systems that do not have a binary named "python" [PR #1115 and some before it]
- Added Russian translation for NUT-Monitor GUI client [PR #806]
- Separated NUT-Monitor UI into two applications, NUT-Monitor-py2gtk2 and NUT-Monitor-py3qt5, suitable for two generations of Python ecosystem with their great differences; `NUT-Monitor` name is retained for wrapper script which calls one of these, such that the current system can execute [PRs #1310, #1354]
- Various USB driver families: expanded device-matching with "device" in addition to "bus" and generic USB fields. This is needed to support multiple attached devices that seem identical by other fields (e.g. same vendor, same model, same USB bus, and no serial number) [PR #974]
- Various USB driver families: Improved HID parsing for byte-stream to number conversions on different CPU architectures [PR #1024]
- Various USB HID driver families: added support for composite devices utilizing interface greater than 0 for the UPS interface [PR #1044]

- usbhid-ups:
  - added generic framework for fixing Report Descriptors which can be used for different manufacturers by adding code to the appropriate subdriver rather than polluting the main code with UPS specific exceptions, and applied fixes for known mistakes in (some releases of firmware for) CyberPower CPS*EPFCLCD [issue #439, PR #1245]
  - added `onlinedischarge` option for UPSes that report `OL+DISCHRG` when wall power is lost [PR #811]
  - changed detection of VendorID 0x06da handling of which is claimed by Liebert/Phoenixtec HID historically, and MGE HID (for AEG PROTECT NAS UPSes) since NUT 2.7.4, so that the higher-priority MGE subdriver would not grab each and all of the devices exposing that ID [PR #1357]
  - CPS HID: add input.frequency and output.frequency
  - OpenUPS2: only check OEM Information string once (fewer log messages)
  - Liebert GXT4 USB VID:PID [10AF:0000]
  - add battery voltage and input/output transfer voltage and frequency in Liebert/Phoenixtec HID mapping, to support PowerWalker VFI 2000 TGS better [PR #564, issue #560]
  - add a little delay between multicommands [PR #1228]
  - fix Eaton/MGE mapping for beeper handling
  - add IBM USB VID
  - add deep battery test for CyberPower OL3000RMXL2U
  - report the libusb version used
  - fixed CPU architecture dependent bitmask math issues, causing wrong numbers interpreted from wire protocol data in Big-Endian LP64 builds (SPARC64, s390x, etc.) [issue #1023, PRs #1024, #1040, #1055, #1226]
  - add Delta UPS Amplon R Series, tested on R1K and R3K model [PR #987]
  - add Delta Minuteman UPS VID/PID [PR #1230, issues #555 and #1227]
  - add AMETEK Powervar UPM [PR #733]
  - add Tripplite AVR750U (ProductID 0x3024) [PR #963]
  - add Arduino HID device support with new arduino-hid subdriver [PR #1044]
  - add new salicru-hid subdriver, tested with Salicru SPS Home 850 VA [PR #1199, issue #732]
  - add new ever-hid subdriver to support EVER UPS devices (Sinline RT Series, Sinline RT XL Series, ECO PRO AVR CDS Series) [PR #431]
  - add ability to set `battery.mfr.date` for APC HID UPS [PR #1318]
- usbhid-ups / mge-shut: compute a realpower output load approximation for Eaton UPS when the needed data is not present
- snmp-ups:
  - APC ePDU MIB support
  - add `input.phase.shift` variable
  - add configurable write-able `ondelay` (`ups.delay.start`) and `offdelay` (`ups.delay.shutdown`) as timeticks support [PR #276]
  - outlet groups
  - fix the rounding / truncation of some values
  - add outlet.N.name for Eaton ePDU
  - add input.bypass.frequency for Eaton 3ph
  - fix support for Eaton 2-phase ("split phase") UPS
  - add flag to list currently loaded MIB-to-NUT mappings
  - fix input.L2.voltage on Eaton G2/G3 PDU
  - update Eaton Aphel Revelation MIB
  - support Raritan Dominion PX2 PDU

  - support Emerson Avocent PM3000 PDU
  - improve ALARM flag handling
  - add firmware version for new HPE Network card
  - add ups.load, battery.charge, input.{voltage,frequency} and output.voltage for CyberPower, as well as shutdown and other instant commands
  - several rounds of updates for Eaton devices, including new ATS and ePDU hardware families
  - fixed bit mask values for flags to surely use different numbers behind logical items (inevitably changing some of those macro symbols) [PR #1180]

- snmp-ups and nut-scanner should now support more SNMPv3 Auth and Priv protocols, as available at NUT build time [PRs #1165, #1172]
- nut-scanner: various improvements, including:
  - detection of libraries at runtime
  - tracing information
  - limiting parallelism (thread count) [PRs #1158, #1164]
- nut-ipmipsu: improve FreeIPMI support to build cleanly against older and newer FreeIPMI versions [PR #1179]
- the powerpanel driver now also supports CyberPower OR1500LCDRTXL2U with serial cable [PR #538]
- powercom driver: implement `nobt` config parameter to skip battery check on initialization/startup [PR #1256]
- netxml-ups:
  - Report calibration status
  - Fix for erroneous battery info (MGEXML/0.30) [PR #1069]
- solis: various improvements and fixes
- liebert-esp2: Correct battery V scaling, update docs, implement split-phase unit support [PR #412]
- tripplite: the "Tripp-Lite SmartUPS driver" as tested with SMART2200NET learned to discover the firmware generation and some device features, and in particular to manage power separately on one or two outlet groups [PR #1048]
- tripplite_usb: updated to recognize the "3005" protocol [PR #584]
- libnutclient: introduce getDevicesVariableValues() to improve performances when querying many devices (up to 15 times faster)
- nut-driver-enumerator: introduced a script for Linux systemd and Solaris/illumos SMF to inspect current NUT configuration in ups.conf file and generate service management instances for each currently tracked power device. Also introduced services to monitor the NUT configuration and react to editions of this file, mostly intended for deployments that do massive monitoring of dynamically changing farms of power devices.
- Fix File descriptors leaks by upsmon and upssched (SELinux errors)
- systemd support improvements:
  - POWEROFF_WAIT
  - reload support for upsd
  - Deliver systemd-tmpfiles config to pre-create runtime locations [PR #1037 for Issue #1030]
  - Update units with SyslogIdentifier=%N for better logging [PR #1054]
- upsrw: display the variable type beside ENUM / RANGE
- Added `PROTVER` as alias to `NETVER` to report the protocol version in use. Note that NUT codebase itself does not use this value and handles commands and reported errors individually [issue #1347]

- Implement status tracking for instant commands (instcmd) and variables settings (setvar): this allows to get the actual execution status from the driver, and is available in libraries and upscmd / upsrw [PR #659]
- Add support for extra parameter for instant commands, both in library and in upscmd
- dummy-ups can now specify `mode` as a driver argument, and separates the notion of `dummy-once` (new default for `*.dev` files that do not change) vs. `dummy-loop` (legacy default for `*.seq` and others) [issue #1385]
- new protocol variables:
  - `input.phase.shift`
  - `outlet.N.name`
  - `outlet.N.type`
  - `battery.voltage.cell.max`, `battery.voltage.cell.min`
  - `battery.temperature.cell.max`, `battery.temperature.cell.min`
  - `battery.status`
  - `battery.capacity.nominal`
  - `battery.date.maintenance` (and clarified purpose of `battery.date`)
  - `battery.packs.external` (and clarified purpose of `battery.packs`)
  - `experimental.*` namespace introduced [PR #1046] to facilitate introduction of NUT drivers and their data points for which we do not yet have concepts, or which the original driver contributors did not map well per suitable NUT standards: this allows to balance having those drivers available in the project vs. least surprise for when the explicitly experimental names are changed to something stable and standardized.
  - Proposed to track Date and Time values (still as "opaque strings") preferably in representations compatible to ISO-8601/RFC-3339 [PR #1076] (standards update; changes to actual codebase to be applied in the future)
    - New routine to convert a US formatted date string "MM/DD/YYYY" to an ISO 8601 Calendar date "YYYY-MM-DD" was added to snmp-ups.c [PR #1078]
- Master/Slave terminology was deprecated in favor of Primary/Secondary modes of `upsmon` client:
  - Respective keywords in the configuration files (`upsd.users` and `upsmon.conf`) are supported as backwards-compatible settings, but the obsoleted values are no longer documented.
  - Protocol keyword support was similarly updated, with `upsmon` now first trying to elevate privileges with `PRIMARY <ups>` request, and falling back to `MASTER <ups>` just in case it talks to an older build of an `upsd` server.
  - For the principle of least surprise, NUT codebase still exposes the `net_master()` (as handler for `MASTER` net command) in header and C code for the sake of existing linked binaries, and returns the `OK MASTER-GRANTED` line to the older client that invoked it.
  - Newly introduced `net_primary()` (as handler for `PRIMARY` net command) calls the exact same application logic, but returns `OK PRIMARY-GRANTED` line to the client.
  - Python binding updated to handle both cases, as the only found in-tree protocol consumer of the full-line text.
  - For more details see issue #840 and several pull requests referenced from it, and discussions on NUT mailing lists.
- Build fixes:
  - In general, numerous fixes were applied to ensure portability and avoid warnings (fixing a number of real bugs that caused them); CI was extended to keep the codebase free of those types of warnings which we have got rid of, requiring builds to succeed cleanly in several dozen combinations of compiler versions, C standard revisions (C99 upwards, though on many OSes with GNU99+ extensions), operating systems and CPU architectures.
  - Public CI introduced to automatically test every contribution (PR) and resulting increment of main NUT codebase, including Travis CI and LGTM.com services, and a Jenkins farm on virtual hardware donated by Fosshost.org; this augments testing earlier provided for some branches by Buildbot.
  - Added cppunit testing with valgrind for the C++ client library
  - Make targets added for shell script syntax checks for helper and service scripts

  - Make targets added for spellcheck and for maintenance of the dictionary, including incremental spellcheck to only parse recently edited text files
  - The AsciiDoc detection has been reworked to allow NUT to be built from source without requiring asciidoc/a2x (using pre-built man pages from the distribution tarball, for instance)
  - Makefile contents rearranged for more resilient out-of-tree and in-tree builds beside those made from the root workspace directory
  - Makefiles are tested with GNU Make and BSD Make to ensure portable recipes
  - More use of `pkg-config` to detect dependencies at configure time, as well as fail-safe detection of presence of pkg-config (and its macros) to survive and build without it too
  - "slibtool" pedantic nuances now supported, allowing an alternative to GNU libtool
  - Build scripts updated to remove obsoleted calls to cleanly work with autoconf-2.70 releases in 2020 (also works with 2.69 which was the earlier release since 2012)
  - Dynamic library loading used in certain programs and use-cases improved, especially for 64-bit vs. 32-bit builds on multiple-bitness OSes
  - Logging routines like `upsdebugx()` were refactored as macros so there is slightly less overhead when logging is disabled [PRs #685 and #1100]
  - Numerous classes of compilation warnings eradicated, many of those being potential issues with implicit data type conversions and varied numeric type width, signedness, string buffer size, uninitialized variables or structure fields; some more in progress
  - Several logical errors found and fixed during this walk over codebase.
  - Cases where compilers were overly zealous and particular code was written the way wit was intentionally, including some comparisons that help with different-bitness builds but indeed seem superfluous in a certain single bitness, were commented and encased in pragmas to disable the warnings
  - Basic coding style (indentations, lack of trailing white space) applied per developer guide, but not automatically enforced/checked yet.

- Due to changes needed to resolve build warnings, mostly about mismatching data types for some variables, some structure definitions and API signatures of several routines had to be changed for argument types, return types, or both. Primarily this change concerns internal implementation details (may impact update of NUT forks with custom drivers using those), but a few changes also happened in header files installed for builds configured `--with-dev` and so may impact `upsclient` and `nutclient` (C++) consumers. At the very least, binaries for those consumers should be rebuilt to remain stable with NUT 2.8.0 and not mismatch int-type sizes and other arguments.
- As usual, more bugfixes, cleanup and improvements, on both source code and documentation.

2.8 Release notes for NUT 2.7.4 - what's new since 2.7.3

- New class of device supported: ATS - Automatic Transfer Switch are now supported in NUT. Eaton ATS are supported, and APC ones should be too. Users are welcomed to test and provide feedback
- NUT command and variable naming scheme:
  - Document battery.charger.status, which will in time replace the historic CHRG and DISCHRG flags published in ups.status
  - Many extensions to support outlets groups, thresholds / alarms (ambient, input, output, outlet and outlet.group)
- support for new devices:
  - AEG PROTECT B / NAS
  - APC ATS AP7724 (should be supported)
  - Asium P700
  - Eaton ATS
  - Eaton 5E 1100iUSB

  - Eaton E Series DX UPS 1-20 kVA
  - Eaton Powerware 9125-5000g
  - Electrys UPS 2500
  - Fideltronic INIGO Viper 1200
  - Legrand Keor Multiplug
  - LYONN CTB-800V
  - Micropower LCD 1000
  - NHS Laser Senoidal 5000VA
  - Sweex model P220
  - TS Shara
  - Various APCUPSD-controlled APC devices
- snmp-ups:
  - Improve automatic detection algorithm
  - Provide access to Net-SNMP timeout and retries
  - Proper handling of integer RW variables
  - Implement support for alarms, through ups.alarm and outlet.n.alarm
  - Improve log/debug output trace
  - Fix loss of precision when setting values, using upsrw
  - Support for outlets group management
  - Many improvements and simplification
  - Add support for Tripplite units using IETF mib
  - Improve communication staleness detection and recovery
  - Add devices MAC address publication
  - Register values enumerations, when available
  - Many improvements and fixes to the SNMP subdriver creation script
- Eaton:
  - 3ph SNMP: Many improvements to Powerware / XUPS MIB, for data and commands Add support for Eaton Power Xpert Gateway UPS Card Improve support for temperature and humidity, including low / high values Alarms handling
  - ePDU (G2 and G3): Improve support for ambient sensor, including thresholds and dry contacts Outlet groups handling, including data, thresholds, settings and commands Alarms handling
  - XML/PDC (netxml-ups): Fix Eaton XML published data Add some settings (R/W flags) on ambient thresholds
- bcmxcp_usb: improvements for device claiming and multi-packets responses
- dummy-ups: allow any variable to be modified
- libnutclient: Fix for reads when the socket was closed by NUT server
- macosx-ups:
  - fix for 10.10 (Yosemite), v1.1
  - gracefully handle disconnection of UPS (return "data stale")
- nutdrv_atcl_usb: point to nutdrv_qx (fuji) for 0001:0000
- nutdrv_qx:
  - Add new *sgs* USB subdriver to support TS Shara units

  - various improvements and simplification, to the code and documentation

- nut-ipmipsu: improve FreeIPMI support
- nut-scanner:
  - Don't depend on development libraries, by looking at some known paths, including the one provided through --libdir, to find the correct libraries
  - Fix a crash on a 2nd call to libnutscan with SNMP method
- powercom: fix the processing of input and output voltage for KIN units
- solis:
  - many improvements and cleanup
  - resync with end-of-packet character
  - fixes for Microsol Back-Ups BZ1200-BR
- tripplitesu: Fix initialization when tripplite firmware is buggy (for Tripplite SU1000RT2U and possibly more)
- usbhid-ups:
  - various minor improvements
  - support for Eaton UPS with dual HID report descriptor in HID Parser
  - handle missing USB strings in APC code
- SSL support through Mozilla NSS: Rework the NSS tests to ensure that NSS is actually installed and usable for enabling SSL support in NUT
- Augeas support: Augeas lens for ups.conf was updated to add various missing global directives and ups fields
- scripts/systemd/nut-server.service.in: Restore systemd relationship since it was preventing upsd from starting whenever one or more drivers, among several, was failing to start
- Fix UPower device matching for recent kernels, since hiddev* devices now have class "usbmisc", rather than "usb"
- Network protocol information: default to type NUMBER for variables that are not flagged as STRING . This point is subject to improvements or change in the next release 2.7.5. Refer to docs/net-protocol.txt for more information
- As usual, more bugfixes, cleanup and improvements, on both source code and documentation.

2.9 Release notes for NUT 2.7.3 - what's new since 2.7.2

- reverted POWERDOWNFLAG to /etc/killpower as in 2.6.5 (packagers may want to put this in another filesystem, though)
- configure/make fixes for `systemdsystemunitdir`
- apcsmart: fix command set parsing for protocol version 4 (e.g. Smart-UPS RT 10000 XL)
- upslog: SIGUSR1 forces an immediate log entry
- riello_usb/_ser: USB interface claim fix; improved error handling
- usbhid-ups: add support for OpenUPS2 (PID: D005), Liebert GXT3 (PID: 0008) APC AP9584 Serial→USB kit (PID: 0000), and some Powercom models (PID: 0001). Fixed scaling for Cyberpower 0764:0501.
- USB core: do not call usb_set_altinterface(0) by default
- nutdrv_qx:
  - added fabula, fuji USB and Voltronic-QS-HEX subdrivers

  - add bestups subdriver to supersede the old standalone bestups driver

- NUT Monitor: added FreeDesktop AppData file (including screenshots)
- renamed udev rules file to 62-nut-usbups.rules (permissions fix)
- added AIX packaging
- asem: added a driver for the UPS in ASEM PB1300 embedded PCs
- solis: updated to support APC Microsol units sold in Brazil
- tripplite_usb:
  - updated to use dv/dq charge calculation for all models (also exposes battery_min and battery max as configuration variables)
  - added binary 3005 protocol support (such as for SMART500RT1U)
- genericups: better debugging while parsing the cable description flags
- all drivers: a new *synchronous* driver flag is available for very verbose units, such as some ePDUs
- Eaton:
  - Add support for EnergySaving features for Eaton UPSs (HID USB/SHUT and XCP USB/serial)
  - Fix and complete Eaton ePDUs G2/G3 support
  - ABM (Advanced Battery Monitoring) support through battery.charger.status in HID (USB and SHUT), XCP (USB and serial) and SNMP (Powerware XUPS MIB)
- support for new devices:
  - APC Back-UPS 1200BR and Back-UPS BZ2200BI-BR (Microsol)
  - ASEM SPA PB1300 UPS
  - Belkin Regulator PRO-USB
  - Cyber Power Systems Value 1500ELCD-RU
  - EUROCASE EA200N 2000VA
  - Fideltronik LUPUS 500
  - Flight Technic & International (FTUPS) FT-1000BS and FT-1000BS(T)
  - Grafenthal PR-3000-HS
  - JAWAN JW-UPSLC02
  - Lacerda New Orion 800VA
  - Mecer ME-1000-WTU
  - NHS Sistemas de Energia Expert C Online 6000/8000/10000
  - NHS Sistemas de Energia Expert S Online 6000/8000/10000
  - Powercom BNT-xxxAP (USB product id: 0001)
  - Rucelf UPOII-3000-96-EL
  - Tripp Lite OMNIVSINT800
  - Voltronic Power Apex 1KVA and Imperial 1KVA

2.10 Release notes for NUT 2.7.2 - what's new since 2.7.1

- This release is the second interim release of the 2.7 testing series.
- libupsclient had undefined references related to functions of libcommon. This issue was reported on Debian (bug #731156) and is now fixed
- support for new devices:
  - CABAC UPS-1700DV2
  - Eaton Powerware 3105
  - Emerson Network Power Liebert PSI 1440
  - MicroDowell B.Box LP 500
  - Numeric Digital 800 plus
  - OptiUPS vs. 575C
  - Tripp Lite SU10KRT3/1X
- FreeDesktop Hardware Abstraction Layer (HAL) support was removed.
- nutdrv_atcl_usb: new driver for *ATCL FOR UPS*
- al175: re-introduced this driver (actually, it was in 2.7.1)
- upsdrvctl now provides retry options for upsdrvctl and driver(s)
- snmp-ups: add support for XPPC-MIB and Tripp Lite SU10KRT3/1X. Also fix erroneous status in HP/Compaq SNMP MIB (with the most recent HP firmware (1.76); improved various MIBs (APC, HP/Compaq, ...)
- nutdrv_qx: add new *fallback* Q1 subdriver, with minimal *Q1* support. General improvements on all subdrivers.
- mge-shut: partially revert PnP/RTS change, for initializing the communication with the UPS. Note that nut-scanner similar function was not modified however.
- FreeBSD DEVD support: generate devd.conf files for USB UPSes This adds a --with-devd-dir=PATH option to ./configure
- The NUT website was moved to a standalone website. A separate code repository and source archive are now available.
- As usual, more bugfixes, cleanup and improvements, on both source code and documentation.

2.11 Release notes for NUT 2.7.1 - what's new since 2.6.5

Note
There was no public NUT 2.7.0 release.

- This release is an interim release, part of the testing series, and the first release after the transition from Subversion to Git. The last release (2.6.5) is almost a year old. A lot of work has been done, but a good amount remains to achieve 2.8.0 goals. Please read the UPGRADING.adoc notes.
- Added support for SSL via the Mozilla NSS library, in addition to the existing OpenSSL support.
- Added a new driver, nutdrv_qx, for Megatec/Qx devices. This driver will eventually replace the blazer_ser and blazer_usb drivers. In particular, it adds support for Voltronic Power devices.
- Increased USB_TIMEOUT to standards-compliant 5.000 seconds in most drivers. This should reduce the number of timeouts on low-speed USB 1.1 devices.
- The jNut Java source has been split into a separate GitHub repository.

- Added many devices to the HCL. Of particular note are many Tripp Lite USB HID PDC models which were tested against NUT by Tripp Lite.
- Reworked some visual elements of the HCL. The output is better tailored for graphical and text-only browsers, but suggestions are welcome for additional accessibility enhancements.
- Also increased timeouts and added redundant commands to improve reliability of mge-utalk driver.
- Added the apcupsd-ups driver to interoperate with apcupsd installations.
- Added documentation on creating subdrivers for snmp-ups and nutdrv_qx.
- Added new drivers for the Riello UPS product line (riello_ser/riello_usb).
- Many improvements to the BCM/XCP drivers have been merged in. This includes an improved data reception loop, and additional mappings.
- Added a few variables to the Powercom HID mappings.
- Updated the apcsmart driver, and renamed the previous driver to apcsmart-old.
- Fixed the battery percentage calculation in the bestfcom driver.
- libnutclient has been added as a C++ alternative to libupsclient.
- Packaging files for Solaris and HP-UX (sponsored by Eaton)
- Fix shutdown of Eaton HID, using usbhid-ups and mge-shut
- usbhid-ups: final fix for APC Back UPS ES. APC Back UPS ES devices have buggy firmware, and this version does not cause a regression. The max_report variable should be set automatically based on the USB identification values.
  - UPDATE: known as `maxreport` flag for `usbhid-ups` driver, and as a `max_report_size` setting in code, as of NUT v2.8.2 release.
- nut-scanner: fix crash
- IPMI support can handle more different versions of FreeIPMI
- Support power supplies scan over the network nut-scanner can now scan for power supplies with IPMI over LAN. This is currently limited to IPMI 1.5 only
- Implement a framework to spell check documentation source files, using Aspell. This includes an interactive build target (make spellcheck-interactive), and an automated one (make spellcheck), mainly for QA / Buildbot purpose. Note that a base NUT dictionary is also available (docs/nut.dict), providing a glossary of terms related to power devices and management
- Improve systemd integration
- snmp-ups: Fixed a crash on outlet management, and added delta_ups MIB support. Also fixed mappings for upsBypassVoltage, upsBypassCurrent, and upsBypassPower in three-phase IETF MIB.

2.12 Release notes for NUT 2.6.5 - what's new since 2.6.4

- This release fixes an important regression in upssched: any upssched.conf command that takes a second argument resulted in a defective frame sent to the parent process. Thus, the command was not executed (report and patch from Oliver Schonefeld)
- Website hosting: free NUT from Eaton website hosting

  NUT website (https://www.networkupstools.org) is no longer hosted by Eaton. Arnaud Quette (NUT project leader) has taken over NUT hosting on his own, to give NUT back some independence.

  This effort is also part of a logic to stop crediting Eaton for contributions from others (especially Arnaud Quette, as an individual). The new hosting service is located, as for Arnaud's blog (http://arnaud.quette.fr) on Gandi servers, using PaaS.

  This will allow more flexibility and automation of the release process.

- macosx-ups: new OS X Power Sources meta-driver
  - Mac OS X provides UPS status information in a format similar to what is shown for laptop batteries. This driver will convert that information into a format compatible with NUT (Charles Lepple).
- support for new devices:
  - Eaton ePDU Switched
  - Online Zinto A (USB ID 0x06da:0x0601)
  - REDi Blazer 400VA / 600VA / 800VA
  - UNITEK Alpha650ipF and Alpha650ipE (USB ID 0x0f03:0x0001)
- mge-shut driver has been replaced by a new implementation (newmge-shut). In case of issue with this new version, users can revert to oldmge-shut. UPDATE: oldmge-shut was dropped between 2.7.4 and 2.8.0 releases.
- First NUT virtualization package: NUT now supports integration with VMware ESXI 5.0, through a native VIB package. This is, for the time being, an external effort from René Garcia (refer to the Download section on NUT website). But work is underway to improve this integration, and include it in the NUT source tree
- IPMI support (nut-ipmipsu driver and nut-scanner): prepare for supporting API changes in upcoming FreeIPMI versions 1.1.x and 1.2.x.
- snmp-ups now supports high precision values for APC, and more variables
- the NUT variables and commands namespace has been fixed and completed, with the known and used variables that were missing.
- more bugfixes, cleanup and improvements, on both source code and documentation.

2.13 Release notes for NUT 2.6.4 - what's new since 2.6.3

- This release fixes an important vulnerability in upsd (CVE-2012-2944: upsd can be remotely crashed)

  NUT server (upsd), from versions 2.4.0 to 2.6.3, are exposed to crashes when receiving random data from the network.

  This issue is related to the way NUT parses characters, especially from the network. Non printable characters were missed from strings operation (such as strlen), but still copied to the buffer, causing an overflow.

  Thus, fix NUT parser, to only allow the subset of ASCII charset from `Space` to `~` (Reported by Sebastian Pohle, Alioth bug #313636, CVE-2012-2944)

  A separate patch, which applies to any faulty version, is also available: http://trac.networkupstools.org/projects/nut/changeset/-3633

  For more information, refer to the Common Vulnerabilities and Exposures: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2012-2944
- A static source code analysis has been done by Michal Hlavinka from RedHat, using Coverity (12 issues fixed).
- Add new "LIST CLIENTS" and "NETVER" commands to NUT network protocol. "NETVER" allows to retrieve the Network protocol version, while "LIST CLIENTS" provides the list of clients connected to a device. Refer to the developer guide, "Network protocol information" section for more information.
- Support of ranges of values for writable variables has been added, to complete the existing enumerated values mechanism. This will start to appear in some drivers soon, beginning with Eaton. Refer to the developer guide, "Creating a new driver..." section for more information.
- PyNUT.py has been updated to version 1.2.2, adding support for LIST CLIENTS, FSD, HELP and VER (Rene Martín Rodríguez)

- support for new devices:
  - AEG Power Solutions PROTECT HOME
  - more APC SNMP cards
  - ATEK Defensor range
  - all Borri models
  - all COVER ENERGY SA
  - CyberPower OR700LCDRM1U, PR6000LCDRTXL5U and CP1000PFCLCD
  - Dell UPS Network Management Card
  - Dynamix 1000VA USB
  - Eaton Management Card Contact (ref 66104)
  - EVER POWERLINE RT / 11 / 31 and DUO II Pro
  - GE Digital Energy GT Series 1000-3000 VA
  - Gtec models
  - all recent HP serial / USB UPS (G2, G3 and R/T models,) and HP UPS Management Module
  - Ippon INNOVA RT
  - KOLFF BLACK NOVA
  - Lexis X-power Tigra 1kVA
  - Microline C-Lion Innova
  - Online Yunto YQ450
  - PowerShield Defender 1200VA
  - PowerWalker Online VFI LCD, Line-Interactive VI LCD and Line-Interactive VI
  - Riello Netman Plus 102 SNMP Card
  - Tripp-Lite OMNISMART500
- apcsmart has received some fixes to work better on Mac OS X, and in general
- bcmxcp has improved support for battery status, and better supports PW9120 units
- bestfortress has improved Best Fortress LI675VA support
- blazer_ser and blazer_usb now try to automatically estimate high and low voltages, to be able to calculate battery charge; support for online Innova UPS (T, RT and 3/1 T) has been added; Best UPS support has been improved, to prepare for superseding bestups driver
- bestups has also received some care, though users are encouraged to switch to blazer_ser, since bestups will soon be deprecated.
- newmge-shut has been heavily improved. However, replacement of the current mge-shut has been postponed to the next release, due to the CVE issue.
- oneac 0.80 improves support for all families of Oneac (EG, ON, OZ and OB), including more data and instant commands (Bill Elliot).
- usbhid-ups: for Eaton devices, ups.start.auto is now automatically adjusted for shutdown.{return,stayoff} to behave as expected; Liebert firmwares with incorrect exponents have also been addressed.
- snmp-ups now provides support for UPS shutdown, based on usbhid-ups mechanisms (composite commands and fallback). Composite commands are also supported now. This means, for example, that if *shutdown.return* is not supported, a combination of *load.off* + *load.on* may be used; Actual validity of instant commands is now tested before commands addition; Eaton/MGE MIB has been cleaned and completed; 3-phases support has been added to Socomec Netvision MIB; HP/Compaq MIB has been completed, with thresholds, nominal values and more commands.
- nut-scanner now also has libupsclient has a weak runtime dependency; more docs and bugfixes have also happened.

- Provide an Uncomplicated Firewall (UFW) profile (nut.ufw.profile)
- Riello protocols have been officially published in NUT protocols library: https://www.networkupstools.org/ups-protocols.html#_riello
- Duplicate instances of upsd / upsmon are now detected upon startup
- NUT variables namespace has been completed with missing variables and commands that are already known and standard
- upslog now comes with a companion file, for logrotate configuration
- more devices embed NUT for power protection, now including Thecus NAS range
- more bugfixes, cleanup and improvements, on both source code and documentation, with a good bunch from Greg A. Woods.

2.14 Release notes for NUT 2.6.3 - what's new since 2.6.2

- nut-scanner is now more portable, and provides more coherent option names. IPMI support has also been added, to discover local power supplies. This version brings weak runtime dependencies in libnutscan, which allows to compile nut-scanner with all options and to run according to the available dependencies (USB, SNMP, IPMI, ...).
- libnutscan now provides pkg-config support and the needed header files are distributed. Some documentation is also available in the developer guide and manual pages have been updated and completed.
- support for new devices:
  - Cyber Power Systems with SNMP RMCARD (100, 201, 202 and 301)
  - Dynamix 650VA USB
  - LDLC UPS-1200D
  - Tecnoware UPS ERA LCD 0.65
  - Powercom BNT-xxxAP (USB ID 0d9f:0004)
  - Various USB devices using UPSilon 2000 software
- apcsmart has received minor correction.
- bcmxcp_usb now handles disconnection issues and reconnection mechanism.
- blazer_usb enables again inclusion of buggy USB Device and Vendor IDs in udev rules file; language ID support has been added for USB units from LDLC, Dynamix and other no names.
- nut-ipmipsu has also received some improvements.
- snmp-ups has fixed outlets reported current in aphel_genesisII MIB; MGE 3 phases handles better low battery condition; support for Cyber Power Systems with SNMP RMCARD has been added; support of the newer Eaton ePDUs has been improved.
- upsd doesn't anymore fail to start if at least one of the listening interface is available. This is needed on systems where either IPv4 or IPv6 is disabled, and no explicit LISTEN directive has been specified.
- Avahi support is now automatically enabled, upon detection
- jNut (NUT Java interface) adds device discovery support, through a nut-scanner wrapper; jNutWebAPI, a HTTP/JSON web service interface, has also been added to interact with upsd and nut-scanner.
- Base files for HPUX packaging have been added. This is still a work in progress.
- Compilation on IBM AIX has been fixed (namespace conflict with ctypes).
- more bugfixes, cleanup and improvements, on both source code and documentation.

2.15 Release notes for NUT 2.6.2 - what's new since 2.6.1

- NUT can now monitor power supply units (PSU) in servers, through IPMI, using the new experimental *nut-ipmipsu* driver. Users are encouraged to test it, and send feedback and debug output to the development mailing list. This requires GNU FreeIPMI (0.8.5 or higher) development package or files. Thanks goes to Al Chu, FreeIPMI project leader, for his help during this development.
- NUT now provides a tool, called *nut-scanner*, to discover supported devices, both local and remote. nut-scanner will help to ease the configuration step, and power infrastructure discovery.

  This development, sponsored by Eaton, supports the following methods:

  – USB,
  – SNMP,
  – XML/HTTP (from Eaton),
  – NUT servers, using the classic connect or Avahi / mDNS methods.

  IPMI support will be added in the next release.

  A separate library, called *libnutscan*, is also available to provide these feature. Future NUT releases will provides binding for the supported languages (Perl, Python and Java).
- NUT now provides a Java interface called *jNut*. This development, sponsored by Eaton, is currently limited to the client interface. But it will be broaden to device discovery and configuration in the future.

  For more info, refer to nut/scripts/java/README, or the developer guide (chapter *Creating new client*). Javadoc documentation is also provided, along with Java archives (.jar) in the Download section.
- support for new devices:

  – Eaton 3S
  – Cyber Power Systems CP1000AVRLCD
  – various APC models equipped with APC AP9618 management card, including APC Smart-UPS RT XL
  – Orvaldi 750 / 900SP
  – POWEREX VI 1000 LED
  – PowerWalker VI 850 LCD
  – SVEN Power Pro+ series (USB ID ffff:0000)
- A regression has been fixed in udev rules file. This previously caused permission issues to owners of some USB devices.
- Avahi support has been added, for NUT mDNS publication, through a static service file (nut/scripts/avahi/nut.service).
- usbhid-ups has had Eaton completion: some features have been improved, such as *output.voltage.nominal*; 3S Eco control support has been added, along with battery.runtime.low and end of battery life (life cycle monitoring) support; new measurements for 5 PX are also supported now (outlet power factor, power, real power and current).
- apcsmart has been updated to support more variables and features; the previous driver is however still available as *apcsmart-old*, in case of issues.
- bcmxcp now supports per outlet startup and shutdown delays setting; shutdown delay is also used, when available, for outlet.n.shutdown.return instead of the default 3 seconds.
- snmp-ups.c has a new initialization method, that uses sysObjectID, which is a pointer to the preferred MIB of the device, to detect supported devices. This speeds up even more init stage and should render void the use of *mib* option. SNMP v3 session initialization has also been fixed, and Eaton PDU support has been completed.
- Initial support has been added for systemd, the System and Service Manager from RedHat.
- The chapter *NUT configuration management with Augeas* of the developer guide has received some completion: a complete Python Augeas example code is now provided.
- Finally, after years of dedication to NUT, Arjen de Korte is now retired. Sincere thanks to you Arjen from us all.

2.16 Release notes for NUT 2.6.1 - what's new since 2.6.0

- the various recent USB regressions have been definitely fixed.
- NUT now propose a variable to expose UPS efficiency (ups.efficiency). Eaton 5 PX already uses it.
- the Perl module from Gabor Kiss (rewritten from Kit Peters') is now distributed with NUT source code.
- support for new devices:
  - Eaton Ellipse ECO, Powerware 9140, Eaton 5 PX, and ambient sensor on Eaton ePDU managed
  - GE EP series
  - Inform Sinus SS 210
  - IPAR Mini Energy ME 800
  - Mustek Yukai PowerMust 1000 USB
  - Numeric 3000 SW
  - SVEN Power Pro+ series (recent models)
  - Vivaldi EA200 LED
- liebert-esp2: Improved Liebert ESP II support, including UPS shutdown (poweroff), 1 and 3-phase input and output variables, and most input / output / bypass / nominal variables. There is also a fix for the USB to serial cable (Farkas Levente and Spiros Ioannou).
- powercom has improved PowerCom BNT 1500A and BNT-other support, along with driver documentation and code conformance to the NUT rules (Keven L. Ates).
- apcsmart has more improved UPS poweroff support and options (Michal Soltys).
- blazer has also seen some improvements.
- usbhid-ups has completed a bit supported variables for APC and Eaton / MGE.
- on the quality assurance side, Eaton has worked on fixing a few non-conformance issues, like C++ style comments and warnings, using a newly developed verification tool (Prachi Gandhi).
- fix remaining references to LIBSSL_LDFLAGS, instead of LIBSSL_LIBS, which cause unresolved symbol on libupsclient users (Fabrice Coutadeur).
- the website has now a better support for Internet Explorer 6.
- graphic illustrations, used for the Features page on the website Features and chapter of the user manual, have been refreshed (courtesy of Eaton).
- more bugfixes, cleanup and improvements, on both source code and documentation.

2.17 Release notes for NUT 2.6.0 - what's new since 2.4.3

Note
Per original semantic versioning, there were no public NUT 2.5.x releases.

- the main focus of this release is the complete documentation revamping, using AsciiDoc. This includes a new website, user manual, developer guide, packager guide and manual pages, available in various formats (single and multiple pages HTML, and PDF at the moment).

  Be sure to check the `--with-doc` option help of `configure` script, and docs/configure.txt for more information.

- Add Augeas support, to provide easy NUT configuration management, through tools and development APIs. For more information, refer to the developer guide, or scripts/augeas/README.adoc in the source directory.
- support for new devices:
  - APC 5G
  - Eaton PowerWare 5119 RM (smart mode using upscode2 driver)
  - Eaton Best Ferrups (using older ConnectUPS card)
  - Eaton 9395 (serial interface)
  - Eaton ConnectUPS X / BD / E Slot
  - HP T1000 INTL
  - HP T1500 INTL
  - HP T750 G2
  - HP R1500 G2 INTL
  - iDowell iBox UPS
  - Tripp Lite SmartOnline SU1000XLA
  - Tripp Lite Smart1000LCD
  - and some more USB/HID devices IDs
  - CyberPower CP1500AVRLCD and CP1350AVRLCD
  - PowerWalker Line-Interactive VI 1400
  - Rocketfish RF-1000VA / RF-1025VA
- usbhid-ups has better support for shutting down APC SmartUPS RM series, and finally fix the "buffer size" issue, which was breaking some devices data retrieval, or truncating some data on others.
- snmp-ups now support SNMP v3 and its security parameters. IETF MIB support has also been extended.
- fix dummy-ups simulation driver status handling bug, and add the capability to remove exposed variables on the fly.
- the belkin driver now support control commands and status reporting for beeper and battery test.
- the powerpanel driver supports more older CyberPower units.
- mge-utalk, upscode2, blazer and liebert-esp2 have also received some care, and been improved.
- NUT-Monitor and the PyNUT client module have been updated to 1.3, adding more features like automatic connection to the first local device and i18n support.
- improve configure time dependencies checking and processing.
- improve older Unix systems support (HP-UX, AIX, ...) for missing functions.
- refresh and improve USB helper files (udev and UPower).
- more generation automation: the ChangeLog file is now generated automatically at distribution time, along with the files needed for the website hardware compatibility list.
- SSL support has also received some improvements.
- tcp-wrapper now allows hostnames in /etc/hosts.allow too (not only IPv4 and/or IPv6 addresses).
- many bugfixes, cleanup and improvements.

2.18 Release notes for NUT 2.4.3 - what's new since 2.4.2

- this is a bugfix release that only solves the regression on IPv6 activation.

2.19 Release notes for NUT 2.4.2 - what's new since 2.4.1

- the general USB support has been vastly improved, including many bug fixes, better OS support, new features and devices.
- NUT now talks to Solar Controller Devices with the new ivtscd driver.
- the snmp-ups driver supports more PDU, with a smaller disk footprint.
- apcsmart supports more older SmartUPS and Matrix units.
- the bestfortress driver is resurrected.
- the virtual driver has been renamed to *clone*.
- the netxml-ups driver has received some care.
- various debugging and development improvements have been done, around driver output; dummy-ups with more interaction and scripting and the device-recorder.sh script.
- the build system has received many bugfixes and improvements.
- the UPower (previously known as DeviceKit-power) rules file is now generated by NUT.
- support for new devices:
  - Apollo 1000A and 1000F
  - various Baytech RPC
  - old Best Power Fortress
  - Cyber Power Systems PR3000E, CP 1500C and OR2200LCDRM2U
  - all the new Dell UPS range (serial, USB and network)
  - Eaton E Series NV and DX UPS, and Powerware 9130
  - older HP T500 and T750, newer T750 INTL (USB) and R1500 G2 (serial)
  - Inform Informer Compact 1000VA
  - many serial and USB devices from Ippon, like Back Comfo Pro, Smart Power Pro and Smart Winner
  - IVT SCD series
  - Liebert GXT2-3000RT230 and PowerSure PSA
  - Mustek PowerMust 424 / 636 / 848 USB
  - all new PowerCOM USB devices with HID PDC interface
  - Tripp-Lite INTERNETOFFICE700, SMART700USB and ECO550UPS
  - UPSonic DS-800 (USB)

2.20 Release notes for NUT 2.4.1 - what's new since 2.4.0

- the microdowell driver has appeared to support various MicroDowell Enterprise units (see the "new devices" list below).
- support for new devices:
  - MicroDowell Enterprise B8, B10, N8, N11, N15, N20, N22, N30, N40, N50, N60 and HiBox ST.
- NUT-Monitor now better handles the ups.status field, and has switched to version 1.1.
- the situation of the build toolchain has been fixed, with regard to the "make clean" target and the wrongly removed generated USB files. This broke further configure call.

2.21 Release notes for NUT 2.4.0 - what's new since 2.2.2

Note
Per original semantic versioning, there were no public NUT 2.3.x releases.

- preliminary support for Power Distribution Units (PDUs): NUT can now support PDUs, either natively (ie using NUT snmp-ups driver), or through a binding to the Powerman daemon. The list of supported PDUs is already quite long, including:
  - Eaton ePDUs (Managed and Monitored),
  - some Aphel models,
  - some Raritan PDUs,
  - and the whole list of Powerman supported devices: http://powerman.sourceforge.net/supported.html
- support for new devices:
  - the various PDUs cited above
  - Chloride Desk Power 650
  - Cyber Power Systems Value 400E/600E/800E (USB models)
  - Delta GES602N
  - Digitus DN-170020
  - the whole Eaton ranges (mostly composed of MGE Office Protection Systems and Powerware units) including BladeUPS
  - Forza Power Technologies SL-1001
  - HP PowerTrust 2997A
  - HP R/T 2200 G2
  - Infosec XP 1000 and XP 500
  - Ippon Back Power Pro (serial and USB)
  - Kebo 1200D/D Series
  - Liebert PowerSure Personal XT
  - MGE Office Protection Systems Protection Station
  - Neus 400va and 600va
  - Phasak 400VA and 600VA
  - Plexus 500VA
  - Powercom Black Knight PRO / King PRO and Imperial
  - PowerKinetics BlackOut Buster
  - Sweex 1000 USB
  - UNITEK Alpha 500
  - WinPower CPM-800
- NUT now embeds Python client support through the PyNUTClient module and the NUT-Monitor application. Both are from David Goncalves, and are still available from http://www.lestat.st. For more information, refer to scripts/python/-README.adoc.
- the dummy-ups driver now supports a "repeater" mode. This allows it to act as a NUT client, and to forward data. This can be useful for supervision and load sharing purposes.
- tcp-wrappers support has been added to the upsd server, to grant users access by source IP for commands that require to be logged into the server. This replaces the previous internal implementation (ACL in upsd.conf).
- the nut.conf file has been introduced to standardize startup configuration across the various systems.

- NUT now ships a bash completion function for *upsc* command (scripts/misc/nut.bash_completion). Simply copy it to /etc/bash_comple
- many internal changes to improve maintainability, while lowering the maintenance cost (thus allowing developers to focus on what matters: the code!). Examples of this are:
- the USB information automatic extraction to generate the various USB helper files,
- the upsdrv_info_t structure to track more driver information, and remove the need for the upsdrv_banner() function
- common USB code refactoring, as it is done for the serial functions.
- tons of bugfixes, cleanup and improvements to make NUT stronger than ever!

2.22 Release notes for NUT 2.2.2 - what's new since 2.2.1

- support for new devices: APC BACK-UPS XS LCD, Atlantis Land, Mustek Powermust Office 650, Oneac XAU models, Powerware PW5115 and PW9120 (USB), Nitram Elite 2005
- Integrated Power Management (NUT HAL integration) has reached a major milestone: it is now the most advanced UPS integration into Power Management layer known in existing OSs. It has received many corrections and improvements, and allows to PowerOff the UPS at the end of a power cycle (which is the most important feature, not supported on other systems). The various files are now installed into the correct location.
- the usbhid-ups driver has received attention. Most notably, the shutdown handling has been reworked, and support for MGE UPS SYSTEMS 3 phases units has been added.
- snmp-ups now supports MGE* Environment Sensor (ref 66 846). The ambient.temperature reporting has also been fixed for units other than APC.
- the netxml-ups driver has appeared to support MGE* network HTTP/XML cards.
- NUT now distributes by default the shared version of libupsclient (version 1.0.0), and use this for the provided clients (upsmon, upsc, upsrw, upscmd). This is part of an effort to reduce NUT's footprint, both on disk and in memory.
- powerpanel has reach a new step toward the replacement of nitram and cpsups drivers. The final step is scheduled for NUT 2.4.
- many changes, cleanup and fixes to the NUT core and various drivers.

2.23 Release notes for NUT 2.2.1 - what's new since 2.2.0

- support for new devices:
  - all MGE Office Protection Systems units
  - Advice TopGuard 2000
  - Belkin F6H375-USB
  - Dynamix UPS1700D
  - Effekta RM2000MH,
  - Jageson Technology Jasuny USPS
  - Powercom SMK-1500A and SXL-1500A
  - PowerWalker Line-Interactive VI 400/800 and 600
  - Powerware 9110
  - UNITEK Alpha 2600
  - UPSonic CXR1000
  - some vintage serial APC UPSs

- the usbhid-ups driver has been improved, and fixed in many areas, through a backport of the development (trunk) version.
- the udev rules, for Linux hotplug support of the USB UPSs, has been updated to support kernel newer than 2.6.22.
- the megatec and megatec_usb drivers have also been backported from the development (trunk) version.
- the client development files have also received some care: the upsclient pkg-config file has been fixed, and the upsclient.h file allows older NUT clients to continue using the UPSCONN structure.

2.24 Release notes for NUT 2.2.0 - what's new since 2.0.5

Note
Per original semantic versioning, there were no public NUT 2.1.x releases.

- The new build infrastructure, using automake, is now used. This has major impact on the compilation and installation procedures, and thus on the NUT packaging. For more information, refer to UPGRADING.adoc and packaging/debian/ for an example of migration.
- NUT now provides support for FreeDesktop Hardware Abstraction Layer (HAL) which brings full Plug And Play experience to USB UPS owners. For more information, refer to docs/nut-hal.txt.
- support for new devices:
  - Ablerex 625L
  - ActivePower 400VA, 2000VA;
  - Belkin Home Office F6H350-SER, F6H500-SER, F6H650-SER
  - Belkin Office Series F6C550-AVR
  - Belkin Universal UPS F6C100-UNV (USB), F6C1100-UNV (USB), F6C1200-UNV (USB), F6H350deUNV (serial), F6H350ukUN (serial), F6H650ukUNV (serial)
  - Compaq R3000h
  - Cyber Power Systems PR2200
  - Dynex DX-800U
  - Geek Squad GS1285U
  - Krauler UP-M500VA
  - Mecer ME-2000
  - MGE UPS SYSTEMS Ellipse MAX
  - Online Zinto D
  - PowerTech SMK-800
  - SVEN Power Pro+ series
  - Power Smart RM 2000
  - Tripp-Lite SmartOnline SU1500RTXL2ua, smart2200RMXL2U.
- added IPv6 support,
- the newmge-shut driver has appeared. This one uses the same HID core as usbhid-ups, but communicate over a serial link. It will eventually replace the current mge-shut driver.
- client commands (upsc, upsrw and upscmd): hostname is now optional, and defaults to "localhost"
- many drivers have been improved and have received bug fixes: powerpanel, megatec, megatec_usb, safenet, tripplite_usb, gamatronic,
- the hotplug and udev scripts, in charge of setting the right permissions on the USB devices, are now installed automatically when appropriate.
- more generally, the NUT core and documentation, including the manpages, have been improved and updated.

2.25 Release notes for NUT 2.0.5 - what's new since 2.0.4

This release is a backport of the development version. Many changes have already been backported previously. Thus it is more a synchronization release, though it includes many bugfixes and support for new models.

- support for new devices:
  - APC Smart-UPS with 6TI firmware
  - Belkin Small Enterprise F6C1500-TW-RK
  - Compaq R3000 XR, R5500 XR
  - Cyber Power 550SL, 725SL, 685AVR, 800AVR, 1200AVR, AE550
  - Eltek
  - Inform GUARD
  - Microsol Rhino
  - Opti-UPS PowerES 420E
  - PowerMan RealSmart, BackPro
  - Powerware PW9315 3-phase
  - SOLA 305
  - Tripp-Lite SMART550USB, SMART2200RMXL2U, OMNI1000LCD, OMNI900LCD, OMNI650LCD, 1500 LCD, AVR550U
  - Viewsonic PowerES 420E
- bcmxcp: added 3-phase support
- megatec: better hardware support, more instant commands
- mge-hid: support more instant commands
- newhidups: fixed APC and Tripp Lite bugs, various memory bugs, improved report buffering, improved Solaris support, added *-x explore* option for easy diagnosis of new devices
- solis: shutdown programming, support new cables, Solaris support
- tripplite_usb: updated SMARTPRO support, fixed OL/OB reporting, better error handling, some memory bugs
- new dummy-ups driver simulator
- added HTML interface for access to CGI scripts

2.26 Release notes for NUT 2.0.4 - what's new since 2.0.3

- The newhidups critical bug (segmentation fault) has been fixed. It has also received some more care, like bugfixes and new models support and enhancement for Solaris. [Peter Selinger and Arnaud Quette]
- A bug has been fixed in NUT core to support resuming from suspend-to-disk. This should also fix other similar issues, like time synchronization through the NTP - Network Time Protocol. [Arjen de Korte]
- The mge-shut driver now better detects the Low Battery status, support new models and fixes some wrong status and data. It also fixes some issue where the UPS wasn't restarting (refer to mge-shut manpage). [Arnaud Quette]
- The genericups custom configuration through ups.conf is working again [Arjen de Korte]
- The genericups driver type 22 also support CyberPower 725SL (and maybe others SL models) [David Kaufman]
- The new megatec driver, which will replace a bunch of drivers by nut 2.2 (refer to docs/megatec.txt and UPGRADING.adoc) has been backported from the trunk (Development tree). The powermust driver has also received some attention. [Carlos Rodrigues]

- The new rhino driver was added to support Microsol Rhino UPS hardware The solis has also been improved for Solaris compatibility, and internal / external shutdown programming. solis can now save external shutdown programming to ups, and support new cables for solis 3 [Silvino B. MagalhÃ£es]
- Several fixes and improvements have been made to upsrw, upsset, cpsups, tripplite_usb and the FAQ. [Arjen de Korte and Charles Lepple]

2.27 Release notes for NUT 2.0.3 - what's new since 2.0.2

- The recent and major newhidups changes have been backported from the Development tree. It now:
- supports models from MGE UPS SYSTEMS, APC and Belkin. Mustek and Unitek units are also recognized for development purpose,
- handles better device reopening, after a disconnection,
- handles multiple devices, with several parameters to find the right UPS. [Peter Selinger, Charles Lepple and Arnaud Quette]
- The bcmxcp_usb driver has been added to support Powerware USB units. [Wolfgang Ocker and Kjell Claesson]
- The tripplite_usb driver has been added to support Tripp Lite USB units. [Charles Lepple]
- The sec driver is back as gamatronic [Gamatronic, Nadav Moskovitch]
- The genericups driver has received official care from Gamatronic to add support for the Gamatronic UPS with alarm interface. [Gamatronic, Nadav Moskovitch]
- The powermust driver now supports Soyntec Sekury C 500 and C 800 units. [Hanno Borns]
- The mge-shut driver has received a bit of attention too, and enhance ups.model retrieval for some specific case (release 0.65)
- The drivers don't change to the "statepath" directory anymore at initialization time if called using -k. This avoid unneeded failure to poweroff the UPS if /var is already unmounted. [Gaspar Bakos]
- The belkinunv driver now supports Belkin F6C1100-UNV [Dave Breiland]
- The isbmex driver has been upgraded to version 0.05, which fixes various errors in formulas, add shutdown capability and revert back baudrate to B9600 (instead of B2400), as it broke the communication [Ricardo Martinezgarza]
- The support of Sysgration UPGUARDS Pro650 in fentonups has been fixed [Simon J. Rowe]
- The packaging files for Red Hat have received various fixes [Thomas Jarosch]
- The solis driver has been fixed to avoid a naming collision and compile on Solaris [Paweł Kierdelewicz]
- The snmp-ups driver has corrected the problem when exposing certain time data.

2.28 Release notes for NUT 2.0.2 - what's new since 2.0.1

- the newhidups USB driver has been improved a lot and is no more experimental. It also now has a basic APC support, which will soon replace the legacy hidups driver.
- The mge-utalk driver has improved its support for old units.
- The mge-shut driver has been improved for restart/shutdown sequences which was previously blocking the serial port.
- The general MGE support has been added Pulsar EXtreme C / EX RT, Comet EX RT, Pulsar SV, Pulsar PSX, Ellipse Office and NOVA AVR USB.
- The genericups driver now supports Generic RUPS 2000, AEC MiniGuard UPS 700 (using Megatec M2501 cable), and Powerware 3110. [Nick Barnes, Paul Andreassen]

- The powermust driver now supports SquareOne Power QP1000, Mustek PowerMust 1400VA Plus and 2000VA USB. [Carlos Rodrigues]
- The fentonups driver has been enhanced and now supports Sysgration UPGUARDS Pro650. [Michel Bouissou, Simon J. Rowe]
- The cpsups driver now supports MicroDowell B.Box BP 500/750/1000/1500. [Armin Diehl]
- The snmp-ups driver now supports Socomec SNMP devices (Netvision MIB), and Powerware ConnectUPS SNMP cards. [Thanos Chatziathanassiou, Olli Salvia]
- The bcmxcp driver is back with support for Powerware UPSs. [Tore Ørpetveit, Kjell Claesson]
- The cyberpower driver now supports CyberPower 1000AVR. [Dave Huang]
- The new solis driver supports Microsol units: Solis 1.0, 1.5, 2.0 and 3.0. [Silvino B. Magalhaes]
- The apcsmart driver has fixed APC600 support.
- The etapro driver fixes brokenness due to ser_get_line use [Marek Michalkiewicz]
- The new upscode2 driver supports Fiskars, Compaq and Powerware devices. [Niels Baggesen, Havard Lygre]
- The tripplite driver has fixed a battery charge bug [Cedric Tefft]

2.29 Release notes for NUT 2.0.1 - what's new since 2.0.0

- The bestuferrups driver has been forked into the new bestfcom driver which has better handling of the inverter status alarm messages and more. [Kent Hill]
- Mustek UPS support returns with two drivers which have overlapping coverage: mustek and powermust. [powermust: Carlos Rodrigues, mustek: Martin Hajduch]
- Additional CyberPower Systems hardware is supported with the new cpsups driver. Three recognized models are the CPS1500AVR, CPS1100VA, and OP500TE. [Walt Holman, Brad Sawatzky]
- The genericups driver can now generate staleness warnings in specific cases where the UPS provides a way to test for its presence. See the "CON" setting in ups.conf for more details. [stan / saticed.me.uk]
- Documentation for monitoring a Back-UPS RS 500 on a system without USB ports has been added to the cables directory. [Martin Edlman]
- The everups driver now supports types 73-76 (NET 700/1000/1400/500-DPC) [hunter]
- The new metasys driver supports Meta System models: Line, HF Millennium, HF Top Line, ECO Network, ECO, Ally HF, Megaline [BlaXwan]
- The ippon driver now allows user-defined settings for the delay before switching off, and the delay before powering on. [Yuri Elizarov]
- The victronups driver is now at version 0.1.9, which adds many instant commands: calibration control, battery and front panel tests, and bypass control. [Gert Lynge]
- The tripplite driver has received a major overhaul to bring it up to working condition for the 2.0 tree, including code cleanups, several new variables, commands, and user-definable parameters. See ChangeLog for more. [Nicholas J Kain]
- The mge-utalk driver has been upgraded to version 0.81, which fixes the lack of read-write variables and loss of sync on models which don't support restoring settings. [Arnaud Quette]
- The Micro Ferrups model RE is now supported by the bestuferrups driver. The driver will also now read the ambient temperature and will no longer constantly report the data as stale. [Tim Thompson]
- The fentonups driver's init sequence has been reworked to work better with some hardware, including a fix to the parser code. [MLH]

- A workaround has been added to the hidups driver to avoid variables which are stuck by calling HIDIOCINITREPORT in every poll. [Stuart D. Gathman]
- SOLA 610 UPS hardware and others which do not support the ID command may now be monitored by the bestups driver after forcing ID= in ups.conf. [Jason White]
- "pollinterval" is now available via driver.parameter for consistency. [Arnaud Quette]
- The mge-shut and newhidups drivers, along with the supporting hidparser/libhid code have received many updates, including lowering USB bandwidth consumption, driver unbinding (only in Linux), code cleanups, and more which can be seen in the ChangeLog file. [Arnaud Quette]
- The fentonups driver now recognizes several more Megatec protocol units:
  - SuperPower HP360, Hope-550 [Denis Zaika]
  - Unitek Alpha 1000is [Antoine Cuvellard]
- Some variables like uc_sigmask were renamed to avoid clashes with symbols on systems like HP/UX.
- All man pages have been reworked to switch literal "-" characters to hyphens or "\-" as appropriate. [Shaul Karl]
- upssched's CANCEL events were broken following the change to text-based socket messages in 1.5 and have been fixed. [Steven Schoch]
- Calls to varargs functions with raw strings from the config files without an intervening "%s" have been fixed in upsmon, upssched, snmp-ups and upsd. [Ulf Harnhammar]

2.30 Release notes for NUT 2.0.0 - what's new since 1.4.x

- The new naming scheme for variables and commands (introduced in 1.4) is now mandatory. The 1.4 tree supported both the old (STATUS) and the new (ups.status) as a transitional release, and now that time is over.

  This means that 2.0 is generally smaller than 1.4 code, since the interim compatibility hacks have been removed.
- New serial handling code has been added, with greatly simplified operations. The old mess involving repeated calls to sigaction, alarm, and read has been condensed to a select-read loop.

  This change allows drivers which don't do any serial communications at all (hidups, snmp-ups) to drop that baggage, so they are a bit smaller when compiled.
- The drivers now recognize "chroot=' and *user*= in the global section of ups.conf. This means you don't have to use -r and -u when starting upsdrvctl.
- upsmon now supports the -K argument to check for the presence of the POWERDOWNFLAG file. If it exists and contains the magic string, then upsmon will exit(EXIT_SUCCESS). Otherwise, it will exit(EXIT_FAILURE).

  This feature can be used to simplify shutdown scripts, since now you don't have to keep the script in sync with the upsmon.conf.
- Many small things like signed value comparisons, int vs. size_t and proper use of const/struct were fixed throughout the source. These were mostly for correctness, but a few potential bugs involving very big or very small numbers were fixed at the same time.
- The access control system in upsd.conf has been reworked and simplified. Since access levels have become meaningless in recent releases, the new system is just ACCEPT <acl> or REJECT <acl>.

  If you are upgrading from a previous version of the software, you will have to edit your upsd.conf to use this method. See the UPGRADING.adoc file for more details.
- The build process now halts when make fails in one of the subdirectories. [Petter Reinholdtsen, Charles Lepple]
- Helper data for using upsclient via pkgconfig is now created if pkgconfig is detected when configure runs. [Arnaud Quette]

- The polling interval in drivers may now be set with *pollinterval* in ups.conf. [Gabriel Faber]
- Blazer UPS equipment is now supported with the blazer driver. [Phil Hutton]
- Energizer USB UPS hardware is now supported on Linux with a new experimental driver. [Viktor T. Toth]
- The newhidups driver has been merged as the first step towards portable USB UPS support. This will eventually replace the old Linux-only hidups driver. The newhidups driver is tagged experimental since it is under active development. [Arnaud Quette, Charles Lepple]
- The newapc driver has been renamed to apcsmart, replacing the old driver with that name. If you used the newapc driver, be sure to delete the old binary and fix your ups.conf.
- The apcsmart driver now supports asynchronous notification data from the hardware, which means it can wake up as soon as something happens. This affects the OL/OB/LB/RB data in ups.status, and generally reduces the latency in dispatching status changes by a few seconds.
- The apcsmart driver can now support quirky hardware which does not provide the usual listing of valid command characters. This feature is necessary to monitor new models like the APC CS 350 and old ones like the Matrix 5000. It also now has sdtype=4 to handle the strange shutdown behavior on the CS series.
- The belkin driver now works around broken firmware version 001, avoiding a lengthy delay at startup. It also implements the shutdown sequence differently, and should actually work on more hardware now.
- The bestups driver has been slowed down to play nicer with the hardware, and is much more reliable as a result. Among other things, it should always detect the UPS on the first try, meaning no more "dot dot dot" when it starts.
- The cyberpower driver is no longer tagged experimental, and now supports powering off the load. It also supports battery tests via instcmds.
- Effekta MT 2000 RM hardware is now supported by the fentonups driver. [Christoph Moar]
- The new safenet driver supports UPS hardware that uses the protocol of the same name. This includes models from many manufacturers, including Fairstone, Fenton, Gemini, Powerwell, Repotec, Soltec and Sweex. See the README or driver.list for the full details. [Arjen de Korte]
- The genericups driver now has type 20 to monitor the Powerware 5119 RM. See http://lists.exploits.org/ups/Oct2003/00052.html. [Daniel Thompson]
- The belkinunv driver has been added to allow monitoring Belkin Universal UPS hardware. [Peter Selinger]
- Cyber Power Systems 1100AVR hardware which has a different protocol than the existing binary type (supported by *cyberpower*) is now supported by the experimental cyberpower1100 driver. [Walt Holman]
- upsdrvctl now returns success or failure information in the exit code. Any failure during a requested operation will result in a nonzero value (specifically EXIT_FAILURE).

2.31 Release notes for NUT 1.4.0 - what's new since 1.2.x

- The drivers and upsd now communicate over Unix domain sockets instead of state files, shared memory, or state files with mmap. This change makes many things possible, including the new dynamic variable and command naming scheme described below.

  There is a new development tool called sockdebug in the server directory for debugging driver-server communications on the sockets.

- The old static variable scheme has been replaced by a new dynamic implementation. Vague names have been turned into meaningful names that fit into an organized system. UTILITY is now input.voltage. OUTVOLT is now output.voltage.

  This also applies to the names of instant commands. BTEST1 is test.battery.start, and BTEST0 is test.battery.stop.

  The old names are still supported for compatibility with older clients. This compatibility mode will be maintained throughout the 1.4 series, and will be gone by the release of 2.0. Users with older clients are encouraged to upgrade their software during this time.

- The network protocol has been expanded to handle these new names. Older functions which only apply to the old names will continue to be supported through the 1.4 series.
- The drivers and server (upsd) can now change their user ids and chroot themselves with the new -u and -r arguments. This lets you create a "chroot jail" with the bare minimum components.

  This technique is used to provide a higher degree of security. If someone exploited upsd to get a shell somehow, they would be stuck in the jail.
- upssched now explicitly confirms reception of timer commands before exiting. This was done to avoid a race where one process would exit right when another one was starting. The second one would believe its command had been handled when it had been lost.
- upslog has been reworked to use standard getopt parsing to provide the monitoring settings. The old way of specifying arguments is still supported for backwards compatibility.

  upslog has also been changed to only parse the format string once, rather than doing it every time through the loop. This should provide a minuscule drop in CPU utilization.
- Usernames are now required in upsmon and upsd. This means that you must add a username to your MONITOR lines in upsmon.conf and then create a matching user in upsd.users.

  Installations from the 1.2 era probably already used usernames, so this mostly affects those from 1.0 and before.
- Drivers are now pinged regularly by upsd when they aren't posting updates about the UPS status. This provides another check in the data validation process. If upsd fails to get a response within a few seconds, the UPS will be marked stale.
- A few minor memory leaks were discovered with valgrind and squashed.
- upsstats now reuses connections to upsd when cycling through multiple entries in the hosts.conf. This makes things a bit faster and avoids some of the noise in the syslog.

  This only applies to entries that are adjacent. To take advantage of this feature, you may have to rearrange them per example below.

  Connection reuse for nonadjacent entries may be considered in the future.

```
MONITOR ups-1@host-1 ...
MONITOR ups-1@host-2 ...
MONITOR ups-2@host-2 ...
MONITOR ups-3@host-3 ...
```

- upsd now warns about insecure configuration files at startup. These files (upsd.conf, upsd.users, and the certfile) should only be readable by upsd. Never make them world-readable.
- The programs no longer print "shutting down" when they are just exiting. This was changed to avoid confusion about the term, since "shutting down" has a special meaning in UPS software.
- Signal handlers no longer do any significant amount of work. Some of the programs used to do numerous things in there, raising concerns about reentrancy. They now set flags and allow the main loop to do the actual work from there.
- A bug in upsmon where NOTIFYFLAG settings could be ignored was fixed.
- Group handling has been changed. configure no longer accepts --with-group, and the programs no longer setgid() to a hard-coded value. They now setgid() to the primary group of whatever the user value may be.

  This may be compiled in with --with-user as before, and many programs accept -u to override it at runtime.
- The state path is no longer created during *make install*. Users are now expected to create it themselves. This removes a lot of evil complexity from the build and install sequences.

- upsd no longer implements the DROP access command, as it could confuse the clients by getting them out of sync. DROP is now implemented as DENY, which sends an error message. If you use DROP, you should change it to DENY rather than relying on this compatibility measure.
- The belkin driver no longer reports OFF and OL at the same time.
- The bestups driver no longer sleeps during polls, which makes it more responsive to things like instant commands.
- The cyberpower driver now has much better hardware detection code and no longer freezes at startup under some conditions. It also now supports the shutdown function. Instant commands for shutdowns and battery tests were also added.
- The dummyups testing driver has been removed. The dummycons testing driver can do everything that dummyups once did and much more. dummycons is also now built by default for easier testing.
- The newapc driver has been reworked to take advantage of the new internal driver state functions. Some variables without an obvious purpose were dropped.
- The newapc driver now sends all five bytes when using sdtype 1. Previously it didn't send the entire string, and it didn't work. [Don Lewis]
- The hidups driver has been expanded to allow for setting variables, a shutdown sequence, and more. [Arnaud Quette]
- The mge-utalk driver had trouble establishing communications in some cases due to the RTS line being set. This has been fixed.

  The mge-shut driver has been added to the tree, and has replaced the older mge-ellipse driver. [Arnaud Quette, Philippe Marzouk]
- Outlet-level control has been defined in the variable tree, and will be added to drivers where the hardware supports it. This can be used to shut down some components earlier than others to prolong your runtime on battery.

  This is supported in the mge-shut driver now, and may show up in others before long. [Arnaud Quette]
- KIN-2200AP hardware is now recognized by the powercom driver. This change may also support other KIN-xxxxAP equipment. [Preston A. Elder]
- The 1.1kVA UPS is now supported by the bestuferrups driver. This driver was also changed to allow easy addition of more models in the future. [Bob Apodaca]
- The fentonups driver can now handle devices which implement the "I" detection differently, and now supports the Giant/WELI 500 as a result. [Joon Guillen]
- The serial number of the UPS being monitored can now be specified with serial= in ups.conf in the genericups driver. [Shaul Karl]
- The newapc driver now sends ESC to break out of menus when the initial detection fails. Some new APC models have interactive menus on the serial port, and the driver couldn't handle them before.
- The snmp-ups driver now reports ambient temperature and humidity data for APC equipment. It also now supports the shutdown.reboot and shutdown.reboot.graceful commands. [Dmitry Frolov]
- The list of supported variables and commands in the snmp-ups driver has been expanded. [Arnaud Quette, J.W. Hoogervorst]
- Various drivers now report bypass mode with the BYP status word. [Arnaud Quette]
- Energy Sistem equipment is now supported with the esupssmart driver. [Antonio Trujillo Coronado]
- The Tripp-Lite SU series (SmartOnline) is supported with the new tripplitesu driver. [Allan Hessenflow]
- The HP PowerTrust A2994A is now recognized by the hp driver. [Jan Sporbeck]
- Many drivers were cleaned up to perform basic sanity checks on the status data before using it.
- An explicit cleanup function has been added to the driver core to ensure that all dynamic resources are freed before exiting. This is part of the larger process to check for memory leaks and other bad things. [Arnaud Quette]

- upsd now provides variable descriptions from an auxiliary file. This file is optional, which allows for a smaller memory footprint. It can also be edited for localization or other customizations.
- upsimage and upsstats can now render BATTVOLT data. [Andrew R. Ghali]
- String handling has been cleaned up throughout the tree. Calls to functions like strcpy and strcat were either replaced with other (range-checking) functions or were rewritten to avoid it.
- Many compile-time defaults may now be overridden at runtime. In the environment NUT_CONFPATH and NUT_STATEPATH may be used. upsdrvctl has been changed to execve to pass these along to the drivers. ups.conf now supports driverpath=, and upsd.conf supports DATAPATH. [Bryan Henderson]
- The configure --with-gd switches now actually do something useful when gd has been installed outside the default search directories. [Patrik Schindler]
- The inline keyword is now handled properly on systems which do not support it or have it specified as another name. This was breaking compiles on some systems. [Petter Reinholdtsen]

2.32 Release notes for NUT 1.2.2 - what's new since 1.2.1

- The snmp-ups driver has been upgraded and expanded. It now supports multiple MIBs, meaning it can handle RFC 1628, APCC, and MGE equipment. You can pick the right one with "mibs=" in ups.conf.

  Support for setting variable and instant commands is also available. [Arnaud Quette and Dmitry Frolov]
- The powernet driver has been upgraded. It now supports more variables, has cleaner logging, and may now be considered stable. [Dmitry Frolov]
- The hidups driver now supports physical port IDs. This avoids most of the problems where the hiddev* names can jump around too easily. It will now stay in the same place as long as you keep it plugged into the same physical port. See the ChangeLog file for more details. [David Brownell]
- The hidups driver now also supports the MFR variable on APC Back-UPS ES equipment. [Jonathan A. Davis]
- The sms driver has been updated to version 0.70. [Marcio Gomes]
- The bestups driver now recognizes Best Power Axxium Rackmount equipment. [Ales Casar]
- The liebert driver now uses O_NONBLOCK, and should now work consistently on OpenBSD as a result. [Alex Cichowski]
- The liebert driver also now uses debouncing logic on the status lines. It was possible to get false readings that would start a shutdown or just annoy users with excessive onbatt/online notify messages. The new code forces the status to settle down for 3 polls before accepting the new value.

  This means that very short power events may not be detected. The alternative is having your machine shut down just because it decided to wiggle over to OB LB for a few seconds.
- upsmon has had the disconnect logic fixed so the "communications lost" (COMMBAD) notify will actually go out when the connection fails. [Steve Monett]
- upssched now uses a lock file to prevent a race where two could start at the same time. The second upssched would "win", and the first one would be unreachable. This had the side-effect of not being able to cancel timers on the first one.

  If you use upssched, you must define the LOCKFN directive when upgrading to this version, or it will not work. [Gaspar Bakos]
- The packaging and scripts for Red Hat systems have been updated. [Antonino Albanese]
- upsd is now a bit more lenient about access levels in the *numlogins* check, which is what caused the problem in upsmon (next item).

- upsmon no longer gets stuck in slavesync() when upsd is configured to drop certain queries. This usually happened at the worst possible time: in the middle of a shutdown. [John David Garza]
- The upsclient functions now do more sanity checking on data from upsd so a short read won't return garbage to the callers.
- upsset now works properly with ENUM/VARTYPE values for multiple UPSes on a single upsd. [Dmitry Frolov]
- Various portability fixes for building on SGI were applied. [Andrea Suatoni]
- upsd no longer tries to reference a deleted client structure if the client disconnects at the wrong time. Previously, it tried to use that pointer after the sendback() function had already failed on write and deleted the client. This could cause upsd to segfault depending on what areas were accessed. [Patrik Schindler]

2.33 Release notes for NUT 1.2.1 - what's new since 1.2.0

- The sms driver is back, with support for Microlink Manager III hardware. [Marcio Gomes]
- Fideltronik Ares Series hardware is now supported as genericups type 19. [Tomek Orzechowski and Arkadiusz Mikiewicz]
- The drivers no longer silently drop instant commands or set commands from upsd that happen to get fragmented in transit. [linux@horizon.com]
- The old multilink driver is back with a new name: liebert. It supports Liebert UPStation GXE hardware with the contact-closure cable. This is currently an experimental driver as there is no way to power down the load.
- configure now picks up the right flags for gd automatically if gd 2.0.8 or higher is installed. This greatly simplifies the CGI build process for most users.
- Shutdowns on FreeBSD using the genericups driver should work again. [Petri Riihikallio]

2.34 Historic releases

More ancient history is not covered in detail here.

You can see docs/history.txt for more details.

3 NUT Upgrading Notes

This file lists changes that affect users who installed older versions of this software, or third-party integrations and library or data consumers. When upgrading from an older NUT version, be sure to check this file to see if you need to make changes to your system.

We welcome feedback from package maintainers — if you had to patch something out, or work around something in NUT code or recipes, please let us know in the issue tracker. Chances are, other distributions feel your pain, and some generalized solution belongs in the upstream project as an easy to use build configuration toggle to be shared by all interested downstream projects.

Note

For packaging (OS distribution or in-house) it is recommended to primarily `./configure --with-all` and then excise `--without-something` explicitly for items not supported on your platform, so you do not miss out on new NUT features as they come with new releases. Some may require that you update your build environment with new third-party dependencies, so a broken build of a new NUT release would let you know how to act.

This is a good time to point out that for stricter packaging systems, it may be beneficial to add `--enable-option-checking=fatal` to the `./configure` command line, in order to quickly pick up any other removed option flags.

3.1 Changes from 2.8.5 to 2.8.6

- PLANNED: Keep track of any further API clean-up?

- The Perl `UPS::Nut` module now decodes NUT quoting and backslash escapes in query/list values and `nutauth.conf`, including include paths. Remove manual decoding of these returned values and pass literal credentials to `Authenticate()`. Single quotes in configuration are now literal, as in the C parser; replace single-quoted values with double-quoted values where grouping was intended. The legacy `Set()` argument contract is unchanged. [PR #3634]

- `nut-scanner` now prefers the `SONAME` of each dynamically loaded library (such as `libusb-1.0.so.0`) over the fully versioned file name it saw at build time (such as `libusb-1.0.so.0.6.0`). Packagers no longer need to rebuild NUT after an ABI-compatible update of libusb, Net-SNMP, neon, Avahi, GLib or FreeIPMI which only changed that file name. The detected values are reported by `./configure` as `SONAME_LIB*` next to the existing `SOPATH_LIB*` and `SOFILE_LIB*` values. [issue #3474]

- The common configuration and protocol parser now accepts literal # characters inside double quotes. An unescaped # outside double quotes still starts a comment. Existing `\#` escapes remain supported and are still needed when sharing configuration files with older NUT versions. Encoded protocol output continues to escape hashes for older peers; default `upsc` text output is unchanged. [issues #607, #1306, PR #3633]

- Correct malformed numeric values in `upsd.conf` when upgrading: `MAXAGE`, `TRACKINGDELAY`, `MAXCONN` and numeric `CERTREQUEST` values are now checked in full and must fit their destination ranges. Values such as `15seconds` were previously read as `15`; these and overflowing values are now logged and ignored, retaining the previous setting. Valid decimal values, leading zeroes and trailing whitespace remain accepted. [issue #676, PR #3631]

- Potentially a breaking change for C++ clients that rushed to use the new SSL support options in `libnutclient`: for Mozilla NSS setup, the way to provide an expected `CERTHOST` address was missing. Now the API is fixed in this regard, at the cost of adding arguments to methods introduced in the previous release. [issue #3331, PR #3408]

- Potentially a breaking change for existing deployments with a (large) `MAXCONN` setting in `upsd.conf`: now this value is checked against the `getrlimit()` (e.g. `ulimit -n`) setting of the operating system for this daemon, where available, and the `upsd` data server would refuse to start if the requested value is larger than what is allowed (minus some reserve for configuration files and other use-cases). [issue #3365]

- Potentially a breaking change for existing deployments with `upsmon` and `upssched` integration: the `CMDSCRIPT` and `NOTIFYCMD` definitions were revised for the first token to mean an exact program name (possibly with spaces in the value), not a shell scriptlet. Documentation in earlier NUT releases implied that arguments could be passed here — this is no longer true. If you need to pass special arguments to a notification program, please use additional separately quoted tokens, or write a small helper script to do so and specify its path here. The `SHUTDOWNCMD` is still a scriptlet, as far as specifying command arguments is concerned, but it also newly supports parsing multiple tokens as additional arguments. [PR #3499]

- Potentially a breaking change for existing deployments with some flaw in configuration of the `upsd` data server: If SSL configuration was provided, but the server failed to apply some aspect of that, it should now abort with an explanation (and not proceed with insecure start-up like it could do before). [issue #3331, PR #3435]

- Enabled support for `nutauth.conf` files to provide credentials and/or SSL settings in clients which previously only did best-effort attempts at secure communications without an individual certificate, and only anonymously for reading like `upsc`.

  The new `-A filename` option defaults to trying to use a `nutauth.conf` file (if NOT specified here, but rather found in one of the default locations) but not failing if one is not usable; any specific values can require use of such a file or to not even try reading one (*none* as the legacy default). See the updated manual pages for more details. [issues #3329, #3411]

  > **Note**
  > In a future NUT release, the default `configure` script handling for the `--with-ssl-client-validation=auto` option may be changed from `no` to `yes`.

- The `upsmon` and `upsstats.cgi` clients capable of connections to multiple NUT data servers, potentially in separately managed security realms, now support an `AUTHCONF` option in their configuration files to specify the SSL details to be used with each such server. Legacy `upsmon` configuration keywords like `CERTPATH`, `CERTFILE`, `CERTIDENT`, `CERTHOST`,

CERTVERIFY and FORCESSL act as a "global defaults" section in nutauth.conf terms, and any actual nutauth.conf file entries gets loaded on top of these data. The default behavior (when AUTHCONF remains unspecified) is a best-effort attempt to load the file from one of the standard locations. If your NUT deployment defines an e.g. /etc/nut/nutauth.conf file which would be loaded automatically, but its contents conflict with what upsmon needs to know, please specify AUTHCONF = none in upsmon.conf. [issue #3439]

- nut-driver-enumerator now preserves quotes and backslashes in normalized configuration output and decodes values once using NUT token syntax. Value queries omit trailing comments, take the first value token and treat single quotes literally. This corrected parsing can change media dependencies. Section checksums may change on upgrade and trigger the usual reconciliation, including driver restarts; subsequent checks of unchanged configuration remain stable. [PR #3635]

- The powervar_cx_usb, tripplite_usb drivers used a built-in limit on reconnection attempts after which they exited (*60* and *10* respectively). This was revised to follow the new common setting reconnect_max_tries, which defaults to trying indefinitely now. [#3541]

- Deployments using nutdrv_qx driver with cypress subdriver may be impacted by a fix which added a special quirk, enabled by default for devices with USB ID 0665:5161 (Belkin, Voltronic, VOLT Polska), to drain stale bytes from USB buffers of the device before our driver sends queries, and use a longer timeout. If this change breaks your device, please add cypress_drain_quirk = 0 in your ups.conf and let us know. [issue #2534, PR #3563] With this quirk, a device which keeps sending data for longer than a reply takes is now reset over USB and reconnected; after 3 such resets in a row the driver exits ("Replies from the UPS stayed out of step with commands"). The quirk (and its default by USB ID) now also applies with an explicit subdriver = cypress setting, which used to turn it off. [PR #3720]

- The huawei-ups2000 driver fixed two serious problems that affected all previous NUT versions. All huawei-ups2000 users are strongly recommended to upgrade to NUT 2.8.6, all previous versions are considered too unreliable to use. Due to a Modbus compatibility problem, it was not possible to **write to any registers at all** on some UPS models, meaning that changing any **variable** or sending any **instant command** was impossible. Furthermore, due to a broken instant command handler, some instant commands would always fail with a reg1 is negative error (even with a compatible UPS), including bypass.start, shutdown.return, shutdown.reboot, and shutdown.reboot.graceful. [issue #3593, #3603, PR #3604].

- The NUT-Monitor GUI client application has historically branched from an implementation with Python 2 syntax and GTK 2 backend for GUI, into another script with Python 3 syntax and QT5 backend, and later another with QT6. Those two latter scripts have now been re-merged into one with Python 3 syntax and support for both QT5 and QT6 backends, one selected at run-time by the executable script name. Packaging-wise this should remain identical to earlier releases (also for distributions which delivered more than one implementation), however dependency rules can be now collapsed to requiring any of the QT5/QT6 backends for essentially the same client script. Delivery of the two filenames for same contents (to pick the backend at run-time) may also be optimized with symbolic or hard links in packaging. [PR #3596]

- Manual page name for generic_gpio_libgpiod driver was fixed to match its name (previously the document base file name could have been generic_gpio). This can impact packaging recipes. [PR #3693]

3.2 Changes from 2.8.4 to 2.8.5

- Added a configure script option to --enable-shared-private-libs which allows to deliver NUT common libraries as shared objects used by different NUT binaries, rather than linking just the used bits into each binary. This has some potential both for mayhem (so disabled by default) and for significant reduction of installation footprint. If enabled with a NUT packaging build, note there would be several more NUT shared library files to deliver with the packages (formally versioned and named by NUT release semantic version triplet). [issue #2800]

- Related to the above, libupsclient will remove the exported symbol for nut_debug_level variable in a later NUT release, and now introduces the upscli_upslog_set_debug_level() and upscli_upslog_get_debug_level() methods. [PR #3330]

- For ages, most recipes for building NUT had customized the sysconfdir to be /etc/nut, which is not exactly the **system** configuration directory. This is finally deprecated, with new --with-confdir configuration option taking over the role of a full path to configuration files (by default ${sysconfdir}${confdir_suffix}), and new --with-confdir-suffix allowing to specify just /nut or /ups that would be tacked onto the default ${sysconfdir} to resolve the ${confdir}. Default behavior should be same as with previous builds: if sysconfdir is customized (or prefix is kept at built-in

default), the `confdir_suffix` will default to empty; otherwise it assumes the value of `/${PACKAGE_NAME}` to become `/nut` in most cases. A `--with-confdir-examples` option was also introduced, to help distributions that place `*.conf.sample` files into docs or other locations. [#3131]

- Source directory `data/html` was renamed to `data/htmlcgi` in order to better reflect its contents and purpose, compared to documentation-oriented `html*` directories (and recipe variable names). This may impact some packaging recipes which do not rely on `make install` alone. Note that more files feature in the installation, e.g. a NUT logo rendition to use as favicon. [#3049, #3249]

- Some fixes were applied to HTML templates for NUT CGI clients. It can be useful for NUT deployments being upgraded to compare the files they have installed (and possibly customized) with the `*.html.sample` files delivered by the new build. [PR #3180]

- Introduced a `@NUT_UPSSTATS_TEMPLATE@` command which the NUT CGI template files now MUST start with (safety check that we are reading a template). While the delivered `upsstats*.html.sample` files would include the change, ultimate `upsstats*.html` templates deployed for end-users MUST be updated. [issue #3252, PR #3249]

- Dropped the `compile` script from Git sources. It originates from automake and is added to work area (if missing) during `autogen.sh` rituals anyway (as `make` says, `'automake --add-missing' can install 'compile'` when you run without it). It is still distributed as part of `make dist` tarball. We are open to any feedback whether this removal impacts NUT rebuilds on any systems. [#1209]

- Default `PIDPATH` is now more strictly `/var/run`, unless building on a system conforming to FHS-3.0 standard where that location is absent or is a symlink, while `/run` exists and is a true directory. Package recipes usually override `--with-pidpath` anyway to provide a dedicated location for NUT `root`-owned daemon PID files. [#3099]

- The autoconf managed macro for `--with-port` option was renamed from plain `PORT` to less ambiguous `NUT_PORT` to avoid potential conflicts with any third-party code. If anyone did use this macro in their custom builds or integrations, some renaming (or stubbed definitions to the effect of `PORT=NUT_PORT`) may be needed. [#3238]

- In drivers using the common `main.c` and `main.h` framework, introduced a `void upsdrv_tweak_prognames(void)` required method (may be no-op) to optionally tweak `prognames[]` entries now that there is certain support to accept program name aliases, not just one hard-coded string value. Any third-party drivers may require rebuilding to extend with this method. Any drivers that would undergo promotion with renaming can take advantage of this new facility to keep working under both old and new file names. [#3101]

- The `upsdrvquery*` semi-internal API for talking on the driver socket was updated with default handling for SIGPIPE. If your consumer wants to catch it themselves, instead of using `errno=EPIPE` after a failed call, you can use `upsdrvquery_NOSIGP` to disable neutering of the signal inside the API itself. [PR #3277]

- Added new API methods and defined bitmap values for `libupsclient` C binding to query and report SSL capabilities of the library build (none, OpenSSL, Mozilla NSS): `upscli_ssl_caps_descr()` and `upscli_ssl_caps()`. [PR #33xx]

- Fixed man page naming for `nutdrv_siemens-sitop(.8)` (dash vs. underscore) to match the driver program name. Packaging recipes may have to be updated. Follow-up from slightly botched renaming in original contribution. [PR #545]

- The `configure` script would now probe (if it can) the operating systems for more user and group account names, such as `upsmon`, `nutmon`, `ups`, `nut` (last hit wins, separately for user and groups accounts) settling on one of those if detected instead of `nobody` (and optionally `nogroup`). It would also warn if `nobody` or `nogroup` end up being used for a build. This is not likely to affect package builds (which usually define the accounts to use), but may (hopefully positively) affect rebuilds of NUT on systems where an older version is already deployed. [#3173]

- The `configure` script should now try harder to report specifically the "purelib" location as `PYTHON*_SITE_PACKAGES`. Packaging recipes may have to be updated. [#1209]

- New supported values were introduced for `--with-python{,2,3}=auto-prio=NUM`: this allows to auto-prioritize some **one** available Python interpreter. Default settings were changed from plain `auto` (allowing for multiple `python*` interpreters and `site-package` locations to be used), to prefer a `python3` if available, else `python2`, else `python`, and forget about others even if discovered. Packaging recipes may have to be updated, either to explicitly package the binding for several versions, or to not-deliver files no longer installed into the prototype area. [#1792]

- Added a `configure` script option to (primarily) `--disable-threading` for systems with detected but broken `libpthread` support, which can cause `nut-scanner` to crash while loading system libraries. [#3300]
- Introduced a new driver category for interaction with OS-reported UPS devices via D-Bus, with the `nut-upower` driver as the first implementation. This driver requires `glib-2.0` and `gio-2.0` development libraries to be available at build time. For packagers, this adds new dependencies if the build should include this driver. [PR #3279]

3.3 Changes from 2.8.3 to 2.8.4

- Introduced handling (possibly rewriting) for man page section "Overviews, conventions, and miscellaneous" (commonly number 7), to deliver support for `man nut` queries (NUT overview manual page also created). A new `configure --with-docs-ma` option was introduced so that directories for man page sections can now be automatically named as either "base" number of the section (e.g. `man1`) or by full section name (`man1m`), as different OS distributions have different preferences in this regard. As a packager, you may have to update the recipes accordingly (especially if your platform rewrites section numbers). [#2945, #2950]
- Introduced `powervar-cs` (Serial) and `powervar-cu` (USB) drivers which share a `powervar-cx` manual page, rendered once, with pages named after the driver aliased to it in post-processing. This may impact packaging, depending on its technology. Feel free to deliver file copies, symbolic or hard links as seems right for your platform. [#2988]
- Option to `configure --enable-docs-man-for-progs-built-only` was added, to differentiate NUT builds that deliver man pages for only built programs (legacy default) or for all of them (as needed for docs sites). [#2976]
- Added APC BVKxxxM2 to list of devices where `lbrb_log_delay_sec=N` may be necessary to address spurious LOWBATT and REPLACEBATT events. [issue #2942]
- In `nutdrv_qx`, updated `megatec` protocol subdriver for more detailed responses to `I` query which may return `ups.serial` (after a shorter `device.mfr`) and the `battery.runtime` (after a shorter `device.model`). Note that the expected response is shorter than in other dialects (38 vs. 39 bytes), so if this change breaks anything for your UPS that reported the values above correctly (e.g. the `ups.firmware` version becomes shorter or none of these are reported), please let NUT developers know. [#2980]

3.4 Changes from 2.8.2 to 2.8.3

- NUT development snapshots can now have more version components than the standard semantic versioning triplet, optionally adding the amount of commits on the development trunk since previous release, and the amount of commits on a feature branch that are unique to it. Release artifacts that have zeroes in both positions would have them stripped and still have the standard "semver" exposed, but the development snapshots can now be more reasonably upgraded with automated tooling. A copy of the current version information would be embedded into "dist" archives as a `VERSION_DEFAULT` file, so it can be used without git. Certain distros can benefit from a `VERSION_FORCED` file or a `NUT_VERSION_FORCED` environment variable exported from their build system, e.g. via `echo NUT_VERSION_FORCED=1.1.1 > VERSION_FORCED`. Unfortunately, some appliances tag all software the same with their firmware version; if this is required, a (NUT_)VERSION_FORCED_SEMVER envvar or file can help identify the actual NUT release version triplet used on the box. Please use it, it immensely helps with community troubleshooting! Documentation about this would be maintained in `docs/nut-versioning.adoc` [issue #1949]
- When packaging or custom-building for (Linux) distributions with systemd, you can now take advantage of `nut-systemd.preset` file to enable or disable certain NUT units by default; its comments document each choice. [issue #2721]
- A `nut-udev-settle.service` was introduced to replace dependency on the `systemd-udev-settle.service` which is deprecated and causes warnings on some systems. It was shown to benefit NUT use-cases however. [#2638]
- Reference packages prepared by `make package` should now separate the `PACKAGE_VERSION` from the platform-dependent prefix by a dash character in the ultimate package file name. Previously they were glued together for some platform targets (HPUX, Solaris). Solaris SVR4 package file names should new differentiate `i386` vs. `amd64` and `sparc` vs. `sparcv9`, depending on `target_cpu` of the build. If you had any scripts relying on the older pattern, they may have to be updated.

- The `make dist` goal now takes more care to require availability of the man pages to put into the prepared distribution archive. Development and CI builds on platforms unable to fulfill this goal can use `make distcheck-ci` (and `make dist-ci`) to fake presence of pre-built man pages with placeholder files, to complete other aspects of `distcheck` validation. [#2842]

- The PyPI distribution of the `PyNUTClient` module tarball should now use a lower-cased file name (and immediate versioned directory name inside) to match the requirements of PEP-0625. The Python module name (and its directory) should remain camel-cased. OS distribution package recipes that deliver the module separately (e.g. as part of Python ecosystem rather than NUT) may have to adjust. [#2773]

- The NUT-Monitor GUI application localization was updated; `make install` should now deliver also `xx_YY.UTF-8` pattern named symbolic links to the short-named directories and files involved, since some platforms insist on having those for translations to be found — this should be reflected in OS packaging recipes as well. [#2845]

- New `libupsclient` API methods added:

  - `upscli_str_add_unique_token()` and `upscli_str_contains_token()`, to help C NUT clients process `ups.status` and similarly structured strings same way as NUT core code base. [#2852, #2859]
  - `upscli_set_default_connect_timeout()` to modify the internal timeout used by `upscli_connect()` (default 0 still means blocking connections, positive values should time-limit the connection attempts), and `upscli_get_default_` to retrieve its copy. [#2847]

- API versions of `libupsclient` and `libnutscan` export more symbols now, and so were bumped to new "current" numbers; this may impact the naming of shared object files to be delivered by updated packaging. [#2895]

- Standard NUT clients including `upsc`, `upscmd`, `upsrw`, `upslog`, `upsmon`, `upsimage`, `upsset` and `upsstats`) were updated to default with a 10-second connection establishment timeout in case of name resolution lags or unresponsive hosts (notably a problem with `upsmon` contacting many remote systems at once). This may potentially impact NUT deployments which somehow relied on the blocking behavior of these clients; you can use the `NUT_DEFAULT_CONNECT_TIMEOUT` environment variable to fix this. [#2847]

- Several NUT clients including `upscmd`, `upsrw`, `upsimage`, `upsset`, `upsstats`, and `upslog` (during reconnection), did not `UPSCLI_CONN_TRYSSL` so went plaintext even when secure connections were possible. Fixed to at least try being secure, same way as `upsc` does for a long time. This may cause console or log messages when SSL can not be initialized, you can use the `NUT_QUIET_INIT_SSL` environment variable to suppress them where the cryptography is known to be not set up, so the warnings bring no value. [#2847]

- `lib/*.pc.in`: propagate `-R/PATH` (or equivalent — as detected by the `configure` script for the currently used compiler and linker toolkits) in `pkg-config` metadata pointing to NUT library installation location (by default not in system prefix) to help third-party clients link with us automatically. If this causes issues, `--disable-ldflags-nut-rpath(-cxx)` options (or `--enable...="..."` with specific linker arguments) can help. [#2782, #2865]

- Updated man page generation with `configure` script options to specify that manual sections on the target platform differ from (Linux-based) defaults hard-coded into page sources; this should allow to simplify NUT packaging recipe maintenance in distributions (no more updating patches for changed or added documentation sources)

- `upsmon` should now integrate natively with systemd-driven OS sleep events (built with systemd version 221 or newer "inhibitor interface"), so various hacks previously packaged into `/usr/lib/systemd/system-sleep/` scripts or units requiring/conflicting with the `sleep.target` may be obsolete. For fallback with older systemd, a `nut-sleep.service` is provided now. [#1070, #2596, #2597]

- Added systemd and SMF service integration for `upslog` as a `nut-logger` service (disabled by default, needs a `upslog.conf` file to deliver the `UPSLOG_ARGS=...` setting for actual monitoring and logging). [#1803]

- Handling of per-UPS `ALARM` state was introduced to `upsmon`, allowing it to optionally treat it as a factor in deciding that the device is in a "critical" state (polled more often, assumed dead if communications are lost). Since it is up to devices and their NUT drivers what they would raise as an alarm (might be something as mundane as ECO mode being active), some alarms can contribute to unwanted/early shutdowns. For this reason a `0|1` setting `ALARMCRITICAL` was introduced into `upsmon.conf` (default is `1`), for such users to be able to prevent their `upsmon` from treating the `ALARM` status as overly severe when it is not in fact. [#2658, #415]

- usbhid-ups and netxml-ups updated to handle "No battery installed!" alarm also to set the RB (Replace Battery) value in ups.status. This may cause dual triggering of notifications (as an ALARM generally and as an important REPLBATT status in particular) in upsmon, but better safe than sorry. [#415]

- usbhid-ups subdriver PowerCOM HID seemingly sent UPS shutdown and stayoff commands in wrong byte order, at least for devices currently in the field. Driver now sends the commands in a way that satisfies new devices; just in case a flag toggle powercom_sdcmd_byte_order_fallback was added to set the old behavior (if some devices do need it). [PR #2480]

- usbhid-ups subdriver CyberPower HID default pollfreq sped up to 12 seconds (common default is 30 seconds). Feedback is welcome if this improves connection stability or overwhelms the UPS controller instead. [issue #1689, PR #2718]

- usbhid-ups subdriver CyberPower HID default offdelay is set to 60 and ondelay to 120 seconds, in accordance with man page suggestions; users with custom settings not divisible by 60 will be loudly warned. [#1394]

- snmp-ups subdriver netvision-mib: synchronized netvision_output_info with the currently available SOCOMECUPS-M this can impact some other devices using that MIB (negatively, if the older mappings were indeed correct for any practical cases, and were not a typo). [#2803]

- nutdrv_qx fixed hunnox_protocol() to honour the optional novendor setting for devices that are confused by such query (e.g. DEXP LCD EURO 1200VA); it may be remotely possible that some other devices could begin to misbehave due to this fix — please let us know then. [#2839]

- mge-utalk driver will no longer set non-standard status values COMMFAULT and ALARM (for a specific status bit); instead, it will set modern ups.alarm with values COMMFAULT and/or DEVICEALARM (and raise an ALARM in ups.status for either, as standard alarms go). If your clients (e.g. custom parsing scripts) for devices supported by this driver depended on those non-standard tokens in ups.status, they would have to be updated to handle the new token values in ups.alarm instead. [#2708]

- Added support for lbrb_log_delay_sec=N setting to delay propagation of LB or LB+RB state (buggy with APC BXnnnnMI devices/firmwares issued circa 2023-2024 which flood the logs with spurious LOWBATT and REPLACEBATT events). This may work better for some devices when combined with flags like onlinedischarge_calibration and lbrb_log_delay_w [#2347]

- Enabled installation of built PDF and HTML (including man page renditions) files under the configured docdir. It seems previously they were only built (if requested) but not installed via make, unlike the common man pages which are delivered automatically. Packaging recipes can likely be simplified now. [#2445]

- A NUT_DEBUG_SYSLOG environment variable was introduced to tweak activation of syslog message emission (and related detachment of stderr when daemons are backgrounding), which can be useful for systemd service units. It can be set via nut.conf file for all standard consumers, or patched/dropped-in to systemd unit definitions specifically (less recommended, but may be easier to package). The positive effect would be avoiding duplicate logging as both syslog and stderr ending up in the same journal. [#2394]

- A CHANGELOG_REQUIRE_GROUP_BY_DATE_AUTHOR setting was added (for make calls and used by tools/gitlog2change script), and it defaults to true allowing for better ordered documents at the cost of some memory during document generation. Resource-constrained builders (working from a Git workspace, not tarball archives) may have to set it to false when calling make for NUT. [#2510]

- Drivers should now be able to set STATEPATH via ups.conf to match upsd custom configuration ability; in fact, the data server would prefer the value from ups.conf over the one in upsd.conf, if both are present. Note that NUT_STATEPATH environment variable trumps both. [issue #694]

- NUT products like nut-scanner, which dynamically load shared libraries at run-time without persistent pre-linking, should now know the library file names that were present during build (likely encumbered with version suffixes), and prefer them over plain libname.so patterns used previously (which on some platforms are only delivered by development packages as symlinks). Packaging recipes can likely be simplified now: some distros certainly did patch NUT source to similar effect). [#2431]

- Numerous changes to nut-scanner and symbols that its libnutscan.so delivers have caused a library version bump. New methods have been added and one structure (nutscan_ipmi_t) updated in a (hopefully) backwards compatible manner. [PR #2523, issue #2244 and numerous PRs for it]

- The `nutconf` tool added to main codebase with NUT v2.8.2 release could be packaged as a single program (with just a dependency on `libnutscan`), e.g. the library code with configuration file processing logic was built into it. Starting with NUT v2.8.3, the `libnutconf` may optionally be built as a standalone shared library, to deliver for development of integrations using `--with-dev-libnutconf` option. In this case the `nutconf` tool program would also depend on it for run-time linking. This may have to be considered in packaging recipes. [#2828]

- Internal API change for `sendsignalpid()` and `sendsignalfn()` methods, which can impact NUT forks which build using `libcommon.la` and similar libraries. Added new last argument with `const char *progname` (may be `NULL`) to check that we are signalling an expected program name when we work with a PID. With the same effort, NUT programs which deal with PID files to send signals (`upsd`, `upsmon`, drivers and `upsdrvctl`) would now default to a safety precaution — checking that the running process with that PID has the expected program name (on platforms where we can determine one). This might introduce regressions for heavily customized NUT builds (e.g. embedded in NAS or similar devices) whose binary file names differ significantly from a `progname` defined in the respective NUT source file, so a boolean `NUT_IGNORE_CHECKPROCNAME` environment variable support was added to optionally disable this verification. Also the NUT daemons should request to double-check against their run-time process name (if it can be detected). [issue #2463]

- More environment variable support was added to NUT programs, primarily aimed at wrappers such as init scripts and service unit definitions, allowing to tweak what (and whether) they write into debug traces, and so "make noise" or "bring invaluable insights" to logs or terminal; they can generally be used for services and init scripts via `nut.conf`:

  - See `NUT_IGNORE_CHECKPROCNAME` and `NUT_DEBUG_SYSLOG` above. [#1915]
  - See `NUT_DEFAULT_CONNECT_TIMEOUT` above. [#2847]
  - A `NUT_QUIET_INIT_BANNER` envvar (presence or "true" value) prevents tool name and NUT version banner from being printed out when programs start. [issues #1789 vs. #316]

- A `configure` script option to build `--with-modbus+usb` was added to let the caller insist on the use of USB-capable libmodbus (or fail the NUT build attempt). Certain build arguments can default this option to become enabled (implicitly): `configure --with-modbus --with-usb` and either `--with-drivers=*apc_modbus*` (actually implies `--with-modbus`) or `--with-modbus-includes=... --with-modbus-libs=...` as a way to avoid surprises with custom NUT builds aiming to have an USB-capable `apc_modbus` driver (currently this requires a custom-built libmodbus, can be a static build to avoid conflicts with OS). [#2666]

- A `configure` script option to `--enable-NUT_STRARG-always` was added to enable the `NUT_STRARG` macro (to handle `NULL` string printing) even if system libraries seem to safely support this behavior natively. This should primarily help against overly zealous static analysis tools in recent compiler generations. [#2585]

3.5 Changes from 2.8.1 to 2.8.2

- Builds requested with a specific C/C `language standard revision via CFLAGS and CXXFLAGS should again be honoured. There was a mishap with the m4 scripting for autoconf which could have caused use of C11/C11` if compiler supported it, regardless of a request. [PR #2306]

- Added generation of FreeBSD/pfSense quirks for USB devices supported by NUT (may get installed to `$datadir` e.g. `/usr/local/share/nut` and need to be pasted into your `/boot/loader.conf.local`). [#2159]

- nut-scanner now does not propose active `bus`, `busport` and `device` values when generating device configurations by default. They may appear as comments, or enabled by specifying the `-U` command-line option several times. [#2221]

- The `tools/gitlog2changelog.py.in` script was revised, in particular to convert section titles (with contributor names) into plain ASCII character set, for `dblatex` versions which do not allow diacritics and other kinds of non-trivial characters in sections. A number of other projects seem to use the NUT version of the script, and are encouraged to look at related changes in `configure.ac` and `Makefile.am` recipes. [PR #2360, PR #2366]

3.6 Changes from 2.8.0 to 2.8.1

- NUT documentation recipes were revised, so many of the text source files were renamed to `*.adoc` pattern. Newly, a `release-notes.pdf` and HTML equivalents are generated. Packages which deliver documentation may need to update

the lists of files to ship. [#1953] Developers may be impacted by new `configure --enable-spellcheck` toggle (should add spelling checks to `make check` by default, if tools are available) to facilitate quicker acceptance of contributions. Packaging systems may now want to explicitly disable it, if it blocks package building (pull requests to update the `docs/nut.dict` are a better and welcome solution). [#2067]

- Several improvements regarding simultaneous support of USB devices that were previously deemed "identical" and so NUT driver instances did not start for all of them:
  - Some more drivers should now use the common USB device matching logic and the 7 `ups.conf` options for that [#1763], and man pages were updated to reflect that [#1766];
  - The `nut-scanner` tool should suggest these options in its generated device configuration [#1790]: hopefully these would now suffice for sufficiently unique combinations;
  - The `nut-scanner` tool should also suggest sanity-check violations as comments in its generated device configuration [#1810], e.g. bogus or duplicate serial number values;
  - The common USB matching logic was updated with an `allow_duplicates` flag (caveat emptor!) which may help monitor several related no-name devices on systems that do not discern "bus" and "device" values (although without knowing reliably which one is which... sometimes it is better than nothing) [#1756].
- Work on NUT for Windows branch led to situation-specific definitions of what in POSIX code was all "file descriptors" (an `int` type). Now such entities are named `TYPE_FD`, `TYPE_FD_SER` or `TYPE_FD_SOCK` with some helper macros to name and determine "invalid" values (closed file, etc.) Some of these changes happened in NUT header files, and at this time it was not investigated whether the set of files delivered for third-party code integration (e.g. C/C++ projects binding with `libnutclient` or `libupsclient) is consistent or requires additional definitions/files. If something gets broken by this, it is a bug to address in future [#1556]
- Further revision of public headers delivered by NUT was done, particularly to address lack of common data types (`size_t`, `ssize_t`, `uint16_t`, `time_t` etc.) in third-party client code that earlier sufficed to only include NUT headers. Sort of regression by NUT 2.8.0 (note those consumers still have to re-declare some numeric variable types used) [#1638]
  - For practical example of NUT consumer adaptation (to cater to both old and new API types) please see https://github.com/-collectd/collectd/pull/4043
- Added support for `make install` of PyNUT module and NUT-Monitor desktop application — such activity was earlier done by packages directly; now the packaging recipes may use NUT source-code facilities and package just symlinks as relevant for each distro separately [#1462, #1504]
- The `upsd.conf` listing of `LISTEN` addresses was previously inverted (the last listed address was applied first), which was counter-intuitive and fixed for this release. If user configurations somehow relied on this order (e.g. to prioritize IPv6 vs. IPv4 listeners), configuration changes may be needed. [#2012]
- The `upsd` configured to listen on IPv6 addresses should handle only IPv6 (and not IPv4-mappings like it might have done before) to avoid surprises and insecurity — if user configurations somehow relied on this dual support, configuration changes may be needed to specify both desired IP addresses. Note that the daemon logs will now warn if a host name resolves to several addresses (and will only listen on the first hit, as it did before in such cases). [#2012]
- A definitive behavior for `LISTEN *` directives became specified, to try handling both IPv4 and IPv6 "any" address (subject to `upsd` CLI options to only choose one, and to OS abilities). This use-case may be practically implemented as a single IPv6 socket on systems with enabled and required IPv4-mapped IPv6 address support, or as two separate listening sockets - logged messages to this effect (e.g. inability to listen on IPv4 after opening IPv6) are expected on some platforms. End-users may also want to reconfigure their `upsd.conf` files to remove some now-redundant `LISTEN` lines. [#2012]
- Added support for `make sockdebug` for easier developer access to the tool; also if `configure --with-dev` is in effect, it would now be installed to the configured `libexec` location. A man page was also added. [#1936]
- NUT software-only drivers (dummy-ups, clone, clone-outlet) separated from serial drivers in respective Makefile and configure script options - this may impact packaging decisions on some distributions going forward [#1446]
- GPIO category of drivers was added (`--with-gpio` configure script option) - this may impact packaging decisions on some (currently Linux released 2018+) distributions going forward [#1855]

- An explicit `configure --with-nut-scanner` toggle was added, specifically so that build environments requesting `--with-all` but lacking `libltdl` would abort and require the packager either to install the dependency or explicitly forfeit building the tool (some distro packages missed it quietly in the past) [#1560]
- An `upsdebugx_report_search_paths()` method in NUT common code was added, and exposed in `libnutscan.so` builds in particular - API version for the public library was bumped [#317]
- Some environment variable support was added to NUT programs, primarily aimed at wrappers such as init scripts and service unit definitions, allowing to tweak what (and whether) they write into debug traces, and so "make noise" or "bring invaluable insights" to logs or terminal:
  - A `NUT_DEBUG_LEVEL=NUM` envvar allows to temporarily boost debugging of many daemons (`upsd`, `upsmon`, drivers, `upsdrvctl`, `upssched`) without changes to configuration files or scripted command lines. [#1915]
  - A `NUT_DEBUG_PID` envvar (presence) support was added to add current process ID to tags with debug-level identifiers. This may be useful when many NUT daemons write to the same console or log file, such as in containers/plugins for Home Assistant, storage appliances, etc. [#2118]
  - A `NUT_QUIET_INIT_SSL` envvar (presence or "true" value) prevents `libupsclient` consumers (notoriously `upsc`) from reporting whether they have initialized SSL support. [#1662]
  - A `NUT_QUIET_INIT_UPSNOTIFY` envvar (presence or "true" value) prevents daemons which can notify service management frameworks (such as systemd) about passing their lifecycle milestones, to not report loudly if they could not do so (e.g. running on a system without a framework, or misconfigured so they could not report and the OS would restart the false-positively "unresponsive" service). [#2136]
- `configure` script, reference init-script and packaging templates updated to eradicate `@PIDPATH@/nut` ambiguity in favor of `@ALTPIDPATH@` for the unprivileged processes vs. `@PIDPATH@` for those running as root [#1719]
- The "layman report" of NUT configuration options displayed after the run of `configure` script can now be retained and installed by using the `--enable-keep_nut_report_feature` option; packagers are welcome to make use of this, to better keep track of their deliveries [#1826, #1708]
- Renamed generated nut-common.tmpfiles(.in) ⇒ nut-common-tmpfiles.conf(.in) to install a /usr/lib/systemd-tmpfiles/*.conf pattern [#1755]
  - If earlier NUT v2.8.0 package recipes for your Linux distribution dealt with this file, you may have to adjust its name for newer releases.
  - Several other issues have been fixed related to this file and its content, including #1030, #1037, #1117 and #1712
- Extended Linux systemd support with optional notifications about daemon state (READY, RELOADING, STOPPING) and watchdog keep-alive messages. Note that `WatchdogSec=` values are currently NOT pre-set into systemd unit file templates provided by NUT, this is an exercise for end-users based on sizing of their deployments and performance of monitoring station [#1590, #1777]
- snmp-ups: some subdrivers (addressed using the driver parameter `mibs`) were renamed: `pw` is now `eaton_pw_nm2`, and `pxgx_ups` is `eaton_pxg_ups` [#1715]
- The `tools/gitlog2changelog.py.in` script was revised, in particular to generate the `ChangeLog` file more consistently with different versions of Python interpreter, and without breaking the long file paths in the resulting mark-up text. Due to this, a copy of this file distributed with NUT release archives is expected to considerably differ on first glance from its earlier released versions (not just adding lines for the new release, but changing lines in the older releases too) [#1945, #1955]

3.7 Changes from 2.7.4 to 2.8.0

- Note to distribution packagers: this version hopefully learns from many past mistakes, so many custom patches may be no longer needed. If some remain, please consider making pull requests for upstream NUT codebase to share the fixes consistently across the ecosystem. Also note that some new types of drivers (so package groups with unique dependencies) could have appeared since your packaging was written (e.g. with modbus), as well as new features in systemd integration (`nut-driver@instances` and the `nut-driver-enumerator` to manage their population), as well as updated Python 2 and Python 3 support (again, maybe dictating different package groups) as detailed below.

- Due to changes needed to resolve build warnings, mostly about mismatching data types for some variables, some structure definitions and API signatures of several routines had to be changed for argument types, return types, or both. Primarily this change concerns internal implementation details (may impact update of NUT forks with custom drivers using those), but a few changes also happened in header files installed for builds configured `--with-dev` and so may impact `upsclient` and `nutclient` (C++) consumers. At the very least, binaries for those consumers should be rebuilt to remain stable with NUT 2.8.0 and not mismatch int-type sizes and other arguments.
- libusb-1.0: NUT now defaults to building against libusb-1.0 API version if the configure script finds the development headers, falling back to libusb-0.1 if not. Please report any regressions.
- apcupsd-ups: When monitoring a remote apcupsd server, interpret "SHUTTING DOWN" as a NUT "LB" status. If you were relying on the previous behavior (for instance, in a monitor-only situation), please adjust your upsmon settings. Reference: https://github.com/networkupstools/nut/issues/460
- Packagers: the AsciiDoc detection has been reworked to allow NUT to be built from source without requiring asciidoc/a2x (using pre-built man pages from the distribution tarball, for instance). Please double-check that we did not break anything (see docs/configure.txt for options).
- Driver core: options added for standalone mode (scanning for devices without requiring ups.conf) - see docs/man/nutupsdrv.txt for details.
- oldmge-shut has been removed, and replaced by mge-shut.
- New drivers for devices with "Qx" (also known as "Megatec Q*") family of protocols should be developed as sub-drivers in the `nutdrv_qx` framework for USB and Serial connected devices, not as updates/clones of older e.g. `blazer` family and `bestups`. Sources, man pages and start-up messages of such older drivers were marked with "OBSOLETION WARNING".
- liebert-esp2: some multi-phase variable names have been updated to match the rest of NUT.
- netxml-ups: if you have old firmware, or were relying on values being off by a factor of 10, consider the `do_convert_deci` flag. See docs/man/netxml-ups.txt for details.
- snmp-ups: detection of Net-SNMP has been updated to use `pkg-config` by default (if present), rather than `net-snmp-config` (-script(s) as the only option available previously. The scripts tend to specify a lot of options (sometimes platform-specific) in suggested `CFLAGS` and `LIBS` compared to the packaged `pkg-config` information which also works and is more portable. If this change bites your distribution, please bring it up in https://github.com/networkupstools/nut/issues or better yet, post a PR. Also note that `./configure --with-netsnmp-config(=yes)` should set up the preference of the detected script over `pkg-config` information, if both are available, and `--with-netsnmp-config=/path/name` would as well.
- snmp-ups: bit mask values for flags in earlier codebase were defined in a way that caused logically different items to have same numeric values. This was fixed to surely use different definitions (so changing numbers behind some of those macro symbols), and testing with UPS, ePDU and ATS hardware which was available did not expose any practical differences.
- usbhid-ups: numeric data conversion from wire protocol to CPU representation in GetValue() was completely reworked, aiming to be correct on all CPU types. That said, regressions are possible and feedback is welcome.
- nut-scanner: Packagers, take note of the changes to the library search code in common/common.c. Please file an issue if this does not work with your platform.
- dummy-ups can now specify `mode` as a driver argument, and separates the notion of `dummy-once` (new default for `\*.dev` files that do not change) vs. `dummy-loop` (legacy default for `*.seq` and others) [issue #1385]
  - Note this can break third-party test scripts which expected `*.dev` files to work as a looping sequence with a `TIMER` keywords to change values slowly; now such files should get processed to the end once. Specify `mode=dummy-loop` driver option or rename the data file used in the `port` option for legacy behavior. Use/Test-cases which modified such files content externally should not be impacted.
- Python: scripts have been updated to work with Python 3 as well as 2.
  - PyNUT module (protocol binding) supports both Python generations.

  - NUT-Monitor (desktop UI client) got separated into two projects: one with support for Python2 and GTK2, and another for Python3 and Qt5. On operating systems that serve both environments, either of these implementation should be usable. For distributions that deprecated and removed Python2 support, it is a point to consider in NUT packages and their build-time and installation dependencies. The historic filenames for desktop integration (`NUT-Monitor` script and `nut-monitor.desktop`) are still delivered, but now cover a wrapper script which detects the environment capabilities and launches the best suitable UI implementation (if both are available).

- apcsmart: updates to CS "hack" (see docs/man/apcsmart.txt for details)
- upsdebugx(): added `[D#]` prefix to log entries with level > 0 so if any scripts or other tools relied on parsing those messages making some assumptions, they should be updated
- upsdebugx() and related methods are now macros, optionally calling similarly named implementations like s_upsdebugx() as a slight optimization; this may show up in linking of binaries for some customized build scenarios
- libraries, tools and protocol now support a `TRACKING` ID to be used with an `INSTCMD` or `SET VAR` requests; for details see docs/net-protocol.txt and docs/sock-protocol.txt
- upsrw: display the variable type beside ENUM / RANGE
- Augeas: new `--with-augeas-lenses-dir` configure option.

3.8 Changes from 2.7.3 to 2.7.4

- scripts/systemd/nut-server.service.in: Restore systemd relationship since it was preventing upsd from starting whenever one or more drivers, among several, was failing to start
- Fix UPower device matching for recent kernels, since hiddev* devices now have class "usbmisc", rather than "usb"
- macosx-ups: the "port" driver option no longer has any effect
- Network protocol information: default to type NUMBER for variables that are not flagged as STRING . This point is subject to improvements or change in the next release 2.7.5. Refer to docs/net-protocol.txt for more information

3.9 Changes from 2.7.2 to 2.7.3

- The nutdrv_qx(8) driver will eventually supersede bestups(8). It has been tested on a U-series Patriot Pro II. Please test the new driver on your hardware during your next maintenance window, and report any bugs.
- If you are upgrading from a new install of 2.7.1 or 2.7.2, double-check the value of POWERDOWNFLAG in $prefix/etc/upsmon.conf - it has been restored to /etc/killpower as in 2.6.5 and earlier.
- If you use upslog with a large sleep value, you may be interested in adding `killall -SIGUSR1 upslog` to any OB/OL script actions. This will force upslog to write a log entry to catch short power transients.
- Be sure that your SSL keys are readable by the NUT system user. The SSL subsystem is now initialized after `upsd` forks, to work around issues in the NSS library.
- The systemd nut-server.service does not Require nut-driver to be started successfully. This was previously preventing upsd startup, even for just one driver failure among many. This also matches the behavior of sysV initscripts.

3.10 Changes from 2.7.1 to 2.7.2

- upsdrvctl is now installed to $prefix/sbin rather than $driverexec. This usually means moving from /bin to /sbin, apart from few exceptions. In all cases, please adapt your scripts.
- FreeDesktop Hardware Abstraction Layer (HAL) support was removed. Please adapt your packaging files, if you used to distribute the nut-hal-drivers package.
- This is a good time to point out that for stricter packaging systems, it may be beneficial to add "--enable-option-checking=fatal" to the ./configure command line, in order to quickly pick up any other removed option flags.

3.11 Changes from 2.6.5 to 2.7.1

- The apcsmart(8) driver has been replaced by a new implementation. There is a new parameter, *ttymode*, which may help if you have a non-standard serial port, or Windows. In case of issues with this new version, users can revert to apcsmart-old.
- The nutdrv_qx(8) driver will eventually supersede blazer_ser and blazer_usb. Options are not exactly the same, but are documented in the nutdrv_qx man page.
- Mozilla NSS support has been added. The OpenSSL configuration options should be unchanged, but please refer to the upsd.conf(5) and upsmon.conf(5) documentation in case we missed something.
- upsrw(8) now prints out the maximum size of variables. Hopefully you are not parsing the output of upsrw - it would be easier to use one of the NUT libraries, or implement the network protocol yourself.
- The jNut source is now here: https://github.com/networkupstools/jNut

3.12 Changes from 2.6.4 to 2.6.5

- users are encouraged to update to NUT 2.6.5, to fix a regression in upssched.
- mge-shut driver has been replaced by a new implementation (newmge-shut). In case of issue with this new version, users can revert to oldmge-shut. UPDATE: oldmge-shut was dropped between 2.7.4 and 2.8.0 releases.

3.13 Changes from 2.6.3 to 2.6.4

- users are encouraged to update to NUT 2.6.4, to fix upsd vulnerability (CVE-2012-2944: upsd can be remotely crashed).
- users of the bestups driver are encouraged to switch to blazer_ser, since bestups will soon be deprecated.

3.14 Changes from 2.6.2 to 2.6.3

- nothing that affects upgraded systems.

3.15 Changes from 2.6.1 to 2.6.2

- apcsmart driver has been replaced by a new implementation. In case of issue with this new version, users can revert to apcsmart-old.

3.16 Changes from 2.6.0 to 2.6.1

- nothing that affects upgraded systems.

3.17 Changes from 2.4.3 to 2.6.0

- users of the megatec and megatec_usb drivers must respectively switch to blazer_ser and blazer_usb.
- users of the liebertgxt2 driver are advised that the driver name has changed to liebert-esp2.

3.18 Changes from 2.4.2 to 2.4.3

- nothing that affects upgraded systems.

3.19 Changes from 2.4.1 to 2.4.2

- The default subdriver for the blazer_usb driver USB id 06da:0003 has changed. If you use such a device and it is no longer working with this driver, override the *subdriver* default in *ups.conf* (see man 8 blazer).
- NUT ACL and the allowfrom mechanism has been replaced in 2.4.0 by the LISTEN directive and tcp-wrappers respectively. This information was missing below, so a double note has been added.

3.20 Changes from 2.4.0 to 2.4.1

- nothing that affects upgraded systems.

3.21 Changes from 2.2.2 to 2.4.0

- The nut.conf file has been introduced to standardize startup configuration across the various systems.
- The cpsups and nitram drivers have been replaced by the powerpanel driver, and removed from the tree. The cyberpower driver may suffer the same in the future.
- The al175 and energizerups drivers have been removed from the tree, since these were tagged broken for a long time.
- Developers of external client application using libupsclient must rename their "UPSCONN" client structure to "UPSCONN_t".
- The upsd server will now disconnect clients that remain silent for more than 60 seconds.
- The files under scripts/python/client are distributed under GPL 3+, whereas the rest of the files are distributed under GPL 2+. Refer to COPYING for more information.
- The generated udev rules file has been renamed with dash only, no underscore anymore (i.e. 52-nut-usbups.rules instead of 52_nut-usbups.rules)

3.22 Changes from 2.2.1 to 2.2.2

- The configure option "--with-lib" has been replaced by "--with-dev". This enable the additional build and distribution of the static version of libupsclient, along with the pkg-config helper and manual pages. The default configure option is to distribute only the shared version of libupsclient. This can be overridden by using the "--disable-shared" configure option (distribute static only binaries).
- The UPS poweroff handling of the usbhid-ups driver has been reworked. Though regression is not expected, users of this driver are encouraged to test this feature by calling "upsmon -c fsd" and report any issue on the NUT mailing lists.

3.23 Changes from 2.2.0 to 2.2.1

- nothing that affects upgraded systems. (The below message is repeated due to previous omission)
- Developers of external client application using libupsclient are encouraged to rename their "UPSCONN" client structure to "UPSCONN_t" since the former will disappear by the release of NUT 2.4.

3.24 Changes from 2.0.5 to 2.2.0

- users of the newhidups driver are advised that the driver name has changed to usbhid-ups.
- users of the hidups driver must switch to usbhid-ups.
- users of the following drivers (powermust, blazer, fentonups, mustek, esupssmart, ippon, sms) must switch to megatec, which replaces all these drivers. Please refer to doc/megatec.txt for details.

- users of the mge-shut driver are encouraged to test newmge-shut, which is an alternate driver scheduled to replace mge-shut,
- users of the cpsups driver are encouraged to switch to powerpanel which is scheduled to replace cpsups,
- packagers will have to rework the whole nut packaging due to the major changes in the build system (completely modified, and now using automake). Refer to packaging/debian/ for an example of migration.
- specifying *-a <id>* is now mandatory when starting a driver manually, i.e. not using upsdrvctl.
- Developers of external client application using libupsclient are encouraged to rename the "UPSCONN" client structure to "UPSCONN_t" since the former will disappear by the release of NUT 2.4.

3.25 Changes from 2.0.4 to 2.0.5

- users of the newhidups driver: the driver is now more strict about refusing to connect to unknown devices. If your device was previously supported, but fails to be recognized now, add *productid=XXXX* to ups.conf. Please report the device to the NUT developer's mailing list.

3.26 Changes from 2.0.3 to 2.0.4

- nothing that affects upgraded systems.
- users of the following drivers (powermust, blazer, fentonups, mustek, esupssmart, ippon, sms, masterguard) are encouraged to switch to megatec, which should replace all these drivers by nut 2.2. For more information, please refer to doc/megatec.txt

3.27 Changes from 2.0.2 to 2.0.3

- nothing that affects upgraded systems.
- hidups users are encouraged to switch to newhidups, as hidups will be removed by nut 2.2.

3.28 Changes from 2.0.1 to 2.0.2

- The newhidups driver, which is the long run USB support approach, needs hotplug files installed to setup the right permissions on device file to operate. Check newhidups manual page for more information.

3.29 Changes from 2.0.0 to 2.0.1

- The cyberpower1100 driver is now called cpsups since it supports more than just one model. If you use this driver, be sure to remove the old binary and update your ups.conf *driver=* setting with the new name.
- The upsstats.html template page has been changed slightly to reflect better HTML compliance, so you may want to update your installed copy accordingly. If you've customized your file, don't just copy the new one over it, or your changes will be lost!

3.30 Changes from 1.4.0 to 2.0.0

- The sample config files are no longer installed by default. If you want to install them, use *make install-conf* for the main programs, and *make install-cgi-conf* for the CGI programs.
- ACCESS is no longer supported in upsd.conf. Use ACCEPT and REJECT.
  - Old way:

```
ACCESS grant all adminbox
ACCESS grant all webserver
ACCESS deny all all
```

– New way:

```
ACCEPT adminbox
ACCEPT webserver
REJECT all
```

– Note that ACCEPT and REJECT can take multiple arguments, so this will also work:

```
ACCEPT adminbox webserver
REJECT all
```

- The drivers no longer support sddelay in ups.conf or -d on the command line. If you need a delay after calling *upsdrvctl shutdown*, add a call to sleep in your shutdown script.
- The templates used by upsstats have changed considerably to reflect the new variable names. If you use upsstats, you will need to install new copies or edit your existing files to use the new names.
- Nobody needed UDP mode, so it has been removed. The only users seemed to be a few people like me with ancient asapm-ups binaries. If you really want to run asapm-ups again, bug me for the new patch which makes it work with upsclient.
- *make install-misc* is now *make install-lib*. The misc directory has been gone for a long time, and the target was ambiguous.
- The newapc driver has been renamed to apcsmart. If you previously used newapc, make sure you delete the old binary and fix your ups.conf. Otherwise, you may run the old driver from 1.4.

3.31 File trimmed here on changes from 1.2.2 to 1.4.0

For information before this point, start with version 2.4.1 and work back.

4 Project history

This page is an attempt to document how everything came together.

The Network UPS Tools team would like to warmly thank Russell Kroll.

Russell initially started this project, maintaining and improving it for over 8 years (1996 — mid 2005).

4.1 Prototypes and experiments

4.1.1 May 1996: early status hacks

APC's Powerchute was running on kadets.d20.co.edu (a BSD/OS box) with SCO binary emulation. Early test versions ran in cron, pulled status from the log files and wrote them to a .plan file. You could see the results by fingering `pwrchute@kadets.d20.co` while it lasted:

```
Last login Sat May 11 21:33 (MDT) on ttyp0 from intrepid.rmi.net
Plan:
Welcome to the UPS monitor service at kadets.d20.co.edu.
The Smart-UPS attached to kadets generated a report at 14:24:01 on 05/17/96.
During the measured period, the following data points were taken:
Voltage ranged from 115.0 VAC to 116.3 VAC.
The UPS generated 116.3 VAC at 60.00 Hz.
```

```
    The battery level was at 27.60 volts.
    The load placed on the UPS was 024.9 percent.
    UPS temperature was measured at 045.0 degrees Celsius.
    Measurements are taken every 10 minutes by the upsd daemon.
    This report is generated by a script written by Russell Kroll<rkroll@kadets>.
    Modified for compatibility with the BSD/OS cron daemon by Neil Schroeder
```

This same status data could also be seen with a web browser, since we had rigged up a CGI wrapper script which called finger.

4.1.2 January 1997: initial protocol tests

Initial tests with a freestanding non-daemon program provided a few basic status registers from the UPS. The 940-0024C cable was not yet understood, so this happened over the [attachment:apcevilhack.jpg evil two-wire serial hack].

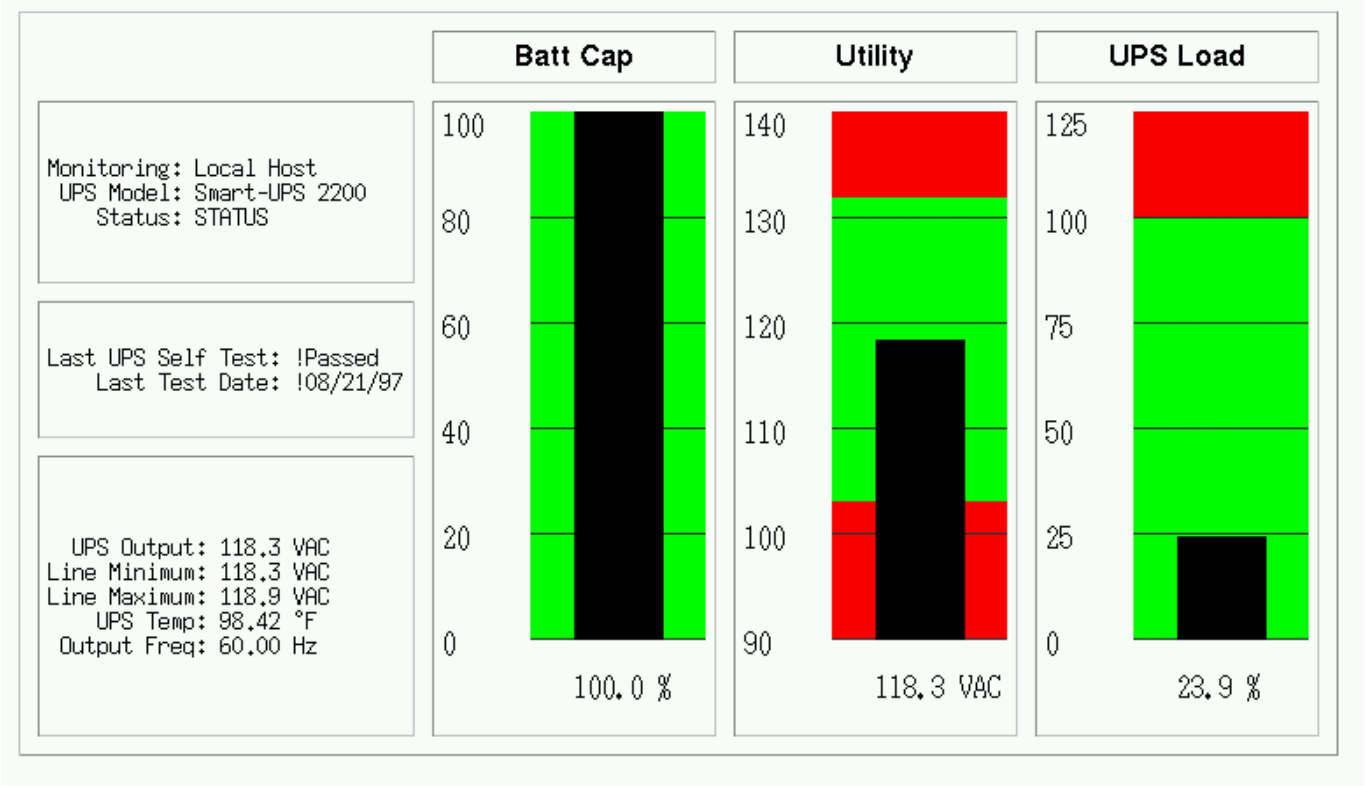
```
    Communicating with SMART-UPS 700 S/N WS9643050926 [10/17/96]
    Input voltage range: 117.6 VAC - 118.9 VAC
    Load is 010.9% of capacity, battery is charged to 100.0% of capacity
```

Note that today's apcsmart driver still displays the serial number when it starts, since it is derived from this original code.

4.1.3 September 1997: first client/server code

The first split daemon/client code was written. upsd spoke directly to the UPS (APC Smart models only) and communicated with upsc by sending binary structures in UDP datagrams.

The first CGI interface existed, but it was all implemented with shell scripts. The main script would call upsc to retrieve status values. Then it would cat a template file through sed to plug them into the page.



upsstats actually has since returned to using templates, despite having a period in the middle when it used hardcoded HTML.

The images were also created with shell scripts. Each script would call upsc to get the right value (utility, upsload, battcap). It then took the value, plugged it into a command file with sed, and passed that into *fly*, a program which used an interpreted language to create images. fly actually uses gd, just like upsimage does today.

This code later evolved into Smart UPS Tools 0.10.

4.2 Smart UPS Tools

4.2.1 March 1998: first public release

Version 0.10 was released on March 10, 1998. It used the same design as the pre-release prototype. This made expansion difficult as the binary structure used for network communications would break any time a new variable was added. Due to byte-ordering and struct alignment issues, the code usually couldn't talk over the network to a system with a different architecture. It was also hopelessly bound to one type of UPS hardware.

Five more releases followed with this design followed. The last was 0.34, released October 27, 1998.

4.2.2 June 1999: Redesigned, rewritten

Following a long period of inactivity and two months of prerelease testing versions, 0.40.0 was released on June 5, 1999. It featured a complete redesign and rewrite of all of the code. The layering was now in three pieces, with the single driver (smartups) separate from the server (upsd).

Clients remained separate as before and still used UDP to talk to the server, but they now used a text-based protocol instead of the brittle binary structs. A typical request like "REQ UTILITY" would be answered with "ANS UTILITY 120.0".

The ups-trust425-625 driver appeared shortly after the release of 0.40.0, marking the first expansion beyond APC hardware.

Over the months that followed, the backupspro driver would be forked from the smartups driver to handle the APC Back-UPS Pro line. Then the backups driver was written to handle the APC Back-UPS contact-closure models. These drivers would later be renamed and recombined, with smartups and backupspro becoming apcsmart, and backups became genericups.

The drivers stored status data in an array. At first, they passed this data to upsd by saving it to a file. upsd would reread this file every few seconds to keep a copy for itself. This was later expanded to allow shared memory mode, where only a stub would remain on the disk. The drivers and server then passed data through the shared memory space.

upsd picked up the ability to monitor multiple drivers on the system, and the "upsname@hostname" scheme was born. Access controls were added, and then the network code was expanded to allow TCP communications, which at this point were on port 3305.

4.3 Network UPS Tools

4.3.1 September 1999: new name, new URL

Several visitors to the web page and subscribers to the mailing lists provided suggestions to rename the project. The old name no longer accurately described it, and it was perilously close to APC's "Smart-UPS" trademark. Rather than risk problems in the future, the name was changed. Kern Sibbald provided the winner: Network UPS Tools, which captures the essence of the project and makes for great short tarball filenames: nut-x.y.z.tar.gz.

The new name was first applied to 0.42.0, released October 31, 1999. This is also when the web pages moved from the old `http://www.exploits.org/~rkroll/smartupstools/` URL to the replacement at `http://www.exploits.org/nu` to coincide with the name change.

More drivers were written and the hardware support continued to grow. upsmon picked up the concepts of what is now known as "primary" and "secondary", and could now handle environments where multiple systems get power from a single UPS.

Manager mode was added to allow changing the value of read/write variables in certain UPS models.

4.3.2 June 2001: common driver core

Up to this point, all of the drivers compiled into freestanding programs, each providing their own implementation of main(). This meant they all had to check the incoming arguments and act uniformly. Unfortunately, not all of the programs behaved the same way, and it was hard to document and use consistently. It also meant that startup scripts had to be edited depending on what kind of hardware was attached.

Starting in 0.45.0, released June 11, 2001, there was a new common core for all drivers called `main.c`. It provided the main function and called back to the `upsdrv_*` functions provided by the hardware-specific part of the drivers. This allowed driver authors to focus on the UPS hardware without worrying about the housekeeping stuff that needs to happen.

This new design provided an obvious way to configure drivers from one file, and so `ups.conf` was born. This eventually spawned upsdrvctl, and now all drivers based on this common core could be started or stopped with one command. Startup scripts now could contain "upsdrvctl start", and it didn't matter what kind of hardware or how many UPSes you had on one system.

Interestingly, at the end of this month, Arnaud Quette entered the UPS world, as a subcontractor of the now defunct MGE UPS SYSTEMS. This marked the start of a future successful collaboration.

4.3.3 May 2002: casting off old drivers, IANA port, towards 1.0

During the 0.45.x series, both the old standalone drivers and the ones which had been converted to the common core were released together. Before the release of 0.50.0 on May 24, 2002, all of the old drivers were removed. While this shrank the list of supported hardware, it set the precedent for removing code which isn't receiving regular maintenance. The assumption is that the code will be brought back up to date by someone if they actually need it. Otherwise, it's just dead weight in the tree.

This change meant that all remaining drivers could be controlled with the `upsdrvctl` and `ups.conf`, allowing the documentation to be greatly simplified. There was no longer any reason to say "do this, unless you have this driver, then do this".

IANA granted an official port number to the project, and the network code switched to port 3493. It had previously been on 3305 which is assigned to `odette-ftp`. 3305 was probably picked in 1997 because it was the fifth project to spawn from some common UDP server code.

After 0.50.1, the 0.99 tree was created to provide a tree which would receive nothing but bug fixes in preparation for the release of 1.0. As it turned out, very few things required fixing, and there were only three releases in this tree.

4.4 Leaving 0.x territory

4.4.1 August 2002: first stable tree: NUT 1.0.0

After nearly 5 years of having a 0.x version number, 1.0.0 was released on August 19, 2002. This milestone meant that all of the base features that you would expect to find were intact: good hardware support, a network server with security controls, and system shutdowns that worked.

The design was showing signs of wear from the rapid expansion, but this was intentionally ignored for the moment. The focus was on getting a good version out that would provide a reasonable base while the design issues could be addressed in the future, and I'm confident that we succeeded.

4.4.2 November 2002: second stable tree: NUT 1.2.0

One day after the release of 1.0.0, 1.1.0 started the new development tree. During that development cycle, the CGI programs were rewritten to use template files instead of hard-coded HTML, thus bringing back the flexibility of the original unreleased prototype from 5 years before. The `multimon` was removed from the tree, as the new `upsstats` could do both jobs by loading different templates.

A new client library called upsclient was created, and it replaced upsfetch. This new library only supported TCP connections, and used an opaque context struct to keep state for each connection. As a result, client programs could now do things that used multiple connections without any conflicts. This was done primarily to allow OpenSSL support, but there were other benefits from the redesign.

upsd and the clients could now use OpenSSL for basic authentication and encryption, but this was not included by default. This was provided as a bonus feature for those users who cared to read about it and enable the option, as the initial setup was complex.

After the 1.1 tree was frozen and deemed complete, it became the second stable tree with the release of 1.2.0 on November 5, 2002.

4.4.3 April 2003: new naming scheme, better driver glue, and an overhauled protocol

Following an extended period with no development tree, 1.3.0 got things moving again on April 13, 2003. The focus of this tree was to rewrite the driver-server communication layer and replace the static naming scheme for variables and commands.

Up to this point, all variables had names like STATUS, UTILITY, and OUTVOLT. They had been created as drivers were added to the tree, and there was little consistency. For example, it probably should have been INVOLT and OUTVOLT, but there was no OUTVOLT originally, so UTILITY was all we had. This same pattern repeated with ACFREQ — is it incoming or outgoing? — and many more.

To solve this problem, all variables and commands were renamed to a hierarchical scheme that had obvious grouping. STATUS became ups.status. UTILITY turned into input.voltage, and OUTVOLT is output.voltage. ACFREQ is input.frequency, and the new output.frequency is also now supported. Every other variable or command was renamed in this fashion.

These variables had been shared between the drivers and upsd as values. That is, for each name like STATUS, there was a #define somewhere in the tree with an INFO_ prefix that gave it a number. INFO_STATUS was 0x0006, INFO_UTILITY was 0x0004, and so on, with each name having a matching number. This number was stored in an int within a structure which was part of the array that was either written to disk or shared memory.

That structure had several restrictions on expansion and was dropped as the data sharing method between the drivers and the server. It was replaced by a new system of text-based messages over Unix domain sockets. Drivers now accepted a short list of commands from upsd, and would push out updates asynchronously. upsd no longer had to poll the state files or shared memory. It could just select all of the driver and client fds and act on events.

At the same time, the network protocol on port 3493 was overhauled to take advantage of the new naming scheme. The existing "REQ STATUS@su700", "ANS STATUS@su700 OL" scheme was showing signs of age, and it really only supported the UPS name (@su700) as an afterthought. The new protocol would now use commands like GET and LIST, leading to exchanges like "GET VAR su700 ups.status" and "VAR su700 ups.status OL". These responses contain enough data to stand alone, so clients can now handle them asynchronously.

4.4.4 July 2003: third stable tree: NUT 1.4.0

On July 25, 2003, 1.4.0 was released. It contained support for both the old "REQ" style protocol (with names like STATUS), and the new "GET" style protocol (with names like ups.status). This tree is provided to bridge the gap between all of the old releases and the upcoming 2.0.

2.0 will be released without support for the old REQ/STATUS protocol. The hope is that client authors and those who have implemented their own monitoring software will use the 1.4 cycle to change to the new protocol. The 1.4 releases contain a lot of compatibility code to make sure both work at the same time.

4.4.5 July 2003: pushing towards 2.0

1.5.0 forked from 1.4.0 and was released on July 29, 2003. The first changes were to throw out anything which was providing compatibility with the older versions of the software. This means that 1.5 and the eventual 2.0 will not talk to anything older than 1.4.

This tree continues to evolve with new serial routines for the drivers which are intended to replace the aging upscommon code which dates back to the early 0.x releases. The original routines would call alarm and read in a tight loop while fetching characters. The new functions are much cleaner, and wait for data with select. This makes for much cleaner code and easier strace/ktrace logs, since the number of syscalls has been greatly reduced.

There has also been a push to make sure the data from the UPS is well-formed and is actually usable before sending updates out to upsd. This started during 1.3 as drivers were adapted to use the dstate functions and the new variable/command names. Some

drivers which were not converted to the new naming scheme or didn't do sanity checks on the incoming UPS data from the serial port were dropped from the tree.

This tree was released as 2.0.0.

4.5 Backwards and Forwards Compatibility (NUT v1.x vs. v2.x)

The old network code spans a range from about 0.41.1 when TCP support was introduced up to the recent 1.4 series. It used variable names like STATUS, UTILITY, and LOADPCT. Many of these names go back to the earliest prototypes of this software from 1997. At that point there was no way to know that so many drivers would come along and introduce so many new variables and commands. The resulting mess grew out of control over the years.

During the 1.3 development cycle, all variables and instant commands were renamed to fit into a tree-like structure. There are major groups, like input, output and battery. Members of those groups have been arranged to make sense - input.voltage and output.voltage compliment each other. The old names were UTILITY and OUTVOLT. The benefits in this change are obvious.

The 1.4 clients can talk to either type of server, and can handle either naming scheme. 1.4 servers have a compatibility mode where they can answer queries for both names, even though the drivers are internally using the new format.

When 1.4 clients talk to 1.4 or 2.0 (or more recent) servers, they will use the new names.

Here's a table to make it easier to visualize:

	Server version			
Client version	1.0	1.2	1.4	2.0+
1.0	yes	yes	yes	no
1.2	yes	yes	yes	no
1.4	yes	yes	yes	yes
2.0+	no	no	yes	yes

Version 2.0, and more recent, do not contain backwards compatibility for the old protocol and variable/command names. As a result, 2.0 clients can't talk to anything older than a 1.4 server. If you ask a 2.0 client to fetch "STATUS", it will fail. You'll have to ask for "ups.status" instead.

Authors of separate monitoring programs should have used the 1.4 series to write support for the new variables and command names. Client software can easily support both versions as long as they like. If upsd returns *ERR UNKNOWN-COMMAND* to a GET request, you need to use REQ.

4.6 networkupstools.org

4.6.1 November 2003: a new URL

The bandwidth demands of a project like this have slowly been forcing me to offload certain parts to other servers. The download links have pointed offsite for many months, and other large things like certain UPS protocols have followed. As the traffic grows, it's clear that having the project attached to exploits.org is not going to work.

The solution was to register a new domain and set up mirrors. There are two initial web servers, with more on the way. The main project URL has changed from `http://www.exploits.org/nut/` to https://www.networkupstools.org. The actual content is hosted on various mirrors which are updated regularly with rsync, so the days of dribbling bits through my DSL should be over.

This is also when all of the web pages were redesigned to have a simpler look with fewer links on the left side. The old web pages used to have 30 or more links on the top page, and most of them vanished when you dropped down one level. The links are now constant on the entire site, and the old links now live in their own groups in separate directories.

4.7 Second major version

4.7.1 March 2004: NUT 2.0.0

NUT 2.0.0 arrived on March 23, 2004. The jump to version 2 shows the difference in the protocols and naming that happened during the 1.3 and 1.5 development series. 2.0 no longer ships with backwards compatibility code, so it's smaller and cleaner than 1.4.

4.8 The change of leadership

4.8.1 February 2005: NUT 2.0.1

The year 2004 was marked by a release slowdown, since Russell was busy with personal subjects. But the patches queue was still growing quickly.

At that time, the development process was still centralized. There was no revision control system (like the current Subversion repository), nor trackers to interact with NUT development. Russell was receiving all the patches and requests, and doing all the work on his own, including releases.

Russell was more and more thinking about giving the project leadership to Arnaud Quette, which finally happened with the 2.0.1 release in February 2005.

This marked a new era for NUT...

First, Arnaud aimed at opening up the development by creating a project on the Debian Alioth Forge. This allowed to build the team of hackers that Russell dreamed about. It also allows to ensure NUT's continuation, whatever happens to the leader. And that would most of all boost the projects contributions.